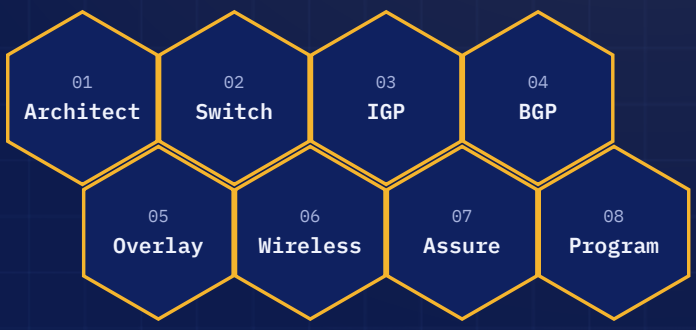
ZCNP

ZyeemZia™ Certified Network Professional

Engineer the enterprise.



ZCNP

Enterprise Fabric

COURSE BOOK · WORKBOOK · EXAM PREPARATION

8 MODULES · 38 LESSONS · 60 EXAM QUESTIONS

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BEFORE YOU BEGIN

How to use this book

A professional certification for engineers who run real enterprise networks: campus architecture, advanced Layer 2, OSPF and BGP, overlays and SD-WAN, enterprise wireless, assurance and model-driven automation.

01
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Work through 38 lessons. Each ends with a knowledge check.

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Eight-week study plan

WHEN	MODULE	DO THIS	DONE
Week 1–2	m1 Architect	Campus design study	☐
Week 3	m2 Switch	MST and protection lab	☐
Week 4–5	m3 IGP	Multi-area OSPF and redistribution lab	☐
Week 6–7	m4 BGP	Dual-ISP BGP lab	☐
Week 8	m5 Overlay	VRF, IPsec and SD-WAN lab	☐
Week 9	m6 Wireless	Enterprise WLAN design	☐
Week 10	m7 Assure	Telemetry and 802.1X lab	☐
Week 11–12	m8 Program	Automation pipeline, capstone, exam	☐

Learner name .....

Start date ..... Target exam date .....

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4. Fabric Campus, WAN and Cloud Edge Placement · 35 min

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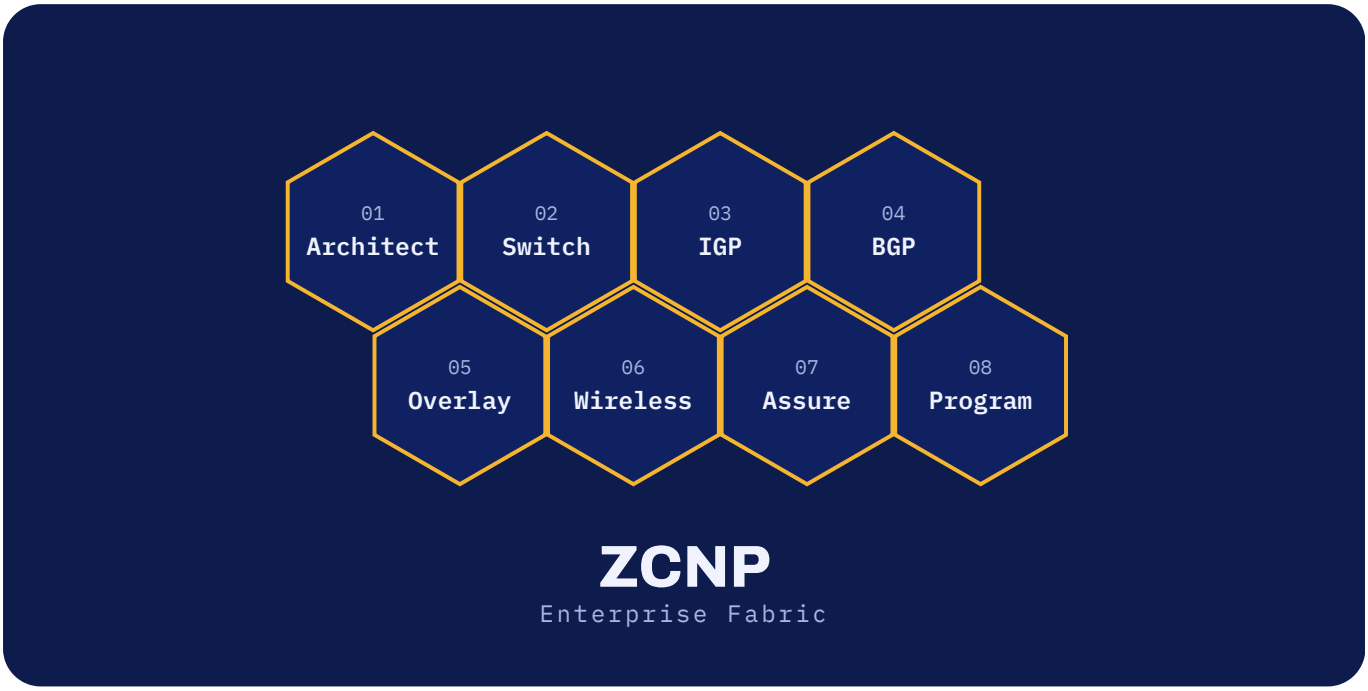
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THE FRAMEWORK

The ZCNP Fabric

Eight interlocking cells that together make an enterprise network: architecture, switching, IGP, BGP, overlays, wireless, assurance and programmability. Strengthen one cell and the whole fabric gets stronger.



01 Architect
Hierarchy, fabrics, high availability and edge placement — the decisions every other cell depends on.

02 Switch
Rapid and multiple spanning tree, protection features, link aggregation, private VLANs and a solid troubleshooting method.

03 IGP
Multi-area OSPF, special areas, OSPFv3, IS-IS fundamentals and safe route control between domains.

04 BGP
Sessions, attributes, policy, route reflection and dual-ISP traffic engineering, with a method for finding why a route is wrong.

05 Overlay
Many logical networks on one underlay: VRFs, tunnels, IPsec, VXLAN EVPN and SD-WAN.

06 Wireless
Design, secure and operate WLANs that carry voice, video and critical devices.

07 Assure
See everything, prove service levels, find root cause fast and protect the network itself.

08 Program
Model-driven APIs, automation code, pipelines and closed-loop operations, with guardrails.

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01

MODULE 1 · ARCHITECT

Enterprise Network Architecture

Hierarchy, fabrics, high availability and edge placement — the decisions every other cell depends on.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design a hierarchical campus and decide when to collapse or separate the core.
- Plan a spine-leaf fabric, including oversubscription and scale-out.
- Choose between Layer 2, multi-chassis and routed access with clear high-availability reasoning.
- Explain fabric campus underlay, overlay and policy, and place WAN, internet and cloud edges.
- Record architecture decisions and trade-offs in a reusable format.

LESSONS

1. The Hierarchical Campus and the ZCNP Fabric · 30 min
2. Spine-Leaf and Data-Centre Fabrics · 30 min
3. Access Designs and High Availability · 35 min
4. Fabric Campus, WAN and Cloud Edge Placement · 35 min

MODULE 1 · ARCHITECT · LESSON 1 · 30 MIN

The Hierarchical Campus and the ZCNP Fabric

Access, distribution and core: why the three-tier model still anchors enterprise design, and when to collapse it.

- Explain the role, functions and failure domain of the access, distribution and core layers.
- Decide when a collapsed-core design is the better choice for a site.
- Use the eight cells of the ZCNP Fabric as a design-review agenda.

A professional network engineer is judged less on whether a single protocol works and more on whether the **whole network behaves predictably** when links fail, traffic doubles or a new building opens. That predictability comes from structure. The hierarchical campus model splits the network into layers with clear jobs, so that each layer can be sized, secured and troubleshot on its own terms.

The ZCNP Fabric — eight interlocking cells

C1
Architect
Topology, layering, redundancy and placement decisions that every other cell depends on.

C2
Switch
Loop-free, resilient Layer 2: RSTP, MST, protection features and link aggregation.

C3
IGP
Fast, scalable internal routing with OSPF or IS-IS, plus redistribution and route control.

C4
BGP
Policy-driven routing at the edges: providers, data centres and large internal domains.

C5
Overlay
VRFs, tunnels, VPNs and fabric encapsulations that separate services over shared infrastructure.

C6
Wireless
Enterprise WLAN design, roaming and security as a first-class access layer.

C7
Assure
Telemetry, flow data, access control and a disciplined troubleshooting method.

C8
Program
Models, APIs and automation that turn designs into repeatable, testable changes.

INSIGHT · CELLS INTERLOCK

A design choice in one cell always lands somewhere else. Choosing Layer 3 access (Architect) removes most spanning-tree work (Switch) but doubles the number of IGP adjacencies (IGP). Review every change against all eight cells, not just the one you are touching.

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Three-tier campus hierarchy

Core	`High-speed Layer 3 transport between distribution blocks`
	`No end-user ports and minimal policy`
	`Fast convergence, ECMP, simple configuration`

Distribution	`Aggregates access switches for one building or zone`
	`Layer 2/Layer 3 boundary in classic designs`
	`Route summarisation, policy, first-hop redundancy`

Access	`Connects users, phones, access points and IoT` `Port security, 802.1X, PoE, QoS marking` `Edge spanning-tree protection`

What each layer is responsible for

LAYER	PRIMARY JOB	TYPICAL FEATURES	DESIGN RULE OF THUMB
Access	Attach endpoints	PoE, VLANs, 802.1X, BPDU guard, QoS trust	Keep it consistent and templated
Distribution	Aggregate and apply policy	SVIs, VRRP, ACLs, summarisation, uplink ECMP	Deploy in redundant pairs
Core	Move packets fast between blocks	Routed point-to-point links, ECMP, fast IGP timers	Never connect hosts or apply heavy policy

A **distribution block** is one pair of distribution switches plus the access switches beneath them. It is also a failure domain: a broadcast storm or misconfiguration inside one block should never spill into the core or into other blocks. Summarising the block's address space at the distribution layer reinforces this — the core sees one route per block, so a flapping access subnet does not ripple through the whole routing domain.

Three-tier versus collapsed core

Three-tier (dedicated core)

- Best for large campuses with many buildings or distribution blocks
- Core can grow independently of the edge
- Fewer links: blocks connect only to the core, not to each other
- Higher equipment and cabling cost

Collapsed core

- Distribution and core functions merged into one redundant pair
- Ideal for a single building or a medium site
- Lower cost and fewer devices to manage
- Becomes a bottleneck when blocks keep being added

Sizing the core decision

MEASURE	FORMULA	WHAT IT TELLS YOU
Full-mesh links between blocks	`links = n × (n − 1) / 2`	Without a core, connecting n distribution pairs directly grows quadratically: 4 blocks need 6 meshes, 8 blocks need 28.
Links with a dedicated core	`links ≈ 2 × n (per core pair)`	Each block uplinks to the two core switches, so cabling grows linearly with the number of blocks.

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PRO TIP · A PRACTICAL THRESHOLD

When a site has more than two or three distribution blocks, or plans to add buildings within the hardware's life cycle, a dedicated core usually pays for itself in simpler cabling and isolated change windows.

Distribution pair with VRRP gateway redundancy

DS1 · IOS-style CLI

```
interface Vlan20
 description USERS-FLOOR2
 ip address 10.20.20.2 255.255.255.0
 vrrp 20 ip 10.20.20.1
 vrrp 20 priority 110
 vrrp 20 preempt
!
DS1# show vrrp brief
Interface   Grp Pri Time  Own Pre State   Master addr   Group addr
Vl20        20  110 3570       Y  Master  10.20.20.2    10.20.20.1
```

VRRP (RFC 5798) gives hosts one virtual gateway. Align the VRRP master with the spanning-tree root for that VLAN so traffic does not take an extra hop across the inter-switch link.

SCENARIO · DECIDE · CHOOSING THE TOPOLOGY

A logistics company is building a single four-storey office for 600 staff, with a small server room. Growth is expected to be modest for the next six years. The architect proposes three tiers with a dedicated core pair. What is the better recommendation?

- (A) Use a collapsed core: one redundant distribution/core pair aggregating the floor switches.
- (B) Keep the dedicated core because three tiers are always the professional standard.
- (C) Remove the distribution layer entirely and connect every access switch straight to the core.
- (D) Use a single large chassis with no redundancy to save money.

Circle your choice. Answer and feedback: module answer key.

Key points

- Hierarchy creates predictable failure domains and lets each layer be sized for its job.
- The core moves packets; it does not host users or heavy policy.
- Collapse the core when the site is small; separate it when blocks multiply.
- Review every design decision against all eight cells of the ZCNP Fabric.

KNOWLEDGE CHECK

1. Which layer of the hierarchical campus is the usual place to summarise a building's subnets toward the rest of the network?
   - (A) The access layer
   - (B) The distribution layer
   - (C) The core layer
   - (D) The internet edge routers
2. A campus has two buildings and no growth plans. Which design is usually most appropriate?
   - (A) A collapsed core with one redundant pair
   - (B) A dedicated three-tier core and distribution
   - (C) A full mesh between every access switch
   - (D) A single non-redundant core chassis

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3. Why should end-user devices not be connected directly to core switches?

(A) Core switches are unable to forward any Layer 2 frames between ports

(B) Core ports are unable to supply Power over Ethernet

(C) Hosts cannot use addresses from the core's subnets

(D) It mixes roles and drags edge policy into the fast transport layer

4. How many links are needed to full-mesh six distribution blocks without a core?

(A) 12

(B) 30

(C) 15

(D) 36

5. On a distribution pair running VRRP, why align the VRRP master with the spanning-tree root for the same VLAN?

(A) To avoid traffic crossing the inter-switch link before routing

(B) Because VRRP refuses to run on the non-root switch

(C) To let both switches answer ARP for the virtual MAC

(D) Because only the root bridge is allowed to hold an SVI for that VLAN

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MODULE 1 · ARCHITECT · LESSON 2 · 30 MIN

Spine-Leaf and Data-Centre Fabrics

Clos topologies, ECMP and oversubscription: the design behind modern data centres and large campuses.

- Describe the spine-leaf topology and the rules that keep it predictable.
- Calculate oversubscription ratios and plan scale-out.
- Contrast a routed spine-leaf underlay with a traditional three-tier data centre.

Traditional data centres were built like campuses: access, aggregation and core, with Layer 2 stretched widely and spanning tree blocking half the links. Modern application traffic is mostly **east-west** — server to server, service to service — and that design gives unequal paths and wasted capacity. The **spine-leaf** topology, a form of folded Clos network, fixes this.

Spine-leaf rules

- Every leaf connects to every spine.
- Leaves never connect to other leaves for data traffic, and spines never connect to other spines.
- Servers, storage, firewalls and border routers attach only to leaves.
- Any leaf-to-leaf path is exactly leaf → spine → leaf: two hops, equal cost.
- Links are routed point-to-point and load-shared with ECMP.

A two-tier spine-leaf fabric

Spine	`4 switches` `Pure transit: no endpoints` `Adds bandwidth when you add a spine`
Leaf	`Top-of-rack switches` `Default gateway (often anycast)` `Border leaves for WAN, internet and firewalls`
Endpoints	`Servers and hypervisors` `Storage` `Service appliances`

Oversubscription and scale

MEASURE	FORMULA	WHAT IT TELLS YOU
Leaf oversubscription	`ratio = total server-facing bandwidth ÷ total spine-facing bandwidth`	48 × 25G down (1,200G) with 6 × 100G up (600G) gives 2:1. Lower is better for east-west heavy workloads.
Maximum leaves	`leaves ≤ ports per spine`	Each leaf consumes one port on every spine, so spine port count caps the fabric's width.
ECMP paths between two leaves	`paths = number of spines (with one link per spine)`	Losing one of four spines removes 25% of fabric capacity, not connectivity.

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Three-tier DC versus spine-leaf

Three-tier data centre

- Optimised for north-south traffic
- Spanning tree blocks redundant links
- Unequal path lengths between racks
- Scale-up: buy bigger aggregation switches

Spine-leaf fabric

- Optimised for east-west traffic
- All links active through Layer 3 ECMP
- Predictable two-hop latency
- Scale-out: add spines for bandwidth, leaves for ports

The **underlay** of a spine-leaf fabric is usually a simple routed network: point-to-point /31 links, loopbacks for each switch, and either OSPF, IS-IS or eBGP as the routing protocol. eBGP underlays, with a private AS number per leaf (or per rack) and one shared AS for the spines, are common in very large fabrics because BGP's policy and scale are well suited to thousands of links. Layer 2 services for workloads are then delivered as an **overlay**, typically VXLAN with an EVPN control plane — covered in the Overlay cell.

Routed leaf uplink in an OSPF underlay

LEAF1 · IOS-style CLI

```
interface Ethernet1/49
 description to SPINE1
 no switchport
 mtu 9216
 ip address 10.0.1.1 255.255.255.254
 ip ospf network point-to-point
 ip ospf 1 area 0
!
interface Loopback0
 ip address 10.255.1.1 255.255.255.255
 ip ospf 1 area 0
```

/31 masks (RFC 3021) save address space on point-to-point links. The jumbo MTU leaves room for overlay encapsulation headers.

WATCH OUT · DO NOT CROSS-CONNECT LEAVES

Adding a leaf-to-leaf link 'for extra resilience' breaks the equal-cost model and can create unexpected traffic patterns. The exception is a dedicated peer link used by a multi-chassis aggregation pair for dual-homed servers, which is not used for routed fabric traffic.

Planning a spine-leaf build

1. **Count endpoints and growth** Estimate server ports per rack and racks per hall for the hardware life cycle.
2. **Pick the leaf model** Choose port speeds and uplink count; this sets the oversubscription ratio.
3. **Size the spines** Spine port count must cover the maximum number of leaves you will ever need.
4. **Design the underlay** Address plan for /31 links and loopbacks, routing protocol, MTU and BFD.
5. **Design the overlay and borders** Tenant VRFs, gateway model and where WAN, internet and firewalls attach (border leaves).

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SCENARIO · DECIDE · CAPACITY ALARM

A fabric has four spines and 20 leaves, each leaf with 4 × 100G uplinks and 48 × 25G server ports. Storage replication jobs saturate uplinks at night. The spines have spare ports. What is the most effective change?

(A) Add more leaves so the replication traffic spreads across more racks.

(B) Connect the busiest leaves directly to each other with 100G links.

(C) Add spines and a matching uplink on each leaf to raise leaf uplink bandwidth.

(D) Enable spanning tree on the spine uplinks so that it blocks the congested paths.

Circle your choice. Answer and feedback: module answer key.

Key points

- Spine-leaf gives every pair of leaves equal-cost, two-hop paths.
- Add spines for bandwidth, leaves for ports; spine port count limits width.
- Oversubscription = downlink bandwidth ÷ uplink bandwidth at the leaf.
- The underlay is routed; workload Layer 2 is delivered as an overlay.

KNOWLEDGE CHECK

6. In a spine-leaf fabric, where should a firewall cluster and the WAN routers attach?

   (A) Directly to the spines
   (B) To a dedicated spine-to-spine link
   (C) To border leaves
   (D) To any access switch in the campus

7. A leaf has 32 × 100G server ports and 8 × 400G uplinks. What is the oversubscription ratio?

   (A) 2:1
   (B) 4:1
   (C) 1:2
   (D) 1:1

8. What primarily limits the maximum number of leaves in a two-tier spine-leaf fabric?

   (A) The number of server ports on each leaf
   (B) The number of ports on each spine switch
   (C) The number of VLANs configured on the leaves
   (D) The maximum MTU supported by the uplinks

9. Why are /31 masks common on spine-leaf links?

   (A) They give two usable addresses on a point-to-point link
   (B) They are required for ECMP load sharing to work
   (C) They prevent the IGP from forming adjacencies
   (D) They let spanning tree keep running on routed uplink ports

10. A fabric with four spines loses one spine. What is the effect on leaf-to-leaf traffic?

    (A) All leaf-to-leaf traffic stops until it is replaced
    (B) Capacity drops by about a quarter; connectivity remains
    (C) Traffic is switched through peer links between the leaves
    (D) Spanning tree reconverges and blocks a spine link

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MODULE 1 · ARCHITECT · LESSON 3 · 35 MIN

Access Designs and High Availability

Layer 2 versus Layer 3 access, redundant links, multi-chassis aggregation, stacking and fast convergence.

- Compare Layer 2 access, Layer 3 (routed) access and multi-chassis designs.
- Select high-availability mechanisms for each layer and explain their failure behaviour.
- Configure a routed access uplink with fast failure detection.

Where you put the Layer 2/Layer 3 boundary is one of the most consequential campus decisions. It decides whether spanning tree is active in the uplinks, whether a VLAN can span more than one access switch, how fast the network recovers from a failure, and how many routing adjacencies you must operate.

Three access models

Layer 2 access (classic)

- Access switches trunk VLANs to a distribution pair
- Distribution holds SVIs and runs VRRP
- Spanning tree blocks one uplink per VLAN
- VLANs can span access switches if needed

Layer 2 with multi-chassis aggregation

- Distribution pair acts as one logical switch for link aggregation
- Each access switch bundles both uplinks into one LAG
- No spanning-tree blocking on uplinks; both links forward
- Requires the pair's peer link and consistent configuration

Layer 3 (routed) access

- Access switch is the default gateway for its VLANs
- Uplinks are routed links with ECMP
- No spanning tree on uplinks; IGP convergence instead
- VLANs are confined to one access switch

Design trade-offs at the access boundary

CRITERION	LAYER 2 ACCESS	MULTI-CHASSIS LAG	ROUTED ACCESS
Uplink utilisation	One uplink blocked per VLAN	Both uplinks active (hash)	Both uplinks active (ECMP)
Failure recovery	RSTP, usually sub-second to a few seconds	Member link loss: sub-second	IGP with BFD: sub-second
VLAN spanning switches	Supported	Supported	Not supported without an overlay
Operational complexity	Spanning tree and VRRP tuning	Pair synchronisation and peer link	More routed devices and adjacencies
Typical fit	Small or legacy sites	Sites needing VLAN mobility	Large, modern campuses

DEFINITION · MULTI-CHASSIS LINK AGGREGATION

Two physical switches present a single LACP system identity to a downstream device, so that device can bundle links to both switches into one logical link. Implementations differ between vendors and are not interoperable across vendors; the concept is the same — two control planes coordinate through a peer link so one switch failure does not take the bundle down.

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Stacking versus multi-chassis pair

Switch stack

- Several switches share one control plane and one management address
- Simple to operate: configured as one device
- Software upgrades often affect the whole stack
- A control-plane fault can affect every member

Multi-chassis pair

- Two independent control planes coordinate state
- Each switch can be upgraded separately in many designs
- More configuration to keep consistent
- Better isolation of software faults

Routed access uplink with BFD

AS1 · IOS-style CLI

```
interface GigabitEthernet1/1/1
 description to DS1
 no switchport
 ip address 10.1.0.1 255.255.255.254
 ip ospf network point-to-point
 ip ospf 1 area 10
 bfd interval 300 min_rx 300 multiplier 3
!
router ospf 1
 router-id 10.255.10.1
 passive-interface default
 no passive-interface GigabitEthernet1/1/1
 no passive-interface GigabitEthernet1/1/2
 bfd all-interfaces
```

BFD (RFC 5880) detects a dead neighbour in about 900 ms with these values, far faster than default OSPF dead timers. Passive-by-default stops the switch forming adjacencies on user VLANs.

Failure detection time

MEASURE	FORMULA	WHAT IT TELLS YOU
BFD detection	`detect ≈ interval × multiplier`	300 ms × 3 ≈ 900 ms before the routing protocol is told the neighbour is down.
OSPF default dead time	`dead = 4 × hello`	With a 10 s hello the dead interval is 40 s — too slow for voice and video if the link does not go physically down.

High availability is more than duplicate boxes. Plan **redundancy at every layer**: dual power supplies on separate feeds, dual uplinks landing on different line cards or switches, a redundant distribution pair, and diverse fibre routes between buildings. Then make sure the control plane reacts quickly: point-to-point network types, BFD, tuned timers and graceful restart or non-stop forwarding where the platform supports it.

> **WATCH OUT · REDUNDANCY YOU CANNOT SEE**
> Two uplinks running through the same conduit, or two power supplies on the same circuit, are one failure. Walk the physical path and the power chain during design reviews.

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SCENARIO · DECIDE · CHOOSING THE ACCESS MODEL

A hospital campus is replacing its access layer. Clinical devices must keep the same subnet when moved between wards on the same floor, and each floor has four access switches. The network team wants both uplinks active and no spanning-tree blocking. Which design fits best?

(A) Routed access with the gateway on each access switch.

(B) Layer 2 access with multi-chassis link aggregation to the distribution pair.

(C) Classic Layer 2 access with RSTP and VRRP, blocking one of the two uplinks per VLAN.

(D) A single access switch per floor with no uplink redundancy.

Circle your choice. Answer and feedback: module answer key.

Key points

- Routed access removes spanning tree from uplinks but confines VLANs to one switch.
- Multi-chassis aggregation keeps VLAN reach and makes both uplinks forward.
- Stacks simplify management but share one control plane.
- Fast detection (BFD, point-to-point links) matters as much as duplicate hardware.

KNOWLEDGE CHECK

11. What is the main limitation of routed (Layer 3) access?

(A) Both uplinks cannot be used at the same time
(B) Spanning tree must run on every routed uplink
(C) VRRP is mandatory on each access switch
(D) A VLAN cannot span multiple access switches

12. BFD runs with a 250 ms interval and a multiplier of 4. Roughly how long does failure detection take?

(A) 250 milliseconds
(B) 4 seconds
(C) 1 second
(D) 40 seconds

13. What does a multi-chassis link aggregation pair present to the downstream switch?

(A) Two separate bundles that spanning tree must block
(B) One LACP partner identity across two physical switches
(C) A routed ECMP path through two gateway addresses
(D) A single stack running one shared control plane for both

14. Which risk is most associated with a switch stack compared with a multi-chassis pair?

(A) Stack members cannot be given redundant power supplies
(B) Stacks cannot be managed as a single device
(C) Stacks do not support link aggregation at all
(D) A shared control-plane fault can affect all members

15. Why configure `passive-interface default` on a routed access switch?

(A) So OSPF only forms adjacencies on the intended uplinks
(B) So OSPF stops advertising the user subnets
(C) So BFD can run without any routing protocol
(D) So the switch always elects itself as the designated router

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · ARCHITECT · LESSON 4 · 35 MIN

Fabric Campus, WAN and Cloud Edge Placement

Overlay and underlay, group-based policy, and where the WAN, internet and cloud connect — with a decision table.

- Explain the underlay, overlay and policy planes of a fabric-based campus.
- Place WAN, internet, DMZ and cloud on-ramp functions in an enterprise topology.
- Use a decision table to justify architecture trade-offs.

A **fabric-based campus** applies data-centre ideas to the enterprise LAN. A simple routed **underlay** provides reachability between switch loopbacks. An **overlay** — typically VXLAN encapsulation (RFC 7348) with a control plane that maps endpoints to their location — carries user traffic, so a user's subnet can appear anywhere without stretching VLANs. A **policy plane** classifies users and devices into groups and enforces rules between groups, regardless of IP address.

Fabric campus planes

Policy	`Groups assigned at authentication (802.1X, MAB)` `Group-to-group rules` `Enforced at fabric edge or border`
Overlay	`VXLAN tunnels between fabric nodes` `Virtual networks (VRFs) for segmentation` `Endpoint-to-location mapping control plane`
Underlay	`Routed point-to-point links` `Loopbacks, IGP (often IS-IS or OSPF)` `Jumbo MTU for encapsulation overhead`
Physical	`Access, distribution/intermediate, border nodes` `Redundant fibre and power`

Fabric roles

ROLE	FUNCTION
Edge node	Connects endpoints, authenticates them, encapsulates their traffic into the overlay
Intermediate node	Pure underlay transit; does not need to understand the overlay
Border node	Connects the fabric to external networks: WAN, data centre, internet, other fabrics
Control-plane node	Holds the endpoint-to-location database used to find where a host lives
Policy server	Authenticates users and devices and assigns them to groups

> **INSIGHT · WHY POLICY FOLLOWS GROUPS**
> IP addresses change with DHCP, roaming and cloud workloads. Writing rules as 'Finance users may reach the finance application group' rather than as long address-based ACLs keeps policy stable while addressing moves. Group tags travel with the traffic inside the overlay header in many implementations.

Beyond the campus, an enterprise has several **edges**, and their placement determines security, latency and cost. Keep each edge in its own module so it can be secured and scaled independently, and connect them to the core or to a dedicated edge distribution pair rather than to user access switches.

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Enterprise edge modules

EDGE	CONNECTS TO	TYPICAL PLACEMENT AND CONTROLS
WAN edge	Branches over MPLS, broadband or SD-WAN	Redundant routers off the core or edge distribution; QoS and routing policy
Internet edge	One or more ISPs	Edge routers, then firewalls; BGP when multihomed; DDoS protection
DMZ	Public-facing services	Separate firewall zones; no direct path to internal VLANs
Cloud edge	Public cloud regions	Private interconnect or IPsec from a colocation or the DC; route filtering both ways
Partner/extranet	Suppliers and banks	Dedicated firewall zone with strict, logged policy

Traffic path from a branch user to a SaaS application (direct internet access)

01 **Branch access**
User authenticated, traffic classified

→

02 **Branch edge**
Application-aware path selection

→

03 **Local internet breakout**
Security inspection or cloud security service

→

04 **Provider network**
Best path to the SaaS front end

→

05 **SaaS application**
Served from the nearest region

Architecture decision table

REQUIREMENT	IF THIS IS TRUE…	PREFER…	ACCEPT THIS TRADE-OFF
VLAN must span switches	Legacy devices need the same subnet in many closets	L2 access with multi-chassis LAG, or a fabric overlay	Pair synchronisation or overlay complexity
Fastest, simplest convergence	No VLAN mobility needed	Routed access with BFD	VLANs confined to one switch
Many buildings	More than 2–3 distribution blocks	Dedicated core	Extra hardware and links
Strong segmentation at scale	Many user/device groups, frequent moves	Fabric campus with group-based policy	New skills, controller dependence
Heavy SaaS and cloud use	Most traffic leaves the enterprise	Local internet breakout with cloud security	Distributed security policy to manage

SCENARIO · DECIDE · BRANCHES DROWNING IN BACKHAUL

A distributor's 60 branches send all traffic, including internet and SaaS, over MPLS to the head office, where it passes through a central firewall. Now that most applications are SaaS, branch links and the head-office internet edge are congested and video calls stutter. What is the best architectural change?

- (A) Double the MPLS bandwidth at every branch and at the head office so backhauled traffic fits.
- (B) Move the SaaS applications back into the head-office data centre to keep traffic internal.
- (C) Block video and voice traffic on branch networks so business applications have enough room.
- (D) Add local internet breakout at branches with cloud-delivered or local security inspection.

Circle your choice. Answer and feedback: module answer key.

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CASE STUDY · NATIONAL BANK: HEAD-OFFICE CAMPUS UPGRADE

A national bank's 14-storey head office runs a flat Layer 2 network: two chassis switches in the basement, 70 access switches, 120 VLANs trunked everywhere, and spanning tree blocking half the riser links. A broadcast storm last quarter stopped the dealing room for 40 minutes. The regulator requires segmentation between treasury, retail operations and guest users. The data centre is in another city, connected by two carrier circuits, and internet access is via a single ISP.

1. Which campus model would you propose, and why?

2. How would you contain failures so a storm on one floor cannot affect the dealing room?

3. How would you meet the segmentation requirement?

4. What edge changes would you recommend?

Model answer: module answer key.

TEMPLATE **Architecture decision record**

Record each significant design choice so reviewers can see what was considered and why.

DECISION	CONTEXT AND REQUIREMENT	OPTIONS CONSIDERED	CHOSEN OPTION	TRADE-OFFS ACCEPTED	FABRIC CELLS AFFECTED
Access model for floors 1–10	*No VLAN mobility; storms must be contained*	*L2 access, multi-chassis LAG, routed access*	*Routed access with BFD*	*VLANs confined to one switch*	*Architect, Switch, IGP, Assure*

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Architecture review checklist

☐ Every layer has a clear role and no hosts attach to the core.

☐ Distribution blocks summarise their address space to the core.

☐ Redundant links follow physically diverse paths and power feeds.

☐ Failure detection uses BFD or point-to-point links, not default dead timers.

☐ VLANs are pruned so none spans blocks without a documented reason.

☐ WAN, internet, DMZ and cloud edges are separate modules with their own policy.

☐ Segmentation is designed (VRFs or groups), not left to flat ACLs.

☐ Oversubscription ratios are calculated and accepted for each layer.

☐ Each decision is recorded with its trade-offs.

REFLECT

Sketch your own organisation's network against the hierarchy. Where does the Layer 2/Layer 3 boundary sit today, and what would you change first if you could redesign one building?

Key terms

Distribution block
A distribution pair and the access switches beneath it, treated as one failure and summarisation domain.

Spine-leaf
A two-tier Clos topology in which every leaf connects to every spine, giving equal-cost two-hop paths.

ECMP
Equal-cost multipath: forwarding traffic over several routes with the same cost, usually hashed per flow.

Multi-chassis link aggregation
Two switches presenting one LACP identity so a device can bundle links to both.

BFD
Bidirectional Forwarding Detection (RFC 5880), a lightweight protocol that detects path failure in milliseconds.

Overlay
A virtual network carried over the underlay, usually by encapsulation such as VXLAN.

Group-based policy
Access rules written between groups of users or devices rather than between IP addresses.

Collapsed core
A design where the distribution and core functions are combined into one redundant pair of switches.

Oversubscription
The ratio of downstream (host-facing) bandwidth to upstream (uplink) bandwidth at a switch.

Routed access
An access design where the access switch is the default gateway and uplinks are Layer 3 links.

Switch stack
Several physical switches operating with one shared control plane and management identity.

Underlay
The physical routed network that provides reachability between fabric nodes.

Border node
A fabric node that connects the overlay to external networks.

KNOWLEDGE CHECK

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16. In a fabric campus, which plane is responsible for reachability between switch loopbacks?

(A) The overlay
(B) The policy plane
(C) The underlay
(D) The management plane

17. Why is policy in a fabric campus usually written between groups instead of IP addresses?

(A) Groups stay stable while addresses change with roaming and DHCP
(B) Groups need fewer switch ports than address-based rules
(C) Firewalls cannot inspect any IP addresses in an overlay
(D) Address-based rules cannot be applied to routed links in a fabric

18. Where should the internet edge connect in an enterprise design?

(A) Directly to the user access switches on every floor
(B) To one of the spine switches in the data centre
(C) To the guest wireless VLAN for simplicity
(D) To the core or a dedicated edge distribution pair

19. Which role in a fabric campus authenticates an endpoint and encapsulates its traffic?

(A) The intermediate node
(B) The edge node
(C) The border node
(D) The control-plane node

20. A site needs the fastest, simplest convergence and has no requirement for VLANs to span switches. Which access model does the decision table point to?

(A) Layer 2 access with RSTP
(B) Multi-chassis link aggregation
(C) Routed access with BFD
(D) A flat network with one VLAN

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 1 · ARCHITECT

Answer key

Knowledge checks

1. B Distribution switches aggregate one block and are the natural Layer 3 boundary, so they summarise the block toward the core. The access layer has no routing view, and the core should stay simple.

2. A With few blocks, merging distribution and core into one redundant pair gives the same resilience at lower cost. A dedicated core only pays off as blocks multiply.

3. D The core should be simple and stable. Attaching hosts adds edge features, policy and failure modes to the layer that every block depends on.

4. C n × (n − 1) / 2 = 6 × 5 / 2 = 15. The quadratic growth is the main argument for a dedicated core in large campuses.

5. A If the gateway sits on one switch and the forwarding path leads to the other, frames take an extra Layer 2 hop. Alignment keeps the path short and predictable.

6. C Spines are pure transit. All services, including external connectivity, attach to leaves — often called border or service leaves.

7. D Down: 32 × 100G = 3,200G. Up: 8 × 400G = 3,200G. The ratio is 3,200 ÷ 3,200 = 1:1, a non-blocking leaf.

8. B Every leaf uses one port on every spine, so the spine's port count sets the ceiling. Beyond that, a three-stage design with super-spines is needed.

9. A RFC 3021 allows /31 on point-to-point links, saving half the space of a /30. It has no effect on ECMP or spanning tree.

10. B Each spine is one of four equal-cost paths. Losing one removes its share of bandwidth while ECMP continues over the remaining three.

11. D In routed access the access switch is the gateway, so each VLAN lives on one switch. ECMP keeps both uplinks active and spanning tree is not used on them.

12. C Detection time ≈ interval × multiplier = 250 ms × 4 = 1,000 ms.

13. B The pair coordinates so the downstream device sees one LACP system and bundles links to both. It is not a stack: each switch keeps its own control plane.

14. D Stack members share one control plane, so a software fault or upgrade can affect the whole stack. Multi-chassis pairs keep two control planes.

15. A Passive interfaces still advertise their subnets but send no hellos, so no adjacency can form on user-facing VLANs. Uplinks are then explicitly made non-passive.

16. C The underlay is the routed network that connects fabric nodes. The overlay rides on top of it, and policy is applied to groups within the overlay.

17. A Group membership follows the user or device identity, so rules remain valid as addresses move.

18. D Edges are separate modules attached to the core or an edge block, so they can be secured and scaled independently of users.

19. B Edge nodes attach endpoints and put their traffic into the overlay. Intermediate nodes are only underlay transit; border nodes connect to external networks.

20. C Without VLAN mobility, routed access gives ECMP uplinks and sub-second detection with BFD, avoiding spanning tree on uplinks.

Scenarios & case studies

> **Scenario · Choosing the topology**
> **A** is best. Correct. One building with modest growth gains nothing from a separate core; a collapsed pair is cheaper, simpler and still fully redundant.

> **Scenario · Capacity alarm**
> **C** is best. Correct. Each new spine adds one more ECMP path per leaf, lowering oversubscription from 3:1 towards 2:1 without redesign.

> **Scenario · Choosing the access model**
> **B** is best. Correct. VLANs can still span the floor's switches, and both uplinks forward as members of one bundle.

> **Scenario · Branches drowning in backhaul**
> **D** is best. Correct. SaaS traffic leaves at the nearest point with consistent security, freeing MPLS and the head-office edge for internal traffic.

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Case study · National bank: head-office campus upgrade

Introduce hierarchy: distribution pairs per group of floors (for example floors 1–5, 6–10, 11–14) and a dedicated core pair, since there are several blocks and growth in branches and services. Prefer **routed access** for most floors so each VLAN is local to a closet and storms are confined to one switch; where legacy systems need stretched subnets, use multi-chassis aggregation within a block only. Prune the 120 VLANs so none spans blocks. Put the dealing room in its own block with its own distribution pair. For segmentation, use VRFs (treasury, retail, guest) carried from access to core, with a firewall between VRFs — or, in a fabric campus, virtual networks plus group-based policy. At the edge, add a second ISP with BGP, separate the internet edge and DMZ from the WAN edge, and terminate the carrier circuits on a redundant WAN edge pair off the core.

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02

MODULE 2 · SWITCH

Advanced Layer 2

Rapid and multiple spanning tree, protection features, link aggregation, private VLANs and a solid troubleshooting method.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Explain and verify RSTP roles, states and fast-convergence mechanisms.
- Design and configure an MST region with load sharing aligned to gateways.
- Apply BPDU guard, root guard, loop guard and storm control to the right ports.
- Tune LACP bundles and design private VLANs for shared segments.
- Troubleshoot Layer 2 incidents with a structured workflow and show output.

LESSONS

1. RSTP Deep Dive · 35 min
2. MST: Regions, Instances and Mapping · 35 min
3. Spanning-Tree Protection · 30 min
4. LACP Tuning and Private VLANs · 35 min
5. Layer 2 Troubleshooting Workflow · 35 min

MODULE 2 · SWITCH · LESSON 1 · 35 MIN

RSTP Deep Dive

Port roles, states, proposal/agreement and topology change — how Rapid Spanning Tree converges in under a second.

- Identify RSTP port roles and states from show output.
- Explain the proposal/agreement handshake and why edge and point-to-point ports converge fast.
- Predict root bridge, root port and designated port selection, including tie-breakers.

Rapid Spanning Tree Protocol (RSTP), originally IEEE 802.1w and now part of IEEE 802.1Q, replaced the timer-driven convergence of classic 802.1D with an **explicit handshake** between neighbours. Where 802.1D waited 30–50 seconds for timers, RSTP moves a point-to-point link to forwarding in well under a second — provided the network is designed so that RSTP can use its fast mechanisms.

RSTP port roles

ROLE	MEANING	STATE IN A STABLE TOPOLOGY
Root	The port with the best path towards the root bridge (one per non-root bridge)	Forwarding
Designated	The best port on a segment for reaching the root; forwards for that segment	Forwarding
Alternate	Receives a better BPDU from another bridge; a ready backup to the root	Discarding
Backup	Receives a better BPDU from the same bridge on a shared segment	Discarding
Disabled	Administratively or physically down	—

States: 802.1D versus RSTP

802.1D

- Disabled
- Blocking
- Listening
- Learning
- Forwarding

RSTP

- Discarding (replaces disabled, blocking and listening)
- Learning
- Forwarding

Root and port selection

1. **Elect the root bridge** Lowest bridge ID wins: priority (multiple of 4,096) plus system ID extension, then lowest MAC address.
2. **Choose each bridge's root port** Lowest root path cost; then lowest sender bridge ID; then lowest sender port ID (priority and number).
3. **Choose a designated port per segment** The port advertising the lowest root path cost on that segment; ties broken by bridge ID, then port ID.
4. **Everything else** Remaining ports become alternate or backup and discard.

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Path cost and bridge ID

MEASURE	FORMULA	WHAT IT TELLS YOU
Long path cost (802.1D-2004 and later)	`cost = 20,000,000,000,000 bit/s ÷ link speed`	1G = 20,000; 10G = 2,000; 25G = 800; 100G = 200. Root path cost is the sum along the path.
Bridge ID	`priority (4 bits × 4,096) + system ID extension (12 bits) + MAC (48 bits)`	Priority 24,576 for VLAN 10 is shown as 24,586 because the VLAN or instance number is added.

Proposal/agreement on a point-to-point link

01 Proposal — Upstream designated port sends a BPDU with the proposal flag

→

02 Sync — Downstream bridge moves its non-edge designated ports to discarding

→

03 Agreement — Downstream root port replies with the agreement flag

→

04 Forward — Upstream port goes straight to forwarding

→

05 Ripple — The handshake repeats on the next link down the tree

WATCH OUT · FAST ONLY WHERE ALLOWED

The handshake works only on **point-to-point** links (normally inferred from full duplex) and edge behaviour only on ports configured as **edge**. A shared (half-duplex) link or a host port left as non-edge falls back to slower timer-based transitions.

Reading RSTP roles and states

SW2 · IOS-style CLI

```
SW2# show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0011.2233.4401
             Cost        2000
             Port        49 (TenGigabitEthernet1/1/1)
  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     0011.2233.4402
Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -----------
Te1/1/1             Root FWD 2000      128.49   P2p
Te1/1/2             Altn BLK 2000      128.50   P2p
Gi1/0/1             Desg FWD 20000     128.1    P2p Edge
```

Many platforms run an instance of RSTP per VLAN (a vendor extension); roles and states are the same as standard RSTP. 'BLK' here means the discarding state.

Topology change is handled differently too. In RSTP only a non-edge port moving **to forwarding** triggers a change; ports going down to discarding do not. The bridge that detects it sets the TC flag in its BPDUs for a short period and flushes MAC addresses learned on its other non-edge ports; neighbours receiving the TC flag do the same and pass it on. Hosts on edge ports are never flushed, which is one more reason to configure edge ports correctly.

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PRO TIP · KEEPALIVES, NOT RELAYS

Every RSTP bridge sends its own BPDUs every hello time (2 s by default), even if it is not the root. A neighbour's information is aged out after three missed hellos (about 6 s), so a silently failed link is detected without waiting for max age.

SCENARIO · DECIDE · SLOW RECOVERY AFTER AN ACCESS SWITCH REBOOT

After an access switch reboots, users on it report that the network takes about 30 seconds to come back, even though the uplinks are full-duplex 10G and RSTP is enabled everywhere. Host ports are plain access ports with no spanning-tree options. What is the most likely cause?

(A) RSTP cannot converge quickly on 10G links and needs 1G uplinks.

(B) The root bridge priority is set too high on the distribution switches, delaying the root election.

(C) Host ports are not configured as edge ports, so they wait through learning before forwarding.

(D) BPDU guard is missing on the uplink ports.

Circle your choice. Answer and feedback: module answer key.

Key points

- RSTP roles: root, designated, alternate, backup; states: discarding, learning, forwarding.
- Fast convergence needs point-to-point links and correctly configured edge ports.
- Proposal/agreement replaces timers with a handshake that ripples down the tree.
- Only non-edge ports moving to forwarding generate topology changes.

KNOWLEDGE CHECK

1. Which RSTP port role provides an immediate replacement if the root port fails?
   (A) Backup (B) Alternate
   (C) Designated (D) Edge
2. Two paths to the root have equal root path cost. What is the next tie-breaker for choosing the root port?
   (A) The highest interface bandwidth (B) The lowest local port number
   (C) The highest sender MAC address (D) The lowest sender bridge ID
3. Using long path costs, what is the root path cost across two 10G links?
   (A) 4,000 (B) 2,000
   (C) 200 (D) 40,000
4. Which event generates an RSTP topology change notification?
   (A) An edge port going down (B) Any port moving to the discarding state
   (C) A non-edge port moving to forwarding (D) A host port coming up
5. During proposal/agreement, what does the downstream bridge do before sending its agreement?
   (A) Puts its non-edge designated ports into discarding (B) Flushes every MAC address learned on all its ports
   (C) Raises its own bridge priority to become the root (D) Waits for the forward delay timer to expire twice

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · SWITCH · LESSON 2 · 35 MIN

MST: Regions, Instances and Mapping

Multiple Spanning Tree for scalable, standards-based load sharing across VLANs.

- Explain what makes switches part of the same MST region.
- Design a VLAN-to-instance mapping that load-shares across a distribution pair.
- Configure and verify MST, and interpret boundary behaviour.

Running one spanning tree per VLAN gives fine-grained load sharing but costs CPU, memory and BPDUs for every VLAN on every trunk. A single tree for all VLANs is efficient but leaves half the links idle. **Multiple Spanning Tree (MST)**, originally IEEE 802.1s and now part of 802.1Q, groups many VLANs into a few **instances**, each with its own topology. Two or three instances usually deliver all the load sharing a campus needs.

MST building blocks

TERM	MEANING
Region	A set of switches with identical MST configuration: name, revision and VLAN-to-instance mapping
IST (instance 0)	The internal spanning tree of a region; the only instance that sends BPDUs and interacts outside the region
MSTI	An additional instance (1, 2, …) carrying a group of VLANs, existing only inside its region
CIST	The common and internal spanning tree spanning all regions and legacy bridges
Boundary port	A port connecting to a different region or to an RSTP/802.1D bridge
M-record	Per-instance information carried inside the single IST BPDU

> DEFINITION · REGION MEMBERSHIP
> Switches are in the same region only if the **name**, **revision number** and the **VLAN-to-instance mapping** all match exactly. The mapping is compared as a digest, so a single VLAN mapped differently splits the region in two.

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MST configuration on a distribution switch

DS1 · IOS-style CLI

```
spanning-tree mst configuration
 name CAMPUS-A
 revision 3
 instance 1 vlan 10-30
 instance 2 vlan 40-60
 exit
spanning-tree mode mst
spanning-tree mst 0-1 priority 4096
spanning-tree mst 2 priority 8192
!
DS1# show spanning-tree mst configuration
Name      [CAMPUS-A]
Revision  3     Instances configured 3
Instance  Vlans mapped
--------  ------------------------------
0         1-9,31-39,61-4094
1         10-30
2         40-60
```

DS2 mirrors this with instance 2 at 4096 and instances 0–1 at 8192, so each switch is root for one group of VLANs and backup root for the other.

MST design principles

D1

Few instances

Group VLANs by the distribution switch that should forward them; two or three instances are typical.

D2

Map everything up front

Map future VLAN ranges now. Changing the mapping later changes the digest and briefly splits the region.

D3

Align roots with gateways

Make the instance root the same switch that is VRRP master for those VLANs.

D4

One region per campus

Keep the configuration identical everywhere; template it and check it with automation.

D5

Avoid legacy boundaries

Interoperation with per-VLAN or 802.1D bridges is possible but adds subtle failure modes.

Inside a region the switches exchange one BPDU per port per hello, carrying the IST information plus an **M-record** per instance. Instead of message age, MST uses a **hop count** (20 by default) that each switch in the region decrements. To the outside world the whole region behaves like **one virtual bridge**: neighbouring regions and RSTP bridges only see the IST/CIST information at boundary ports, and all instances follow the IST's forwarding decision at the boundary.

WATCH OUT · VLAN PRUNING VERSUS INSTANCE TOPOLOGY

MST blocks per instance, not per VLAN. If a trunk that an instance uses as its forwarding path does not carry one of that instance's VLANs, that VLAN can be cut off even though spanning tree reports a loop-free tree. Keep allowed VLAN lists consistent with the instance design.

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Verifying an instance

DS1 · IOS-style CLI

```
DS1# show spanning-tree mst 1
###### MST1    vlans mapped:   10-30
Bridge   address 0011.2233.4401  priority 4097 (4096 sysid 1)
Root     this switch for MST1
Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -----
Te1/1/1          Desg FWD 2000      128.49   P2p
Te1/1/2          Desg FWD 2000      128.50   P2p
Po1              Desg FWD 1000      128.2000 P2p
```

SCENARIO · DECIDE · A REGION THAT KEEPS SPLITTING

After adding VLAN 70 for a new IoT project, the team maps it to instance 2 on DS1 and the floor switches but forgets DS2. Shortly afterwards, show output on DS2 lists several trunks as boundary ports and traffic paths change. What is the correct fix?

- (A) Raise DS2's priority for instance 2 so it stops being root.
- (B) Apply the identical mapping (and revision) to DS2 so all switches share one region again.
- (C) Disable MST on DS2 and run RSTP there instead.
- (D) Map VLAN 70 to instance 0 on the floor switches only, leaving the distribution switches unchanged.

Circle your choice. Answer and feedback: module answer key.

Key points

- A region needs identical name, revision and VLAN-to-instance mapping.
- Instance 0 (IST) is the only instance that sends BPDUs and talks outside the region.
- A region looks like a single bridge to its neighbours.
- Map VLANs early, align instance roots with gateways, and keep trunks consistent.

KNOWLEDGE CHECK

6. Which three items must match for two switches to be in the same MST region?
   - (A) Bridge priority, region name and hello timer
   - (B) Region name, revision number and VLAN-to-instance mapping
   - (C) Revision number, root bridge ID and port costs
   - (D) Instance count, native VLAN and region name on every trunk
7. Which MST instance interacts with switches outside the region?
   - (A) The highest-numbered MSTI
   - (B) Every MSTI independently
   - (C) Instance 0, the IST
   - (D) None; regions do not interact
8. How does MST limit the diameter of a region instead of using message age?
   - (A) With the forward delay timer only
   - (B) By limiting each region to four instances
   - (C) By capping the number of VLANs per instance
   - (D) With a hop count decremented by each switch

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9. Why is it good practice to map future VLAN ranges to instances before they are needed?

   (A) Unmapped VLANs are silently dropped by MST
   (B) MST supports mapping only at initial boot
   (C) Changing the mapping later briefly splits the region
   (D) Unmapped VLANs automatically shut down instance 0 at boot

10. A distribution switch should forward VLANs 10–30 and back up VLANs 40–60. How should its priorities be set?

    (A) Low priority for the instance with 10–30, higher for 40–60
    (B) The same priority for every instance on both switches
    (C) Highest possible priority for every instance
    (D) Low priority for instance 0 only, defaults on other instances

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 2 · SWITCH · LESSON 3 · 30 MIN

Spanning-Tree Protection

Edge ports, BPDU guard and filter, root guard and loop guard — where each belongs and what it does.

- Choose the correct protection feature for each port role in the campus.
- Explain the behaviour of err-disabled, root-inconsistent and loop-inconsistent states.
- Configure and verify protection features and automatic recovery.

Spanning tree trusts whatever BPDUs it receives. A misconfigured switch under a desk, a consumer router bridging two ports, or a fibre that loses one direction can all reshape the tree or create a loop. Protection features turn those trusting defaults into **explicit expectations** for each port.

Protection features at a glance

FEATURE	PLACE IT ON	TRIGGER	ACTION	RECOVERY
Edge (PortFast)	Host-facing access ports	Link up	Skip to forwarding; no TC on flaps	—
BPDU guard	Edge ports	Any BPDU received	Port err-disabled	Manual or err-disable recovery timer
BPDU filter	Rarely; specific edge cases	—	Stops sending (and possibly processing) BPDUs	—
Root guard	Designated ports toward switches that must never be root	Superior BPDU received	Port root-inconsistent (discarding)	Automatic when superior BPDUs stop
Loop guard	Root and alternate ports on switch-to-switch links	BPDUs stop arriving	Port loop-inconsistent (discarding)	Automatic when BPDUs resume

Where each feature belongs

Distribution (root)	`Root guard on downlinks to access switches` `Loop guard on inter-switch links where enabled`
Access uplinks	`Loop guard on root and alternate ports` `No BPDU guard or filter`
Access host ports	`Edge + BPDU guard` `Storm control for broadcast and multicast`

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Applying protection

AS1 and DS1 · IOS-style CLI

```
! AS1 host ports
interface range GigabitEthernet1/0/1 - 46
 switchport mode access
 spanning-tree portfast
 spanning-tree bpduguard enable
 storm-control broadcast level 1.00
! AS1 uplinks
interface range TenGigabitEthernet1/1/1 - 2
 spanning-tree guard loop
! DS1 downlink to AS1
interface TenGigabitEthernet1/0/10
 spanning-tree guard root
! Automatic recovery after 5 minutes
errdisable recovery cause bpduguard
errdisable recovery interval 300
```

What a violation looks like

AS1 · IOS-style CLI

```
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi1/0/12 with BPDU Guard enabled. Disabling
port.
%PM-4-ERR_DISABLE: bpduguard error detected on Gi1/0/12, putting Gi1/0/12 in err-disable state
AS1# show interfaces status err-disabled
Port      Name          Status       Reason
Gi1/0/12  ROOM-412      err-disabled bpduguard
DS1# show spanning-tree inconsistentports
Name         Interface                Inconsistency
VLAN0010     TenGigabitEthernet1/0/10  Root Inconsistent
```

Message formats vary between platforms; the information to look for is the same: which port, which feature, which state.

BPDU guard versus BPDU filter

BPDU guard

- Keeps sending BPDUs normally
- Shuts the port if any BPDU arrives
- Fails safe: a rogue switch is cut off
- Recommended on every host port

BPDU filter

- Suppresses BPDUs on the port
- Applied per interface it can ignore received BPDUs entirely
- Fails open: a loop through the port goes undetected
- Use only with a documented reason, such as a provider hand-off

WATCH OUT · ROOT GUARD AND LOOP GUARD DO DIFFERENT JOBS

Root guard reacts to BPDUs that are **too good** — someone claiming to be a better root. Loop guard reacts to BPDUs that **disappear** — typically a unidirectional fibre fault, where an alternate port would otherwise become designated and start forwarding into a loop. Many platforms do not allow both on the same port.

Unidirectional links deserve special attention. If one fibre strand of a pair fails, the switch on the receiving side stops hearing BPDUs while its own transmissions still arrive at the far end. Without loop guard (or a link-level unidirectional detection mechanism offered by the platform), the alternate port ages out its information,

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becomes designated and forwards — closing a loop. Loop guard holds that port in a loop-inconsistent state until BPDUs return.

SCENARIO · DECIDE · A ROGUE ROOT

A contractor connects a small switch to a meeting-room port. Its priority is 0, so it becomes root for VLAN 10 and traffic between floors starts flowing through the meeting room. The access port had edge configured but nothing else. Which control would have prevented this most directly?

(A) BPDU guard on the meeting-room access port.
(B) Loop guard on the meeting-room access port.
(C) BPDU filter on the distribution switches' uplinks.
(D) Raising the distribution switch priority to 61440.

Circle your choice. Answer and feedback: module answer key.

Key points

- Edge + BPDU guard on every host port; storm control as a second layer.
- Root guard on designated ports toward switches that must never become root.
- Loop guard on root and alternate ports to survive unidirectional failures.
- Avoid BPDU filter unless there is a documented, reviewed reason.

KNOWLEDGE CHECK

11. A port configured with root guard receives a superior BPDU. What happens?
    (A) It is err-disabled until an engineer manually resets it
    (B) It becomes the new root port for that VLAN
    (C) It drops the BPDU but keeps forwarding normally
    (D) It enters root-inconsistent state and stops forwarding

12. Which feature protects against a unidirectional fibre causing an alternate port to start forwarding?
    (A) BPDU guard
    (B) Loop guard
    (C) Root guard
    (D) PortFast

13. Why is BPDU filter considered riskier than BPDU guard on host ports?
    (A) It can hide a loop by ignoring BPDUs
    (B) It shuts ports down far too often
    (C) It raises the switch bridge priority
    (D) It disables storm control on the port

14. Where is root guard normally configured?
    (A) On every access switch host port
    (B) On access uplinks toward the root
    (C) On distribution downlinks toward access switches
    (D) On the root bridge's loopback and management ports

15. What does `errdisable recovery cause bpduguard` with an interval of 300 achieve?
    (A) BPDU guard is disabled for 300 seconds after boot
    (B) BPDUs are rate-limited to 300 per second
    (C) The switch reloads itself after 300 seconds of loops
    (D) Ports shut by BPDU guard re-enable after 5 minutes

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MODULE 2 · SWITCH · LESSON 4 · 35 MIN

LACP Tuning and Private VLANs

Building reliable link bundles and isolating hosts within a shared subnet.

- Configure LACP with appropriate mode, rate, priorities and minimum links.
- Explain hashing behaviour and why bundle size matters.
- Describe private VLAN port types and design a private VLAN for a shared segment.

The **Link Aggregation Control Protocol (LACP)**, defined in IEEE 802.1AX (originally 802.3ad), negotiates bundles of parallel Ethernet links so they act as one logical link. Spanning tree sees one port, so all members forward. LACP's value is not just bandwidth: it verifies that both ends agree before a link joins, preventing the loops and black holes that static bundling can cause.

LACP negotiation modes

LOCAL	REMOTE	RESULT
Active	Active	Bundle forms
Active	Passive	Bundle forms
Passive	Passive	No bundle: neither side starts negotiation
Static (on)	Active or passive	No LACP agreement; mismatch risk, avoid

Tuning parameters

PARAMETER	WHAT IT DOES	GUIDANCE
Rate fast / slow	Asks the partner to send LACPDUs every 1 s (fast) or 30 s (slow); timeout is three intervals	Fast (3 s detection) for links where physical loss may not be signalled
System priority	Lower value decides which members are active when more are configured than allowed	Set deliberately on the device that should be in control
Port priority	Lower value preferred to be active within a bundle	Use with max-bundle to choose standby links
Max-bundle	Limits active members; extra members go hot-standby	Useful to keep hashing even (e.g. exactly 4 active)
Min-links	Takes the whole bundle down if fewer members are up	Lets routing move traffic to a healthier path
Load-balance hash	Fields used to pick a member per flow	Use IP and Layer 4 fields where available

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A tuned LACP trunk

DS1 · IOS-style CLI

```
lacp system-priority 100
port-channel load-balance src-dst-ip
!
interface range TenGigabitEthernet1/1/1 - 4
 channel-group 1 mode active
 lacp rate fast
!
interface Port-channel1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 port-channel min-links 2
!
DS1# show port-channel summary
Flags: P - bundled  s - suspended  I - stand-alone  S - Layer2  U - in use
Group  Port-channel  Protocol  Ports
1      Po1(SU)       LACP      Te1/1/1(P) Te1/1/2(P) Te1/1/3(P) Te1/1/4(s)
```

Te1/1/4 is suspended: its configuration (for example allowed VLANs or speed) differs from the bundle. Fix the member, not the bundle.

Hashing and distribution

MEASURE	FORMULA	WHAT IT TELLS YOU
Member selection	`member = hash(selected header fields) mod buckets`	Every packet of a flow uses the same member, so one large flow never exceeds one link's speed.
Even distribution	`buckets ÷ members should be a whole number`	On platforms with eight hash buckets, 2, 4 or 8 members share evenly; 3 or 6 members do not.

WATCH OUT · CONSISTENCY IS MANDATORY

Members must match in speed, duplex, switchport mode, allowed and native VLANs, and MTU. Mismatched members are suspended or left stand-alone. Configure the logical port-channel interface and let the members inherit, rather than editing members one by one.

A **private VLAN** (PVLAN) splits one VLAN — and one IP subnet — into isolated pieces at Layer 2. The **primary VLAN** carries traffic from the gateway to everyone. **Secondary VLANs** carry traffic from hosts: an **isolated** secondary lets hosts talk only to promiscuous ports, while a **community** secondary lets hosts talk to each other within the community and to promiscuous ports. Typical uses are hosting providers, shared server segments, guest networks and multi-tenant building systems.

Private VLAN port types

PORT TYPE	BELONGS TO	CAN COMMUNICATE WITH
Promiscuous	Primary VLAN (mapped to secondaries)	All ports: isolated, community and promiscuous
Isolated host	Isolated secondary VLAN	Promiscuous ports only
Community host	A community secondary VLAN	Same community and promiscuous ports

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Private VLAN for a shared server segment

SW3 · IOS-style CLI

```
vlan 101
 private-vlan isolated
vlan 102
 private-vlan community
vlan 100
 private-vlan primary
 private-vlan association 101,102
!
interface GigabitEthernet1/0/5
 switchport mode private-vlan host
 switchport private-vlan host-association 100 101
interface GigabitEthernet1/0/48
 switchport mode private-vlan promiscuous
 switchport private-vlan mapping 100 101,102
```

Gi1/0/48 faces the gateway or firewall. Remember that isolation is Layer 2 only: if the gateway routes traffic back between hosts in the same subnet, the isolation can be bypassed unless the gateway blocks it.

SCENARIO · DECIDE · A SLOW BACKUP DESPITE FOUR LINKS

A backup server connects to a four-member LACP bundle of 10G links. A single nightly backup stream to one storage array never exceeds about 10 Gbit/s, although interface counters show all members healthy. What is the explanation?

- (A) LACP is in slow mode, which halves bundle throughput.
- (B) The bundle needs min-links 4 to use every member.
- (C) Spanning tree is blocking three of the four members.
- (D) Per-flow hashing keeps one flow on one member link.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

16. Both ends of a bundle are configured as LACP passive. What happens?
    - (A) The bundle forms at slow rate
    - (B) No bundle forms because neither side initiates
    - (C) The bundle forms but only one link is used
    - (D) The switches silently fall back to static bundling
17. What is the purpose of `port-channel min-links 2` on a four-member bundle?
    - (A) Force at least two members into hot-standby
    - (B) Limit each flow to a maximum of two members
    - (C) Prevent more than two members from joining the bundle at once
    - (D) Bring the bundle down when fewer than two members remain
18. With fast LACP rate, roughly how long before a silent partner failure is detected?
    - (A) About 1 second
    - (B) About 3 seconds
    - (C) About 30 seconds
    - (D) About 90 seconds
19. In a private VLAN, which ports can a host on an isolated port reach at Layer 2?
    - (A) Other isolated ports in the same VLAN
    - (B) Community ports in any community
    - (C) Only promiscuous ports
    - (D) Every port in the primary VLAN

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20. A member of a bundle shows as suspended. What is the most likely cause?

(A) Its configuration differs from the other members
(B) Its LACP port priority is lower than the others
(C) It is the newest member to join the bundle
(D) Its partner port is configured with LACP active mode

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MODULE 2 · SWITCH · LESSON 5 · 35 MIN

Layer 2 Troubleshooting Workflow

A repeatable method, the show commands that matter and how to read them under pressure.

- Apply a structured, bottom-up workflow to Layer 2 incidents.
- Interpret interface, trunk, MAC table and spanning-tree output to localise faults.
- Recognise the signatures of loops, MAC flapping and native VLAN mismatches.

Layer 2 faults are often noisy and widespread: a loop or a pruning error affects many users at once, and symptoms appear far from the cause. A disciplined workflow — define, localise, verify each layer, fix, confirm — prevents the classic mistake of changing several things at once and never knowing which one worked.

ZCNP Layer 2 workflow

1. **Define the problem** Who is affected, which VLANs, since when, what changed? Get one source and one destination to test with.
2. **Follow the frame** Trace the path hop by hop using MAC address tables and LLDP neighbours.
3. **Check the physical layer** Interface state, speed and duplex, error counters, optics levels.
4. **Check VLANs and trunks** VLAN exists and is active, access VLAN correct, VLAN allowed and forwarding on every trunk, native VLAN matches.
5. **Check spanning tree and bundles** Root location, port roles, inconsistent ports, topology change counters, LACP member state.
6. **Fix one thing and confirm** Change one variable, re-test end to end, document the root cause.

Following a MAC address

AS3 · IOS-style CLI

```
AS3# show mac address-table address 0050.56a1.22b3
Vlan    Mac Address       Type        Ports
  20    0050.56a1.22b3    DYNAMIC     Te1/1/1
AS3# show lldp neighbors
Device ID   Local Intf   Hold-time  Capability  Port ID
DS1         Te1/1/1      120        B,R         Te1/0/13
DS2         Te1/1/2      120        B,R         Te1/0/13
```

LLDP (IEEE 802.1AB) tells you which device and port to log into next. Repeat until the MAC points at an access port.

Trunk check

DS2 · IOS-style CLI

```
DS2# show interfaces trunk
Port      Mode  Encapsulation  Status    Native vlan
Te1/0/13  on    802.1q         trunking  1
Port      Vlans allowed on trunk
Te1/0/13  10,30
Port      Vlans in spanning tree forwarding state and not pruned
Te1/0/13  10,30
```

VLAN 20 is not allowed on this trunk, and the native VLAN is 1 while AS3 uses 99 on its side — two faults on one link.

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Symptoms and likely causes

SYMPTOM	LIKELY CAUSE	COMMAND THAT PROVES IT
One VLAN fails across a trunk, others work	VLAN not allowed, pruned or not created	show interfaces trunk, show vlan brief
Untagged traffic leaks between VLANs	Native VLAN mismatch	show interfaces trunk on both ends
CPU high, links saturated, MACs flapping	Layer 2 loop	show interfaces counters, show logging, topology change counters
Host intermittently unreachable	Duplicate MAC or IP, or MAC flapping between ports	show mac address-table, show logging
Rising CRC or input errors	Bad cable, optic or duplex mismatch	show interfaces, optic diagnostics
Port down with err-disabled	BPDU guard, storm control or link-flap protection	show interfaces status err-disabled

Signature of a loop

DS1 · IOS-style CLI

```
%SW_MATM-4-MACFLAP_NOTIF: Host 0050.56a1.22b3 in vlan 20 is flapping between port Te1/0/13 and port
Te1/0/14
DS1# show spanning-tree vlan 20 detail | include changes|from
  Number of topology changes 1843 last change occurred 00:00:02 ago
          from TenGigabitEthernet1/0/14
```

Thousands of topology changes and a recent source port point you to where to look next: follow the 'from' interface downstream.

PRO TIP · STABILISE FIRST, THEN DIAGNOSE

During an active loop, shut the suspected redundant link or the port that topology changes come from to restore service, while capturing logs and counters. Root-cause analysis can follow once users are working again.

SCENARIO · DECIDE · MANAGEMENT TRAFFIC IN THE WRONG VLAN

A new trunk links AS7 to DS2. AS7's untagged management traffic appears in VLAN 1 on DS2, while AS7 expects it in VLAN 99. Tagged user VLANs work normally. What is the right fix?

- (A) Enable BPDU guard on the new trunk so that frames from mismatched VLANs are blocked at once.
- (B) Remove VLAN 99 from AS7 and use VLAN 1 for switch management on every device in the campus.
- (C) Configure the same native VLAN on both ends, then confirm with show interfaces trunk.
- (D) Convert the link into an access port in VLAN 99 on both switches to carry the management VLAN.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · TELECOM REGIONAL OFFICE: THE MONDAY MORNING OUTAGE

A telecom operator's regional office (eight floors, two distribution switches, RSTP per VLAN) suffers a 25-minute outage every few weeks, always on Monday morning. During outages the distribution CPUs reach 100% and logs show MAC flapping between two access uplinks. Floor teams often re-patch desks over weekends. The access ports are configured as edge ports but have no further protection, and storm control is not configured.

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1. What is the most likely cause of the recurring outages?

2. What evidence would you collect during the next event to confirm it?

3. Which configuration changes would prevent it, and on which ports?

4. How would you make sure the fix stays applied across all access switches?

Model answer: module answer key.

TEMPLATE **Spanning-tree design sheet**

Document the intended spanning-tree topology so that show output can be compared against it during troubleshooting.

VLANS / INSTANCE	PRIMARY ROOT (PRIORITY)	SECONDARY ROOT (PRIORITY)	VRRP MASTER	UPLINK PROTECTION	EDGE PORT PROTECTION
MST1: VLANs 10–30	*DS1 (4096)*	*DS2 (8192)*	*DS1*	*Loop guard on access uplinks; root guard on DS downlinks*	*Edge + BPDU guard + storm control*
MST2: VLANs 40–60	*DS2 (4096)*	*DS1 (8192)*	*DS2*	*Loop guard on access uplinks; root guard on DS downlinks*	*Edge + BPDU guard + storm control*

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Layer 2 health checklist

☐ Root bridge for every VLAN or instance is the intended distribution switch.

☐ Secondary root is explicitly configured, not left to chance.

☐ Every host port is an edge port with BPDU guard.

☐ Root guard on distribution downlinks; loop guard on access uplinks.

☐ Trunks carry only required VLANs and native VLANs match on both ends.

☐ All LACP bundles show every member bundled; no suspended members.

☐ Storm control is configured on access ports.

☐ Topology change counters are stable and monitored.

☐ MST configuration is identical on all switches in the region.

REFLECT

Think of the last Layer 2 incident you handled. Which step of the workflow would have shortened it the most, and what check could you automate so it never happens silently again?

Key terms

RSTP
Rapid Spanning Tree Protocol, the handshake-based spanning tree now defined within IEEE 802.1Q.

Edge port
A port expected to connect only to end hosts; it forwards immediately and does not cause topology changes.

MST region
Switches sharing the same MST name, revision and VLAN-to-instance mapping.

BPDU guard
Err-disables an edge port that receives any BPDU.

Loop guard
Blocks a root or alternate port whose BPDUs stop arriving, protecting against unidirectional links.

Min-links
The minimum number of active members a bundle needs to stay up.

Promiscuous port
A private VLAN port that can communicate with all isolated and community ports.

Alternate port
An RSTP port that receives a better BPDU from another bridge and stands by to replace the root port.

Proposal/agreement
The RSTP handshake that moves a point-to-point designated port to forwarding without timers.

IST
Internal spanning tree, MST instance 0; the only instance that sends BPDUs and talks outside the region.

Root guard
Blocks a port that receives a superior BPDU, keeping the root where it was designed.

LACP
Link Aggregation Control Protocol (IEEE 802.1AX) for negotiating link bundles.

Private VLAN
A VLAN subdivided into isolated and community secondary VLANs sharing one subnet.

MAC flapping
A MAC address repeatedly learned on different ports, a common sign of a loop.

KNOWLEDGE CHECK

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21. `show interfaces trunk` shows VLAN 20 is missing from the 'allowed' list on a trunk. What is the impact?
    - (A) VLAN 20 is carried untagged as native
    - (B) VLAN 20 traffic cannot cross that trunk
    - (C) VLAN 20 is load-shared to another trunk
    - (D) Only broadcasts in VLAN 20 are dropped

22. Logs show a host MAC flapping between two uplink ports and topology changes are rising rapidly. What should you suspect first?
    - (A) A duplex mismatch
    - (B) A DHCP scope that is full
    - (C) A native VLAN mismatch
    - (D) A Layer 2 loop

23. Which protocol helps you identify the next-hop switch and port while tracing a MAC address?
    - (A) LLDP
    - (B) LACP
    - (C) VRRP
    - (D) BFD

24. Interface counters show steadily increasing CRC errors on one access port. What is the most likely cause?
    - (A) A VLAN missing from the uplink trunk's allowed list
    - (B) The port being an edge port
    - (C) A physical problem such as cabling or duplex
    - (D) A spanning-tree root election

25. During an active loop affecting the whole office, what is the best first action?
    - (A) Reload both distribution switches at once
    - (B) Isolate the suspected link to restore service, then diagnose
    - (C) Disable spanning tree to stop the BPDUs
    - (D) Wait for spanning tree to detect and resolve the loop by itself

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MODULE 2 · SWITCH

Answer key

Knowledge checks

1. B An alternate port receives a better BPDU from another bridge and can become the root port immediately. A backup port protects a shared segment from the same bridge.

2. D After root path cost, RSTP compares the bridge ID of the neighbour sending the BPDU, then the sender's port ID.

3. A Each 10G link costs 20,000,000,000,000 ÷ 10,000,000,000 = 2,000; two links add up to 4,000.

4. C In RSTP only non-edge ports transitioning to forwarding cause a topology change. Edge port events and ports going to discarding do not.

5. A The sync step blocks downstream designated ports so the newly forwarding upstream link cannot create a loop. The handshake then repeats further down.

6. B Region membership depends on the configuration identifier: name, revision and the mapping digest. Priorities and timers do not define the region.

7. C Only the IST sends BPDUs and represents the region at boundary ports. MSTIs exist only inside the region.

8. D MST uses a maximum hop count (20 by default) inside the region. Message age still applies to CIST information crossing boundaries.

9. C Unmapped VLANs belong to instance 0, which works, but changing the mapping later alters the digest and causes reconvergence as switches are updated one by one.

10. A Lower priority wins root election, so the switch becomes root for the first instance and secondary for the other, with its peer configured the opposite way.

11. D Root guard blocks the port in root-inconsistent state and recovers automatically once superior BPDUs stop.

12. B Loop guard keeps a root or alternate port discarding when BPDUs stop arriving, which is the symptom of a unidirectional link.

13. A Guard fails safe by shutting the port; filter can fail open by suppressing the BPDUs that would reveal a loop.

14. C Root guard belongs on designated ports facing parts of the network that must never contain the root.

15. D The port is automatically brought back after the interval; if the offending switch is still there, BPDU guard will shut it again.

16. B Passive ports only respond to LACPDUs. At least one side must be active.

17. D Min-links lets a degraded bundle go down so routing or spanning tree can choose a better path instead of overloading the remaining link.

18. B Fast rate sends LACPDUs every second and times out after three missed intervals; slow rate takes about 90 seconds.

19. C Isolated hosts can talk only to promiscuous ports, typically the gateway or firewall.

20. A Suspension indicates incompatibility such as different VLANs, speed or mode. Port priority affects active/standby selection, not suspension.

21. B A VLAN that is not allowed is not carried at all on that trunk. Unless another path exists, VLAN 20 is cut off beyond it.

22. D MAC flapping between ports combined with constant topology changes is the signature of a loop.

23. A LLDP (IEEE 802.1AB) advertises device and port identity to neighbours, so you can follow the path hop by hop.

24. C CRC errors are a physical-layer symptom: damaged cabling, bad optics or a duplex mismatch.

25. B Stabilise first: shutting the suspected port stops the loop while evidence is captured. Reloading or disabling spanning tree risks making it worse.

Scenarios & case studies

> **Scenario · Slow recovery after an access switch reboot**
> **C** is best. Correct. Non-edge ports do not get immediate forwarding; with no peer to complete a handshake, they fall back to timer-based transitions of about 30 s.

> **Scenario · A region that keeps splitting**
> **B** is best. Correct. Any difference in name, revision or mapping creates a separate region; matching the configuration restores a single region.

> **Scenario · A rogue root**
> **A** is best. Correct. The first BPDU from the contractor's switch would have err-disabled the port, so it could never take part in the topology.

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Scenario · A slow backup despite four links
D is best. Correct. A single flow hashes to one member, so it is limited to that member's speed. More flows, or a faster single link, are needed for more throughput.

Scenario · Management traffic in the wrong VLAN
C is best. Correct. Untagged frames are placed in each side's native VLAN, so the two ends must agree.

Case study · Telecom regional office: the Monday morning outage
The pattern points to a **Layer 2 loop created at the edge**: weekend re-patching connects two wall ports together, or a small unmanaged switch/hub bridges two ports. Because the ports are edge ports without BPDU guard, they forward immediately and the loop forms. During the next event capture topology-change counters and the 'from' interface, MAC-flap logs, and interface counters to locate the floor and port. Prevent it with **BPDU guard on every edge port** (with err-disable recovery), **storm control** on access ports, **loop guard** on access uplinks, and **root guard** on distribution downlinks. Enforce it with a standard access-port template and an automated compliance check that reports any edge port lacking BPDU guard.

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03

MODULE 3 · IGP

OSPF and Route Control

Multi-area OSPF, special areas, OSPFv3, IS-IS fundamentals and safe route control between domains.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design multi-area OSPF and identify LSA types 1–5 and 7.
- Choose stub, totally stubby and NSSA areas and summarise correctly.
- Configure OSPFv3 and explain IS-IS levels and NET addressing.
- Control routes with prefix lists, route maps, tags and policy-based routing.
- Troubleshoot OSPF adjacencies and missing routes methodically.

LESSONS

1. OSPF Multi-Area Design and LSA Types · 35 min
2. Stub Areas, NSSA and Summarisation · 35 min
3. OSPFv3 and an IS-IS Overview · 35 min
4. Route Control: Prefix Lists, Route Maps, Redistribution and PBR · 40 min
5. Troubleshooting OSPF Adjacencies and Routes · 35 min

MODULE 3 · IGP · LESSON 1 · 35 MIN

OSPF Multi-Area Design and LSA Types

Areas, router roles and the link-state advertisements that describe them.

- Design a multi-area OSPF topology with correct ABR and ASBR placement.
- Identify LSA types 1, 2, 3, 4, 5 and 7 by originator and flooding scope.
- Calculate OSPF interface cost and set a sensible reference bandwidth.

OSPFv2 (RFC 2328) is a link-state protocol: every router in an area holds an identical **link-state database (LSDB)** and runs the shortest-path-first calculation on it. Areas exist to limit how far detail travels. Inside an area, routers know the full topology; between areas, they only learn prefixes and costs. That boundary keeps SPF calculations small, contains flapping, and gives you natural points for summarisation.

Multi-area rules

- Area 0 is the backbone; every other area must attach to it (a virtual link is a temporary exception, not a design).
- An Area Border Router (ABR) has interfaces in area 0 and at least one other area.
- An Autonomous System Boundary Router (ASBR) injects external routes, for example from static routes, BGP or another IGP.
- Inter-area traffic always passes through the backbone, which prevents inter-area routing loops.
- Router IDs must be unique; set them explicitly from a loopback plan.

LSA types you must recognise

TYPE	NAME	ORIGINATED BY	FLOODING SCOPE	DESCRIBES
1	Router LSA	Every router	Its own area	The router's links and their costs
2	Network LSA	Designated router	Its own area	A multi-access segment and the routers attached to it
3	Summary (network) LSA	ABR	Into an area	Prefixes from other areas with their cost
4	ASBR summary LSA	ABR	Into an area	How to reach an ASBR located in another area
5	AS external LSA	ASBR	Whole domain except stub and NSSA areas	External prefixes
7	NSSA external LSA	ASBR inside an NSSA	The NSSA only; translated to type 5 by an ABR	External prefixes injected inside an NSSA

Where each LSA lives

Domain-wide	`Type 5 external routes` `Not in stub, totally stubby or NSSA areas`
Between areas	`Type 3 summaries generated by ABRs` `Type 4 pointers to ASBRs`
Inside one area	`Type 1 router LSAs` `Type 2 network LSAs` `Type 7 in an NSSA`

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OSPF cost

MEASURE	FORMULA	WHAT IT TELLS YOU
Interface cost	`cost = reference bandwidth ÷ interface bandwidth (minimum 1)`	With the common default reference of 100 Mbit/s, every link of 100 Mbit/s or faster costs 1, so OSPF cannot tell 1G from 100G.
Raised reference	`reference 400,000 Mbit/s: 400G = 1, 100G = 4, 10G = 40, 1G = 400`	Set the same reference on every router so costs are consistent across the domain.
Path cost	`path cost = sum of outgoing interface costs along the path`	SPF picks the lowest total; equal totals give ECMP.

ABR configuration

R2 · IOS-style CLI

```
router ospf 1
 router-id 10.255.0.2
 auto-cost reference-bandwidth 400000
 passive-interface default
 no passive-interface TenGigabitEthernet0/0/0
 no passive-interface TenGigabitEthernet0/0/1
!
interface TenGigabitEthernet0/0/0
 ip ospf network point-to-point
 ip ospf 1 area 0
interface TenGigabitEthernet0/0/1
 ip ospf network point-to-point
 ip ospf 1 area 10
```

Reading the LSDB

R2 · IOS-style CLI

```
R2# show ip ospf database
            OSPF Router with ID (10.255.0.2) (Process ID 1)
                Router Link States (Area 0)
Link ID         ADV Router      Age   Seq#       Checksum Link count
10.255.0.1      10.255.0.1      412   0x80000005 0x00A1B2 3
10.255.0.2      10.255.0.2      398   0x80000007 0x00C3D4 2
                Summary Net Link States (Area 0)
Link ID         ADV Router      Age   Seq#       Checksum
10.10.0.0       10.255.0.2      301   0x80000002 0x004455
                Summary ASB Link States (Area 0)
10.255.0.30     10.255.0.3      301   0x80000001 0x0066AA
                Type-5 AS External Link States
Link ID         ADV Router      Age   Seq#       Checksum Tag
0.0.0.0         10.255.0.30     120   0x80000001 0x0077BB 1
```

The type 4 entry is generated by ABR 10.255.0.3 so routers in area 0 can reach ASBR 10.255.0.30, which lives in another area.

OSPF route preference

CODE	ROUTE TYPE	PREFERENCE
O	Intra-area	Most preferred

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CODE	ROUTE TYPE	PREFERENCE
O IA	Inter-area	Second
O E1 / O N1	External type 1: external metric plus internal cost	Third
O E2 / O N2	External type 2: external metric only (common default for redistribution)	Least preferred

INSIGHT · PREFERENCE BEATS METRIC

OSPF chooses by route type first and cost second. An intra-area route with cost 500 beats an inter-area route with cost 20 to the same prefix. Remember this when a 'cheaper' path is ignored.

SCENARIO · DECIDE · PLACING A NEW SITE

A company adds a new regional office with 40 subnets. It will connect to two existing routers that are both in area 0. The team suggests putting the office directly in area 0 'to keep things simple'. What is the better design?

- (A) Create a new area for the office and make the two connecting routers ABRs that summarise its range.
- (B) Put the office in area 0 so it can reach every other area directly.
- (C) Create a new area attached only through a virtual link.
- (D) Run a separate OSPF process at the office and redistribute it into the backbone process on both routers.

Circle your choice. Answer and feedback: module answer key.

Key points

- All areas attach to area 0; ABRs connect them, ASBRs inject external routes.
- Types 1 and 2 stay inside an area; 3 and 4 cross area borders; 5 is domain-wide; 7 lives in an NSSA.
- Raise the reference bandwidth consistently so modern links have meaningful costs.
- OSPF prefers intra-area, then inter-area, then E1, then E2 — regardless of cost.

KNOWLEDGE CHECK

1. Which router originates a type 2 LSA?
   - (A) The designated router on a multi-access segment
   - (B) Every ABR attached to the backbone area
   - (C) Any ASBR that redistributes static routes
   - (D) The router with the highest router ID in area 0
2. What does a type 4 LSA tell routers in an area?
   - (A) Which external prefixes exist in the domain
   - (B) Which router is the DR on each segment
   - (C) Which prefixes belong to the backbone area only
   - (D) How to reach an ASBR located in another area
3. With a reference bandwidth of 100,000 Mbit/s, what is the cost of a 10G interface?
   - (A) 1
   - (B) 100
   - (C) 10
   - (D) 1,000
4. A router has an inter-area route with cost 30 and an E1 route with cost 10 to the same prefix. Which is installed?
   - (A) The E1 route
   - (B) The inter-area route
   - (C) Both, as equal-cost paths
   - (D) Neither; the routes conflict

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5. What is the role of an ABR in an NSSA?

(A) Originating type 2 LSAs for every segment inside the NSSA area

(B) Blocking all type 3 LSAs from the backbone

(C) Translating type 7 LSAs into type 5 for the rest of the domain

(D) Converting type 1 LSAs into type 7 externals

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MODULE 3 · IGP · LESSON 2 · 35 MIN

Stub Areas, NSSA and Summarisation

Shrink the LSDB and the routing table while keeping reachability and control.

- Choose between stub, totally stubby, NSSA and totally NSSA areas.
- Configure inter-area and external summarisation and explain the discard route.
- Calculate a summary prefix from a contiguous address block.

Many areas do not need to know every external or inter-area prefix. A branch region with two exits to the backbone only needs a default route to send everything else upstream. OSPF's special area types filter LSA types at the ABR and replace them with a default route, while **summarisation** replaces many prefixes with one. Both shrink LSDBs, speed up SPF and contain instability.

Area types compared

AREA TYPE	TYPE 3	TYPE 4/5	TYPE 7	DEFAULT ROUTE
Normal	Yes	Yes	No	Only if originated
Stub	Yes	No	No	Type 3 default from ABR
Totally stubby	Default only	No	No	Type 3 default from ABR
NSSA	Yes	No	Yes	Not automatic; ABR can originate a type 7 default
Totally NSSA	Default only	No	Yes	Type 3 default from ABR

WATCH OUT · EVERY ROUTER MUST AGREE

The stub or NSSA setting is carried in hello packets. If one router in the area is configured as stub and its neighbour is not, they will not form an adjacency. The 'no-summary' option that makes an area totally stubby is needed only on the ABR.

Configuring area types and summaries

R2 (ABR) · IOS-style CLI

```
router ospf 1
 area 20 stub no-summary
 area 30 nssa no-summary
 area 10 range 10.10.0.0 255.255.240.0
!
! Every internal router in area 20:
router ospf 1
 area 20 stub
!
! ASBR summarising redistributed routes:
router ospf 1
 summary-address 172.16.0.0 255.255.0.0
```


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Two kinds of summarisation

Inter-area (at the ABR)

- `area X range` command
- Summarises type 1/2 information into one type 3 LSA
- Only on the ABR that connects the area
- Hides flapping subnets from other areas

External (at the ASBR)

- `summary-address` command
- Summarises type 5 or type 7 LSAs
- On the ASBR, or on the NSSA translator ABR
- Reduces imported routes from other domains

Summary calculation

MEASURE	FORMULA	WHAT IT TELLS YOU
Block size	`number of networks = 2^(original length − summary length)`	Sixteen /24 networks fit exactly into one /20, because 2^(24 − 20) = 16.
Summary length	`summary length = original length − log2(number of networks)`	Sixteen contiguous /24s starting at 10.10.0.0 summarise to 10.10.0.0/20 (mask 255.255.240.0).
Alignment test	`the first network must be divisible by the block size`	10.10.16.0–10.10.31.0 summarises to 10.10.16.0/20; 10.10.8.0–10.10.23.0 does not fit a single /20.

Verifying the result

R2 and R21 · IOS-style CLI

```
R2# show ip route 10.10.0.0 255.255.240.0 longer-prefixes | include Null0
O        10.10.0.0/20 is a summary, 00:12:44, Null0
!
R21# show ip route ospf
O*IA  0.0.0.0/0 [110/41] via 10.20.0.1, 00:05:11, TenGigabitEthernet0/0/0
O     10.20.4.0/24 [110/80] via 10.20.0.5, 00:05:11, TenGigabitEthernet0/0/1
```

R2 installs a discard route for the summary so packets to unused parts of the range are dropped rather than looping. R21 in the totally stubby area sees only intra-area routes and one default.

PRO TIP · SUMMARIES AND BLACK HOLES

If a summarising ABR loses its links into the area but still advertises the summary, traffic is attracted and dropped. Use two ABRs with consistent summaries, and design so each ABR keeps a path to every component subnet.

SCENARIO · DECIDE · AN AREA THAT ONLY NEEDS A DEFAULT

A retail region has 120 stores in area 40, connected to the backbone through two ABRs. Stores also use a local internet connection that is redistributed as a static route on each store router for guest traffic only. The store routers are small and struggle with the 9,000 external routes from the data centre. Which area type fits best?

(A) Totally stubby: it removes the most routes from the store routers.

(B) Normal area with `area 40 range` on the ABRs.

(C) Stub: it removes external routes and keeps inter-area detail.

(D) Totally NSSA: stores receive only a default from the ABRs and can still inject their own external routes as type 7.

Circle your choice. Answer and feedback: module answer key.

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Key points

- Stub and totally stubby areas block externals; NSSA variants allow local externals as type 7.
- The stub flag must match on every router in the area; no-summary is configured only on the ABR.
- Summarise at ABRs with `area range` and at ASBRs with `summary-address`.
- Summaries create a discard route; design redundant ABRs to avoid black holes.

KNOWLEDGE CHECK

6. Which area type blocks type 3, 4 and 5 LSAs but still allows redistribution inside the area?

   (A) Totally NSSA
   (B) Totally stubby
   (C) Stub
   (D) Normal

7. Two neighbours in area 20 will not form an adjacency after one was configured as `area 20 stub`. Why?

   (A) The ABR has not yet sent a default route
   (B) Stub areas only support point-to-point network links
   (C) The router IDs must be reset after the change
   (D) The stub flag in the hello packets does not match

8. What is the smallest single summary covering 10.20.32.0/24 through 10.20.47.0/24?

   (A) 10.20.32.0/20
   (B) 10.20.32.0/21
   (C) 10.20.0.0/18
   (D) 10.20.32.0/19

9. Why does a summarising router install a route to Null0 for the summary?

   (A) To advertise the summary to its neighbours over BGP
   (B) To mark the summary as a backup route only
   (C) To make the summary preferred over any default route in the area
   (D) To drop traffic for unused parts of the range and prevent loops

10. Where is `summary-address` used in OSPF?

    (A) On an ABR to summarise inter-area routes
    (B) On an ASBR to summarise external routes
    (C) On any router to summarise its connected links
    (D) On the DR to summarise a broadcast segment

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MODULE 3 · IGP · LESSON 3 · 35 MIN

OSPFv3 and an IS-IS Overview

Running link-state routing for IPv6, and understanding the other major IGP used in large networks.

- Configure OSPFv3 for IPv6 and explain how it differs from OSPFv2.
- Describe IS-IS levels, area boundaries and NET addressing.
- Compare OSPF and IS-IS for an enterprise or service-provider core.

OSPFv3 (RFC 5340) brings the OSPF model to IPv6. The SPF algorithm, areas, LSA flooding and neighbour states are familiar; what changes is how addresses are carried. In OSPFv3, router and network LSAs describe only topology, while prefixes move into new LSA types. With the address-family extension (RFC 5838), one OSPFv3 process can also carry IPv4 routes, although many networks keep OSPFv2 for IPv4.

OSPFv2 versus OSPFv3

ASPECT	OSPFV2	OSPFV3
Carries	IPv4	IPv6 (and IPv4 with address families)
Runs on	IPv4, protocol 89	IPv6, next header 89, using link-local addresses
Router ID	32-bit	Still 32-bit; must be set manually on an IPv6-only router
Prefixes	Inside router and network LSAs	In intra-area-prefix and link LSAs; topology LSAs carry no addresses
Next hop	Neighbour's interface IPv4 address	Neighbour's link-local address
Authentication	Built into the protocol	IPsec, or the authentication trailer (RFC 7166)
Instances per link	One	Several, using an instance ID

OSPFv3 for IPv6

R1 · IOS-style CLI

```
ipv6 unicast-routing
!
router ospfv3 1
 router-id 10.255.0.1
 address-family ipv6 unicast
  passive-interface default
  no passive-interface GigabitEthernet0/0
 exit-address-family
!
interface GigabitEthernet0/0
 ipv6 address 2001:db8:0:12::1/64
 ospfv3 1 ipv6 area 0
!
R1# show ipv6 route ospf
O   2001:DB8:0:20::/64 [110/2]
     via FE80::2, GigabitEthernet0/0
```

The next hop is the neighbour's link-local address. Without an IPv4 address to borrow, the router ID must be configured explicitly.

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IS-IS (ISO/IEC 10589, extended for IP by RFC 1195) is the other major link-state IGP. It is common in service-provider cores and large data-centre and campus fabrics. It was designed for a different network layer, so it runs **directly over Layer 2**, not over IP. That makes it independent of IP addressing and simple to extend: new features are added as **TLVs** (type-length-value fields), which is how IS-IS carries IPv6 (RFC 5308) and wide metrics (RFC 5305) in the same protocol instance.

IS-IS hierarchy

Level 2 (backbone)	`Connects areas` `Must be a contiguous chain of L2-capable routers`
Level 1/Level 2	`Belongs to one area and to the backbone` `Sets the attached bit so L1 routers get a default route`
Level 1	`Knows only its own area` `Sends inter-area traffic to the nearest L1/L2 router`

Reading a NET

MEASURE	FORMULA	WHAT IT TELLS YOU
Structure	`NET = area ID . system ID (6 bytes) . NSEL (00)`	49.0001.0100.0000.0001.00 is area 49.0001, system ID 0100.0000.0001, NSEL 00.
System ID from a loopback	`10.0.0.1 → 010.000.000.001 → 0100.0000.0001`	Pad each octet to three digits and regroup into three blocks of four for a unique, traceable ID.
AFI 49	`49 = private address space`	Like RFC 1918 for IP, 49 is used for networks that do not need globally registered NSAP addresses.

Basic IS-IS

R1 · IOS-style CLI

```
router isis CORE
 net 49.0001.0100.0000.0001.00
 is-type level-2-only
 metric-style wide
 passive-interface Loopback0
!
interface TenGigabitEthernet0/0/0
 ip router isis CORE
 isis network point-to-point
!
R1# show isis neighbors
System Id   Type Interface  IP Address   State Holdtime Circuit Id
R2          L2   Te0/0/0    10.0.12.2    UP    27       00
```

OSPF versus IS-IS

ASPECT	OSPF	IS-IS
Transport	IP	Directly over Layer 2
Area boundary	On the router (interfaces in different areas)	On the link (each router belongs to one area)
Backbone	Area 0	Contiguous Level 2 routers

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ASPECT	OSPF	IS-IS
Database units	Several LSA types	One LSP per router per level, with TLVs
Multi-access election	DR and BDR, non-preemptive	DIS, preemptive, no backup
IPv4 and IPv6	OSPFv2 and OSPFv3 (or v3 address families)	One instance carries both

SCENARIO · DECIDE · CHOOSING AN IGP FOR A NEW CORE

A national ISP-style backbone for a utility company will carry IPv4 and IPv6, connect 60 core routers, and must be easy to extend with segment routing in future. The team knows OSPF well but has some IS-IS experience. Which choice is most defensible?

(A) OSPFv2 for IPv4 and static routes for IPv6.

(B) IS-IS with a single Level 2 domain and wide metrics, carrying IPv4 and IPv6 in one instance.

(C) OSPFv3 for IPv6 and OSPFv2 for IPv4, each designed with ten areas and summarisation at every ABR.

(D) IS-IS with every router Level 1 in separate areas.

Circle your choice. Answer and feedback: module answer key.

Key points

- OSPFv3 runs over IPv6 link-local addresses and moves prefixes into separate LSAs.
- Set the OSPFv3 router ID explicitly on IPv6-only routers.
- IS-IS runs over Layer 2, uses levels instead of area 0, and extends through TLVs.
- A NET encodes area, system ID and NSEL; derive system IDs from loopbacks.

KNOWLEDGE CHECK

11. What next-hop address does OSPFv3 install for an IPv6 route learned from a neighbour?

(A) The neighbour's global unicast address
(B) The neighbour's 32-bit router ID
(C) The neighbour's link-local address
(D) The all-OSPF-routers multicast address

12. Why must the OSPFv3 router ID be configured manually on an IPv6-only router?

(A) The 32-bit ID cannot be derived from IPv6 addresses
(B) OSPFv3 requires the ID to match the loopback IPv6 address
(C) OSPFv3 refuses to run without a configured area range
(D) Automatic IDs would conflict with link-local addresses

13. In the NET 49.0002.0100.0000.0005.00, what is the area?

(A) 0100.0000.0005
(B) 49
(C) 00
(D) 49.0002

14. Where is the area boundary in IS-IS?

(A) On a router with interfaces in two different areas
(B) On the link between routers in different areas
(C) At the designated intermediate system
(D) At the router with the highest system ID

15. How does a Level 1 IS-IS router normally reach destinations outside its area?

(A) Through type 3 summaries advertised by an ABR
(B) Through a static route that must be added by hand
(C) Via a default route towards the nearest attached L1/L2 router
(D) Through Level 2 LSPs that L1/L2 routers flood into the Level 1 area

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MODULE 3 · IGP · LESSON 4 · 40 MIN

Route Control: Prefix Lists, Route Maps, Redistribution and PBR

Select, tag, filter and steer routes and packets without creating loops.

- Write prefix lists with ge/le and predict what they match.
- Build route maps with correct match/set logic.
- Redistribute safely between routing domains using tags.
- Apply policy-based routing to steer selected traffic.

Real networks rarely run one routing protocol cleanly from edge to edge. Mergers, acquisitions, static routes to partners and BGP at the edge all need **route control**: deciding exactly which prefixes cross from one domain to another, with which metrics and markings. The tools are simple — prefix lists, route maps and tags — but the logic must be exact, because a single wrong line can black-hole a network or create a routing loop.

Prefix list matching

ENTRY	MATCHES
permit 10.0.0.0/8	Only the prefix 10.0.0.0/8 itself
permit 10.0.0.0/8 le 24	Any prefix inside 10.0.0.0/8 with length 8 to 24
permit 10.0.0.0/8 ge 24 le 24	Only /24 prefixes inside 10.0.0.0/8
permit 10.0.0.0/8 ge 25	Prefixes inside 10.0.0.0/8 with length 25 to 32
permit 0.0.0.0/0	Only the default route
permit 0.0.0.0/0 le 32	Every IPv4 prefix

DEFINITION · IMPLICIT DENY

A prefix list and a route map both end with an invisible deny. Anything not explicitly permitted is dropped. Add a final permit-all entry only when that is truly what you want.

Route-map logic

R1

Sequences in order

Statements are evaluated by sequence number; the first statement that matches decides the outcome.

R2

AND across match types

Several match commands of different types in one statement must all be true.

R3

OR within one match

Several values in one match command (for example two prefix lists) match if any one matches.

R4

No match clause

A statement without any match matches everything — useful as a final catch-all.

R5

Implicit deny

Routes that match no statement are denied.

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Redistribution copies routes from one routing source into another. Each protocol needs a sensible **seed metric** for imported routes, and each router compares candidates first by **administrative distance**, a router-local trust value. Where two domains are redistributed into each other at **more than one point**, routes can leak back into their origin domain with a better administrative distance or metric, causing suboptimal paths or loops. Tags are the standard defence.

Common default administrative distances

SOURCE	TYPICAL DEFAULT
Connected	0
Static	1
eBGP	20
OSPF	110
IS-IS	115
iBGP	200

Mutual redistribution with tags

R5 (one of two boundary routers) · IOS-style CLI

```
route-map ISIS-TO-OSPF deny 10
 match tag 110
route-map ISIS-TO-OSPF permit 20
 set tag 115
!
route-map OSPF-TO-ISIS deny 10
 match tag 115
route-map OSPF-TO-ISIS permit 20
 set tag 110
!
router ospf 1
 redistribute isis CORE level-2 subnets route-map ISIS-TO-OSPF
router isis CORE
 metric-style wide
 redistribute ospf 1 metric 20 level-2 route-map OSPF-TO-ISIS
```

Routes leaving OSPF are tagged 110 and routes leaving IS-IS are tagged 115. Each direction denies routes carrying the other side's tag, so nothing returns to where it came from. IS-IS needs wide metrics to carry tags.

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Policy-based routing

R7 · IOS-style CLI

```
ip access-list extended GUEST-WEB
 permit tcp 10.50.0.0 0.0.255.255 any eq 443
!
route-map PBR-GUEST permit 10
 match ip address GUEST-WEB
 set ip next-hop 203.0.113.1
!
interface GigabitEthernet0/2
 description Guest VLAN gateway
 ip policy route-map PBR-GUEST
```

PBR is applied to traffic arriving on the interface. Packets that match are sent to 203.0.113.1; everything else is routed normally. Track the next hop's reachability so traffic is not black-holed if it fails.

Route filtering versus PBR

Route policy (prefix lists, route maps)

- Acts on routing updates
- Changes what the routing table contains
- Affects all traffic to a destination

Policy-based routing

- Acts on data packets
- Overrides the routing table for matching traffic
- Can match source, protocol or port

TEMPLATE

Redistribution matrix

Agree, for every boundary router and direction, exactly what is imported and how it is marked.

BOUNDARY ROUTER	FROM → TO	PREFIXES ALLOWED	SEED METRIC	TAG SET	TAGS DENIED
R5	*IS-IS → OSPF*	*10.64.0.0/12 le 24*	*E2 20*	*115*	*110*
R5	*OSPF → IS-IS*	*10.0.0.0/12 le 24*	*20 (wide)*	*110*	*115*

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SCENARIO · DECIDE · A LOOP AFTER A MERGER

After adding a second boundary router between OSPF and IS-IS with mutual redistribution, some IS-IS prefixes are reached through the OSPF domain and back, and traceroutes show loops during failures. Route maps redistribute everything with no filtering. What is the best fix?

(A) Remove the second boundary router to keep one redistribution point.

(B) Lower OSPF's administrative distance below IS-IS on every router.

(C) Tag routes on redistribution and deny the other domain's tag in the opposite direction on both boundary routers.

(D) Use policy-based routing on the boundary routers to steer IS-IS prefixes.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

16. Which prefix list entry matches only /24 prefixes inside 172.16.0.0/12?

(A) permit 172.16.0.0/12 le 24
(B) permit 172.16.0.0/12 ge 24 le 24
(C) permit 172.16.0.0/24
(D) permit 172.16.0.0/12 ge 12

17. A route-map statement contains `match ip address prefix-list A` and `match tag 50`. When does it match?

(A) When the route is in list A or carries tag 50
(B) When the route is in list A but carries no tag at all
(C) Only when neither condition is true
(D) When the route is in list A and carries tag 50

18. What happens to a route that matches no statement in a route map used for redistribution?

(A) It is redistributed with default values
(B) It is redistributed with tag 0
(C) It is not redistributed
(D) It is redistributed as external type 1

19. Why are tags used with mutual redistribution at two boundary routers?

(A) To stop routes being redistributed back into their origin domain
(B) To give redistributed routes a lower metric than internal ones
(C) To make OSPF prefer external over internal routes
(D) To ensure that both boundary routers advertise the same seed metric

20. Policy-based routing is configured on a router's LAN interface. Which traffic does it affect?

(A) Matching packets leaving through that interface
(B) Routing updates received on that interface
(C) Matching packets arriving on that interface
(D) All traffic routed by the router

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MODULE 3 · IGP · LESSON 5 · 35 MIN

Troubleshooting OSPF Adjacencies and Routes

Neighbour states, the parameters that must match and a fast path from symptom to root cause.

- Map each stuck neighbour state to its likely causes.
- Use show and debug output to prove a mismatch.
- Distinguish adjacency faults from route-installation faults.

Most OSPF incidents fall into two families: **neighbours that will not reach Full**, and **neighbours that are Full but routes are missing or wrong**. The first is about hello and database-exchange parameters; the second about area design, LSA filtering, network types, summarisation and route preference. Identify the family first, then the specific cause.

OSPF neighbour states

01 **Down** — No hellos received → 02 **Init** — Hello received, our ID not yet listed → 03 **2-Way** — Bidirectional; DR/BDR election on multi-access → 04 **ExStart / Exchange** — Master/slave chosen; database descriptions swapped → 05 **Loading** — Missing LSAs requested → 06 **Full** — Databases synchronised

Stuck states and likely causes

SYMPTOM	LIKELY CAUSES
No neighbour at all	Interface passive, ACL or filter dropping multicast, wrong subnet, area ID, hello/dead timers, authentication or stub-flag mismatch
Stuck in Init	Hellos only flowing one way (filtering, unidirectional link)
2-Way with some neighbours	Normal between non-DR routers on a broadcast segment
Stuck in ExStart/Exchange	MTU mismatch, duplicate router ID, unicast reachability issues
Full but routes missing	Network-type mismatch, area type filtering, summarisation, distribute lists, route preference

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Proving a timer mismatch

R1 · IOS-style CLI

```
R1# show ip ospf interface GigabitEthernet0/1
GigabitEthernet0/1 is up, line protocol is up
  Internet Address 10.1.12.1/30, Area 0
  Process ID 1, Router ID 10.255.0.1, Network Type POINT_TO_POINT, Cost: 40
  Timer intervals configured, Hello 30, Dead 120, Wait 120, Retransmit 5
  Neighbor Count is 0, Adjacent neighbor count is 0
R1# debug ip ospf hello
OSPF-1 HELLO Gi0/1: Rcv hello from 10.255.0.2 area 0 10.1.12.2
OSPF-1 HELLO Gi0/1: Mismatched hello parameters from 10.1.12.2
OSPF-1 HELLO Gi0/1: Dead R 40 C 120, Hello R 10 C 30
```

R = received from the neighbour, C = configured locally. Debug output is illustrative and differs between platforms; limit debugs to one interface or neighbour on production devices.

Stuck in ExStart

R1 · IOS-style CLI

```
R1# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.255.0.3        1   EXSTART/DR      00:00:36    10.1.13.3       GigabitEthernet0/2
R1# show interfaces GigabitEthernet0/2 | include MTU
  MTU 1500 bytes, BW 1000000 Kbit/sec
R3# show interfaces GigabitEthernet0/2 | include MTU
  MTU 9000 bytes, BW 1000000 Kbit/sec
```

Database description packets carry the interface MTU; a router rejects DBDs advertising a larger MTU than its own. Fix the MTU on both sides; ignoring the MTU check only hides the problem.

> **WATCH OUT · THE SILENT NETWORK-TYPE MISMATCH**
> If one side is point-to-point and the other broadcast, the default timers are the same, so the adjacency can reach Full. But one router describes a point-to-point link and the other a transit network with a DR, so SPF cannot validate the link and routes through it are missing. Always compare network types on both ends.

Fast path from symptom to cause

1. **Is the neighbour listed?** `show ip ospf neighbor`. If absent, check passive interfaces, ACLs and `show ip ospf interface` on both sides.
2. **Compare hello parameters** Area, subnet and mask (on multi-access), timers, authentication, stub flags, network type.
3. **Stuck beyond 2-Way?** Check MTU on both ends and look for duplicate router ID messages.
4. **Full but no route?** Check the LSDB for the LSA, the area type, summaries and filters, then the routing table for a better route from another source.

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SCENARIO · DECIDE · ONE-WAY NEIGHBOUR

On a shared segment, R1 lists R2 in state INIT, while R2 does not list R1 at all. Yesterday a new inbound access list was applied on R2's interface to that segment. What is the most likely fix?

(A) Lower the OSPF priority on R1 so that R2 becomes the designated router for the shared segment.

(B) Permit OSPF (IP protocol 89, including 224.0.0.5 and 224.0.0.6) in R2's new inbound ACL.

(C) Raise the MTU on R1 to match the jumbo setting on R2 so that database descriptions are accepted.

(D) Change the segment's area to a stub area on both routers to reduce flooding between them both.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · HOSPITAL GROUP: MERGING A DIAGNOSTICS NETWORK

A private hospital group acquires a diagnostics chain. The group runs OSPF with area 0 in two data centres and one area per hospital. The diagnostics chain runs IS-IS Level 2 with 30 sites. Integration must be redundant, and both teams want to keep their IGP for a year. In a test, the team connects two boundary routers, redistributes everything both ways, and sees laboratory systems reached through odd paths and occasional loops. One boundary router also fails to form an OSPF adjacency with its data-centre neighbour, which sits in ExStart.

1. What is causing the odd paths and loops?

2. How would you fix the redistribution design?

3. What is the most likely cause of the ExStart problem, and how would you confirm it?

4. How would you document the integration so operators can troubleshoot it later?

Model answer: module answer key.

TEMPLATE **OSPF area plan**

Plan areas, their type, ABRs and summaries before configuring.

AREA	TYPE	SITES / ROLE	ABRS	SUMMARY RANGE	NOTES
0	*Backbone*	*Data centres and core*	*—*	*—*	*Reference bandwidth 400,000 everywhere*
40	*Totally NSSA*	*120 retail stores*	*R40A, R40B*	*10.40.0.0/16*	*Local internet as type 7; both ABRs summarise*

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AREA	TYPE	SITES / ROLE	ABRS	SUMMARY RANGE	NOTES

OSPF troubleshooting checklist

☐ Neighbour listed and in the expected state.
☐ Area ID, subnet and mask match on both ends.
☐ Hello and dead timers match.
☐ Authentication type and key match.
☐ Area type (stub/NSSA) matches on all routers in the area.
☐ Network type is the same on both ends.
☐ MTU matches on both ends.
☐ Router IDs are unique.
☐ Expected LSA is present in the LSDB.
☐ No better route from another source or route type is overriding OSPF.

REFLECT

Which OSPF parameter mismatch have you seen most often in your own networks? What standard configuration template or automated check would make it impossible to repeat?

Key terms

LSDB
Link-state database; identical on every router in an OSPF area.

ABR
Area Border Router, connecting area 0 to one or more other areas.

ASBR
Autonomous System Boundary Router, injecting external routes into OSPF.

Type 3 LSA
Summary LSA generated by an ABR to advertise inter-area prefixes.

Type 5 LSA
AS external LSA flooded throughout the domain except stub and NSSA areas.

Type 7 LSA
NSSA external LSA, translated into type 5 by an ABR.

Totally stubby area
An area receiving only intra-area routes and a default route from the ABR.

NSSA
Not-so-stubby area: blocks type 5 LSAs but allows local redistribution as type 7.

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Reference bandwidth
The value divided by interface bandwidth to calculate OSPF cost.

Attached bit
Set by an IS-IS L1/L2 router so Level 1 routers install a default route towards it.

Route tag
A numeric label attached to a route, used to make policy decisions during redistribution.

NET
Network entity title: the IS-IS address of a router, encoding area and system ID.

Prefix list
An ordered list matching prefixes by address and length, with ge and le options.

Policy-based routing
Forwarding selected packets according to policy rather than the routing table.

KNOWLEDGE CHECK

21. Two routers are stuck in ExStart. What should you check first?
    - (A) Interface MTU on both ends
    - (B) The DR priority values
    - (C) The OSPF process IDs
    - (D) The OSPF reference bandwidth

22. A neighbour is Full but no routes are installed through it. One side is point-to-point and the other broadcast. Why?
    - (A) Point-to-point links cannot carry OSPF routes at all
    - (B) The link is described inconsistently, so SPF cannot use it
    - (C) The adjacency should already have failed because timers differ
    - (D) Broadcast networks only support inter-area routes

23. In debug output, `Dead R 40 C 120` appears. What does it mean?
    - (A) This router uses 40 s, the neighbour uses 120 s
    - (B) Both routers agree on a dead time of 40 s
    - (C) The dead timer will reset after 120 seconds
    - (D) The neighbour uses 40 s, this router uses 120 s

24. Two routers on a broadcast segment remain in 2-Way with each other and Full with the DR. What does this indicate?
    - (A) Normal behaviour for two non-DR routers
    - (B) An MTU mismatch between the two routers
    - (C) An authentication failure on the segment
    - (D) A duplicate router ID on the segment

25. A neighbour never appears in `show ip ospf neighbor`, and `show ip ospf interface` shows the interface is passive. What is the fix?
    - (A) Raise the DR priority on the interface
    - (B) Lower the hello timer on both sides
    - (C) Change the area type to a stub area on both routers involved
    - (D) Make the interface non-passive under the OSPF process

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 3 · IGP

Answer key

Knowledge checks

1. A Network LSAs are generated by the DR to describe a broadcast or NBMA segment and its attached routers.

2. D An ABR generates a type 4 so routers can compute the cost to an ASBR outside their area; the external prefixes themselves are in type 5 LSAs.

3. C 100,000 ÷ 10,000 = 10.

4. B Route type is compared before cost: inter-area is preferred over any external route.

5. C External routes injected inside an NSSA travel as type 7; an ABR translates them into type 5 so other areas learn them.

6. A Totally NSSA receives only a default as type 3 and carries local externals as type 7. Stub and totally stubby areas forbid redistribution.

7. D Area type is negotiated in hellos; all routers in the area must use the same setting.

8. A Sixteen /24s need a /20, and 32 is divisible by 16, so 10.20.32.0/20 covers exactly 10.20.32.0–10.20.47.255.

9. D The discard route catches packets for addresses in the summary that have no more-specific route, so they are not sent back towards a default and looped.

10. B `summary-address` summarises type 5 or type 7 routes at the ASBR (or the NSSA translator). Inter-area summaries use `area range`.

11. C OSPFv3 runs over link-local addresses and uses them as next hops.

12. A The router ID is still 32-bit. Without any IPv4 address to select from, the router has nothing to derive it from.

13. D Everything before the 6-byte system ID and the NSEL is the area ID: 49.0002.

14. B Each IS-IS router belongs to one area; the boundary falls on the links between areas, unlike OSPF where it sits on the ABR.

15. C L1/L2 routers set the attached bit in their L1 LSP, and L1 routers install a default route towards the closest one.

16. B ge 24 le 24 restricts the length to exactly 24 within the /12. `le 24` also matches /12 to /23.

17. D Different match types in the same statement are combined with AND.

18. C Route maps end with an implicit deny.

19. A Tags mark where a route came from so the opposite route map can deny it, preventing feedback loops.

20. C PBR is applied to incoming packets on the interface where the policy is attached; unmatched packets follow the routing table.

21. A Database description exchange fails when MTUs differ. Process IDs are local and need not match; reference bandwidth affects cost only.

22. B Default timers match for both types, so the adjacency forms, but the LSAs describe the link differently and the two-way check in SPF fails.

23. D R is the value received from the neighbour and C the value configured locally.

24. A Non-DR/BDR routers only become fully adjacent with the DR and BDR; 2-Way between them is expected.

25. D Passive interfaces send no hellos, so no adjacency can form on them.

Scenarios & case studies

> **Scenario · Placing a new site**
> **A** is best. Correct. A separate area contains flapping and SPF work, and two ABRs give redundancy while advertising one summary to the backbone.

> **Scenario · An area that only needs a default**
> **D** is best. Correct. It removes type 3, 4 and 5 information from the stores while still allowing local redistribution.

> **Scenario · Choosing an IGP for a new core**
> **B** is best. Correct. A flat L2 design suits 60 core routers, one instance handles both address families, and TLV extensibility suits future features.

> **Scenario · A loop after a merger**
> **C** is best. Correct. Tags stop routes re-entering their original domain at the second boundary point.

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Scenario · One-way neighbour
B is best. Correct. R2 drops R1's hellos, so R1 hears R2 but never sees itself listed: the classic one-way INIT symptom.

Case study · Hospital group: merging a diagnostics network
The loops come from **mutual redistribution at two points without filtering**: routes from IS-IS enter OSPF at one boundary router and return into IS-IS at the other. Fix it with **tags** (set on export, deny the other side's tag on import) on both boundary routers, explicit **prefix lists** defining which ranges each domain may send, and **summarisation** of each domain's ranges at the boundary so only a few stable prefixes cross. Enable wide metrics in IS-IS so tags are carried. ExStart almost always means an **MTU mismatch** (or a duplicate router ID); compare `show interfaces` MTU on both ends and fix the MTU rather than ignoring the check. Document a redistribution matrix and an OSPF area plan, including boundary routers, tags, prefix ranges and seed metrics.

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04

MODULE 4 · BGP

BGP for the Enterprise

Sessions, attributes, policy, route reflection and dual-ISP traffic engineering, with a method for finding why a route is wrong.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Build and troubleshoot eBGP and iBGP sessions.
- Predict BGP best-path selection from attributes and show output.
- Write safe inbound and outbound policy using prefix lists, route maps, AS-path filters and communities.
- Scale iBGP with route reflectors.
- Design dual-ISP multihoming with deliberate inbound and outbound traffic engineering.

LESSONS

1. BGP Sessions: eBGP, iBGP and Neighbour States · 35 min
2. Path Attributes and the Best-Path Decision · 35 min
3. BGP Policy: Prefix Lists, Route Maps and Communities · 35 min
4. Scaling iBGP: Full Mesh and Route Reflectors · 30 min
5. Multihoming to Two ISPs and BGP Troubleshooting · 40 min

MODULE 4 · BGP · LESSON 1 · 35 MIN

BGP Sessions: eBGP, iBGP and Neighbour States

How BGP peers find each other, what changes between external and internal sessions, and how to read session state.

- Explain the differences between eBGP and iBGP sessions.
- Configure eBGP and loopback-based iBGP sessions with an address-family structure.
- Diagnose sessions stuck in Idle or Active from show output.

BGP-4 (RFC 4271) is a **path-vector** protocol built for policy, not for speed. It runs over TCP port 179, exchanges reachability with explicit attributes, and lets each autonomous system (AS) decide what it accepts and what it advertises. Enterprises use it to connect to internet providers and cloud platforms, to exchange routes with partners, and increasingly inside data-centre fabrics and large WANs.

eBGP versus iBGP

eBGP (different AS)

- Peers usually directly connected; packets sent with TTL 1 by default
- Next hop set to the advertising router's own address
- Own AS prepended to AS_PATH on advertisement
- LOCAL_PREF not sent
- Common default administrative distance: 20

iBGP (same AS)

- Peers usually on loopbacks, reachable through the IGP
- Next hop from eBGP is passed on unchanged by default
- AS_PATH not modified inside the AS
- Routes learned from one iBGP peer are not sent to another iBGP peer
- Common default administrative distance: 200

DEFINITION · THE IBGP SPLIT-HORIZON RULE

A BGP speaker does not advertise a route learned from an iBGP peer to another iBGP peer. Because AS_PATH is not changed inside an AS, it cannot detect internal loops, so this rule prevents them. The consequence is that iBGP needs a **full mesh**, or route reflectors or confederations to relax it.

Message types

MESSAGE	PURPOSE
OPEN	Starts a session: version, AS number, hold time, BGP identifier, capabilities
UPDATE	Advertises new paths with attributes or withdraws unreachable prefixes
KEEPALIVE	Confirms the peer is alive between updates
NOTIFICATION	Reports an error and closes the session
ROUTE-REFRESH	Asks a peer to resend its routes so new inbound policy can be applied (RFC 2918)

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Neighbour state machine

01 Idle — Waiting to start; also the state after errors →

02 Connect — Trying to complete the TCP connection →

03 Active — TCP attempt failed; retrying →

04 OpenSent — OPEN sent, waiting for the peer's OPEN →

05 OpenConfirm — OPENs accepted, waiting for KEEPALIVE →

06 Established — Exchanging UPDATEs

Enterprise edge router: one eBGP and one iBGP peer

E1 · IOS-style CLI

```
router bgp 64505
 bgp router-id 10.255.0.1
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 203.0.113.1 remote-as 64500
 neighbor 203.0.113.1 description ISP-A
 neighbor 10.255.0.2 remote-as 64505
 neighbor 10.255.0.2 update-source Loopback0
 address-family ipv4 unicast
  network 198.51.100.0 mask 255.255.255.0
  neighbor 203.0.113.1 activate
  neighbor 10.255.0.2 activate
  neighbor 10.255.0.2 next-hop-self
 exit-address-family
!
ip route 198.51.100.0 255.255.255.0 Null0
```

The `network` command advertises a prefix only if an exact match exists in the routing table, hence the static route to Null0. AS numbers and addresses here come from documentation ranges.

Reading the session summary

E1 · IOS-style CLI

```
E1# show ip bgp summary
BGP router identifier 10.255.0.1, local AS number 64505
Neighbor     V    AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down  State/PfxRcd
10.255.0.2   4 64505     812     809     45   0    0 12:01:44       12
203.0.113.1  4 64500   20411    1830     45   0    0 1d02h           3
```

A number in the last column means the session is Established and shows prefixes received. A word such as Idle or Active means it is not.

Session troubleshooting by state

STATE SEEN	COMMON CAUSES	CHECK
Idle	Neighbour administratively shut down, no route to the peer, maximum-prefix limit exceeded	Config, routing table, logs
Active	TCP not completing: wrong neighbour address, port 179 filtered, wrong update-source, eBGP peer not directly connected without multihop	Ping from the right source, ACLs, firewall logs
Flapping OpenSent/Idle	OPEN rejected: wrong AS number, duplicate router ID, unsupported capability	NOTIFICATION messages in logs

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STATE SEEN	COMMON CAUSES	CHECK
Established, 0 prefixes	Policy filtering in either direction, or peer advertising nothing	Inbound and outbound route maps, received-routes

A wrong-AS notification

E2 · IOS-style CLI

```
%BGP-3-NOTIFICATION: sent to neighbor 192.0.2.9 passive 2/2 (peer in wrong AS) 2 bytes FBFF
E2# show ip bgp summary | include 192.0.2.9
192.0.2.9    4 64510       0       0      1   0    0 never    Idle
```

Error 2/2 is an OPEN message error, 'bad peer AS'. FBFF in hexadecimal is 64511: the provider is using AS 64511 while E2 expects 64510. Confirm the correct AS with the provider.

PRO TIP · PROTECT THE SESSION

Authenticate eBGP sessions where the provider supports it (TCP-AO, RFC 5925, or the older TCP MD5 option), apply the generalised TTL security mechanism (RFC 5082) on directly connected peers, and set a maximum-prefix limit so a provider leak cannot exhaust your router's memory.

SCENARIO · DECIDE · STUCK IN ACTIVE

A new iBGP session between two edge routers' loopbacks stays in Active. Both routers can ping each other's loopback from the CLI. The configuration has `neighbor 10.255.0.2 remote-as 64505` on E1 and the equivalent on E2, with no other neighbour options. What is the most likely fix?

(A) Add `ebgp-multihop 2` on both routers because the loopback peers are not directly connected to each other.

(B) Add `next-hop-self` on both routers.

(C) Change both routers to different AS numbers.

(D) Add `update-source Loopback0` so each router sources the session from the address the other expects.

Circle your choice. Answer and feedback: module answer key.

Key points

- eBGP changes next hop and AS_PATH; iBGP does neither by default.
- Routes from one iBGP peer are not passed to another iBGP peer.
- Active means TCP is failing; Idle often means configuration, routing or max-prefix.
- Use loopbacks with update-source for iBGP and next-hop-self towards internal peers.

KNOWLEDGE CHECK

1. A session shows Active in `show ip bgp summary`. What does this indicate?

   (A) The session is established and actively exchanging routing updates

   (B) The peer has sent a NOTIFICATION and closed it

   (C) The router is retrying a TCP connection that has not succeeded

   (D) The router is waiting for the peer's KEEPALIVE

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2. Why do iBGP routers need a full mesh, route reflectors or confederations?

   (A) iBGP sessions only work between directly connected routers

   (B) iBGP-learned routes are not re-advertised to other iBGP peers

   (C) iBGP sessions cannot carry the LOCAL_PREF attribute

   (D) iBGP peers always change the next hop to their own loopback address

3. What does an eBGP router do to the AS_PATH when advertising a route to its external peer?

   (A) Removes all private AS numbers

   (B) Leaves the AS_PATH unchanged

   (C) Prepends its own AS number

   (D) Replaces it with its own AS only

4. A NOTIFICATION shows error 2/2 (bad peer AS). What is the likely cause?

   (A) The TCP session on port 179 is being blocked by a firewall rule

   (B) The remote-as configured does not match the peer's actual AS

   (C) The peer has exceeded the maximum-prefix limit

   (D) Both routers are using the same hold time

5. The `network 198.51.100.0 mask 255.255.255.0` command is configured but the prefix is not advertised. Why might that be?

   (A) No exact match for the prefix exists in the routing table

   (B) The prefix is in a documentation range

   (C) The network command only works for iBGP peers

   (D) The network command needs a route map attached before it works

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP · LESSON 2 · 35 MIN

Path Attributes and the Best-Path Decision

What each attribute carries and the order in which BGP compares them.

- Classify BGP path attributes as well-known or optional, transitive or not.
- Walk through the best-path decision process in the correct order.
- Predict which path a router selects from show output.

BGP routes are not compared by a single metric. Each path carries **attributes**, and a router compares candidate paths attribute by attribute, in a fixed order, until one path wins. Policy works by changing attributes, so knowing the order is what lets you predict — and control — traffic.

Common path attributes

ATTRIBUTE	CATEGORY	SET BY / SCOPE	PREFERENCE
ORIGIN	Well-known mandatory	The originating router	IGP < EGP < incomplete (lower is better)
AS_PATH	Well-known mandatory	Each AS prepends its number towards eBGP peers	Shorter is better
NEXT_HOP	Well-known mandatory	Usually the eBGP peer's address	Must be reachable
LOCAL_PREF	Well-known discretionary	Inside one AS only; commonly defaults to 100	Higher is better
MULTI_EXIT_DISC (MED)	Optional non-transitive	Sent to a neighbouring AS to suggest an entry point	Lower is better
COMMUNITY	Optional transitive	Tags used for policy (RFC 1997)	No preference by itself
ORIGINATOR_ID, CLUSTER_LIST	Optional non-transitive	Added by route reflectors	Loop prevention and tie-breaking

Best-path decision process (common order)

STEP	PREFER THE PATH WITH…	NOTES
0	A reachable next hop	Paths with unreachable next hops are not candidates
1	Highest local weight	Implementation-specific, local to one router, not a BGP attribute; skip if not supported
2	Highest LOCAL_PREF	The main outbound traffic-engineering knob for the whole AS
3	Locally originated route	Originated by this router with network, aggregate or redistribution
4	Shortest AS_PATH	Main inbound knob, influenced by prepending
5	Lowest ORIGIN	IGP before EGP before incomplete
6	Lowest MED	By default compared only between paths from the same neighbouring AS
7	eBGP over iBGP	Prefer routes learned directly from outside

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STEP	PREFER THE PATH WITH…	NOTES
8	Lowest IGP metric to the next hop	Hot-potato routing: leave the AS at the nearest exit
9	Final tie-breakers	Oldest eBGP path (some implementations), lowest router ID, shortest cluster list, lowest neighbour address

INSIGHT · WHO SEES WHICH ATTRIBUTE

LOCAL_PREF influences how **your** AS sends traffic out; it never leaves your AS. AS_PATH and MED influence how **other** ASes send traffic to you. That asymmetry is the basis of all BGP traffic engineering.

Which path wins?

E1 · IOS-style CLI

```
E1# show ip bgp 0.0.0.0
BGP routing table entry for 0.0.0.0/0, version 18
Paths: (2 available, best #2, table default)
  64510
    10.255.0.2 (metric 2) from 10.255.0.2 (10.255.0.2)
      Origin IGP, metric 0, localpref 80, valid, internal
  64500
    203.0.113.1 from 203.0.113.1 (203.0.113.250)
      Origin IGP, metric 0, localpref 200, valid, external, best
```

Both paths have reachable next hops and equal AS_PATH length, but the decision stops at LOCAL_PREF: 200 beats 80, so ISP-A's default route is used for outbound traffic.

Reading the comparison

MEASURE	FORMULA	WHAT IT TELLS YOU
AS_PATH length	`length = number of AS numbers in the path (an AS_SET counts as one)`	`64500 64500 64500 64496` has length 4; prepending twice adds two.
MED comparability	`compare MED only if first AS in AS_PATH is equal`	Paths from AS 64500 and AS 64510 skip the MED step unless configured otherwise.

Predicting the winner

1. **List all valid paths** Drop any with unreachable next hops or that failed policy.
2. **Walk the order** Compare each attribute in sequence; stop at the first difference.
3. **Name the deciding step** Write down which attribute decided; that is the one to change if the result is wrong.
4. **Check the RIB** Confirm the best path was installed and not displaced by a route with lower administrative distance.

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SCENARIO · DECIDE · UNEXPECTED EXIT

An enterprise receives the same partner prefix from ISP-A (AS path `64500 64496`) and ISP-B (AS path `64510 64496`). Both have local preference 100 and origin IGP. ISP-B's path has MED 10 and ISP-A's has MED 50. The router chooses ISP-A. Why?

(A) MED is not compared between different neighbouring ASes, so a later step such as eBGP over iBGP or IGP cost decided.

(B) Lower MED always wins once LOCAL_PREF and AS_PATH tie.

(C) The router prefers higher MED values.

(D) AS 64500 is numerically lower than AS 64510, so it wins.

Circle your choice. Answer and feedback: module answer key.

Key points

- Next hop must be reachable before any attribute is compared.
- LOCAL_PREF before AS_PATH before ORIGIN before MED before eBGP/iBGP before IGP cost.
- MED is compared only between paths from the same neighbouring AS by default.
- Weight is a local, implementation-specific value, not a path attribute.

KNOWLEDGE CHECK

6. Which attribute is the primary tool for controlling how traffic leaves your AS?

(A) MULTI_EXIT_DISC
(B) AS_PATH
(C) ORIGIN
(D) LOCAL_PREF

7. Two paths tie on LOCAL_PREF and AS_PATH length. One has origin IGP, the other incomplete. Which wins?

(A) The path with origin incomplete
(B) The path with origin IGP
(C) Neither; MED is compared first
(D) Both, installed as equal-cost paths

8. What is the AS_PATH length of `64510 64510 64510 64496`?

(A) 4
(B) 2
(C) 1
(D) 3

9. Why might a router ignore a lower MED from a different provider?

(A) MED is only used for routes learned from iBGP peers
(B) MED is only evaluated after the lowest router ID tie-breaker step
(C) MED values are only used for IPv6 routes
(D) MED is compared only between paths from the same neighbouring AS

10. After LOCAL_PREF, AS_PATH, ORIGIN and MED all tie, which path is preferred?

(A) The path with the higher router ID
(B) The path learned most recently
(C) The eBGP-learned path over the iBGP-learned path
(D) The path carrying the longer CLUSTER_LIST attribute

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP · LESSON 3 · 35 MIN

BGP Policy: Prefix Lists, Route Maps and Communities

Filter what you accept and advertise, and mark routes so policy scales.

- Build inbound and outbound policy for an internet edge.
- Use AS-path filters to prevent becoming a transit network.
- Apply standard, well-known and large communities to signal policy.

Every BGP session needs explicit policy in **both directions**. Inbound policy protects you from bad routes: bogons, your own prefixes coming back, overly specific announcements and route leaks. Outbound policy protects everyone else from you: an enterprise should advertise **only its own prefixes**, never routes learned from one provider to another. Without it, a multihomed enterprise can accidentally become a transit path between two ISPs.

Edge policy baseline

DIRECTION	POLICY	WHY
Outbound	Permit only your own prefixes (prefix list) and only locally originated paths (AS-path `^$`)	Prevents leaking one provider's routes to another
Inbound	Deny your own prefixes	Prevents paths to yourself through the internet
Inbound	Deny private, reserved and documentation ranges (bogons)	These should never appear on the internet
Inbound	Deny IPv4 prefixes longer than /24 (IPv6 longer than /48)	Common global filtering practice
Inbound	Maximum-prefix limit	Protects memory if the peer leaks a large table
Inbound	Route origin validation with RPKI (RFC 6811)	Rejects prefixes whose origin AS is not authorised

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Outbound and inbound edge policy

E1 · IOS-style CLI

```
ip prefix-list OUR-NETS seq 10 permit 198.51.100.0/24
ip as-path access-list 1 permit ^$
!
ip prefix-list FROM-ISP seq 5 deny 198.51.100.0/24 le 32
ip prefix-list FROM-ISP seq 10 deny 10.0.0.0/8 le 32
ip prefix-list FROM-ISP seq 15 deny 172.16.0.0/12 le 32
ip prefix-list FROM-ISP seq 20 deny 192.168.0.0/16 le 32
ip prefix-list FROM-ISP seq 100 permit 0.0.0.0/0 le 24
!
route-map TO-ISP-A permit 10
 match ip address prefix-list OUR-NETS
 match as-path 1
!
router bgp 64505
 address-family ipv4 unicast
  neighbor 203.0.113.1 prefix-list FROM-ISP in
  neighbor 203.0.113.1 route-map TO-ISP-A out
  neighbor 203.0.113.1 maximum-prefix 20000 90
```

The inbound list is shortened for clarity; a production bogon list also covers 0.0.0.0/8, 127.0.0.0/8, 169.254.0.0/16, 100.64.0.0/10, documentation ranges, 224.0.0.0/4 and 240.0.0.0/4.

WATCH OUT · REGULAR EXPRESSIONS ARE EXACT

`^$` matches an empty AS_PATH, which means routes originated inside your own AS. `_64500$` matches paths originated by AS 64500. Test every expression with show commands before applying it, because a wrong expression can withdraw all your prefixes or leak everything.

Communities

TYPE	FORMAT	USE
Standard (RFC 1997)	32-bit, written AS:value, e.g. 64500:120	Tag routes for policy inside your AS or with a provider
NO_EXPORT	Well-known	Do not advertise outside this AS (or confederation)
NO_ADVERTISE	Well-known	Do not advertise to any peer
Extended (RFC 4360)	64-bit, typed	Route targets and other VPN uses
Large (RFC 8092)	96-bit, written ASN:value1:value2	Communities for 32-bit AS numbers

Providers usually publish a **community menu**: values that customers can attach to their advertisements to request actions inside the provider's network, such as lowering local preference, prepending towards certain peers, or not announcing in a region. The values are provider-specific, so always read the provider's documentation and confirm with their support before relying on them.

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Requesting backup treatment from a provider

E2 · IOS-style CLI

```
route-map TO-ISP-B permit 10
 match ip address prefix-list OUR-NETS
 match as-path 1
 set community 64510:70
!
router bgp 64505
 address-family ipv4 unicast
  neighbor 192.0.2.9 route-map TO-ISP-B out
  neighbor 192.0.2.9 send-community
!
E2# clear ip bgp 192.0.2.9 soft out
```

In this example ISP-B documents 64510:70 as 'set local preference 70 in our network'. Many implementations do not send communities unless told to. A soft outbound clear re-sends updates without dropping the session.

Hard reset versus soft reset

Hard reset

- Tears down the TCP session
- All routes withdrawn and relearned
- Traffic disruption
- Rarely needed

Soft reset / route refresh

- Session stays up
- Policy re-applied to stored or re-requested routes
- No disruption
- Standard after a policy change

SCENARIO · DECIDE · THE ACCIDENTAL TRANSIT

A firm multihomed to two ISPs notices its internet links are saturated with traffic that is neither to nor from its own prefixes. Its outbound route maps to both ISPs contain only `permit 10` with no match statements. What happened and what is the fix?

(A) The ISPs are sending too many prefixes; add a maximum-prefix limit inbound.

(B) LOCAL_PREF is too high on one ISP; lower it to balance traffic.

(C) It is advertising routes learned from each ISP to the other; restrict outbound policy to its own prefixes and an empty AS-path.

(D) The firm should stop sending communities to the ISPs.

Circle your choice. Answer and feedback: module answer key.

Key points

- Every session needs inbound and outbound policy.
- Advertise only your own prefixes; use `^$` to allow only locally originated paths.
- Filter bogons, your own prefixes and over-specific routes inbound; set maximum-prefix.
- Communities signal policy; provider community values are provider-specific.

KNOWLEDGE CHECK

11. What does the AS-path regular expression `^$` match?

(A) Every route from every neighbour

(B) Routes originated inside the local AS

(C) Routes with exactly one AS in the path

(D) Routes received from iBGP peers only

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12. Which well-known community stops a route being advertised outside the receiving AS?

(A) NO_ADVERTISE
(B) NO_PEER
(C) LOCAL_PREF
(D) NO_EXPORT

13. Communities are configured in an outbound route map but the provider does not see them. What is the most likely missing step?

(A) Adding a matching inbound community list on the edge
(B) Converting them to large communities
(C) Enabling community sending towards that neighbour
(D) Clearing the session with a hard reset

14. What is the advantage of a soft outbound reset after changing outbound policy?

(A) Updates are re-sent without tearing down the session
(B) All received routes are flushed and relearned
(C) The peer is forced to renegotiate capabilities
(D) The maximum-prefix counter for the peer is set back to zero

15. Why should an enterprise filter IPv4 prefixes longer than /24 from its providers?

(A) BGP cannot carry prefixes longer than /24
(B) Longer prefixes are generally filtered across the internet
(C) Longer prefixes always carry an invalid ORIGIN
(D) Longer prefixes break the route reflection rules inside the AS

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MODULE 4 · BGP · LESSON 4 · 30 MIN

Scaling iBGP: Full Mesh and Route Reflectors

Why full mesh stops scaling, how route reflection works and how to design it safely.

- Calculate iBGP session counts for full mesh and route-reflector designs.
- Explain route-reflector advertisement rules and loop prevention.
- Configure a redundant route-reflector pair.

Because of the iBGP split-horizon rule, every iBGP speaker must normally peer with every other. That is fine for two edge routers, tolerable for five, and unmanageable for fifty. **Route reflectors** (RFC 4456) relax the rule in a controlled way, so that clients peer only with a few reflectors, which pass routes on.

Session counts

MEASURE	FORMULA	WHAT IT TELLS YOU
Full mesh	`sessions = n × (n − 1) ÷ 2`	20 routers need 190 sessions; 50 routers need 1,225.
Two route reflectors	`sessions = 2 × (n − 2) + 1`	20 routers with 2 reflectors: 18 clients × 2 + 1 = 37 sessions.

Route-reflector advertisement rules

ROUTE LEARNED FROM	REFLECTED TO
An eBGP peer	All clients and non-clients (normal behaviour)
A non-client iBGP peer	Clients only (and eBGP peers)
A client	All other clients and all non-clients (and eBGP peers)

Loop prevention and behaviour

- The reflector adds ORIGINATOR_ID (the router ID of the route's originator in the AS) and ignores routes carrying its own ID.
- It prepends its cluster ID to CLUSTER_LIST and discards routes that already contain it.
- A reflector does not change NEXT_HOP by default, so next hops must still be reachable from clients.
- Only the reflector needs special configuration; clients are ordinary iBGP speakers.
- Clients only see the reflector's best path, which can hide alternative exits.

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Route reflector with two clients

RR1 · IOS-style CLI

```
router bgp 64505
 bgp router-id 10.255.0.101
 neighbor 10.255.0.1 remote-as 64505
 neighbor 10.255.0.1 update-source Loopback0
 neighbor 10.255.0.2 remote-as 64505
 neighbor 10.255.0.2 update-source Loopback0
 neighbor 10.255.0.102 remote-as 64505
 neighbor 10.255.0.102 update-source Loopback0
 address-family ipv4 unicast
  neighbor 10.255.0.1 activate
  neighbor 10.255.0.1 route-reflector-client
  neighbor 10.255.0.2 activate
  neighbor 10.255.0.2 route-reflector-client
  neighbor 10.255.0.102 activate
 exit-address-family
```

10.255.0.102 is the second reflector (RR2), peered as a non-client. Clients E1 and E2 peer with both reflectors and need no special configuration.

Redundant reflector design

Reflectors	`RR1 and RR2, often out of the forwarding path` `Peer with each other as non-clients`
Clients	`Edge routers, WAN hubs, data-centre borders` `Each peers with both reflectors`
IGP	`Carries all loopbacks and next hops` `Must converge before BGP can use paths`

PRO TIP · PLACEMENT MATTERS

A reflector chooses its best path from its own position in the IGP. If it sits far from the clients, it may pick an exit that is not the closest for them. Place reflectors centrally, or use mechanisms that advertise more than one path (add-path, RFC 7911) where supported.

Route reflectors versus confederations

Route reflectors

- Change only the reflector configuration
- Most common in enterprises and providers
- Single AS number throughout

Confederations (RFC 5065)

- Split the AS into sub-ASes with eBGP-like sessions between them
- Useful during mergers
- Every router must be reconfigured

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SCENARIO · DECIDE · MISSING ALTERNATIVE PATH

An enterprise uses one route reflector in its head office. Both edge routers receive a default route from their ISP, with equal attributes. Branch routers, all clients, always use the head-office ISP even when the regional edge router is closer to them. What best explains this?

(A) Route reflectors always prefer eBGP paths learned over the oldest session, whichever edge router is nearest.

(B) Branch clients must be fully meshed with the edge routers to receive any routes.

(C) The regional edge router's next hop is unreachable.

(D) The reflector reflects only its own best path, chosen from its position, which favours the head-office exit.

Circle your choice. Answer and feedback: module answer key.

KNOWLEDGE CHECK

16. How many iBGP sessions does a full mesh of 12 routers need?

(A) 66 (B) 24
(C) 132 (D) 144

17. A route reflector learns a route from a non-client iBGP peer. Where does it reflect it?

(A) To all non-client peers (B) To no other iBGP peer
(C) To its clients only (D) Back to the originating peer

18. Which attribute lets a reflector detect a route that has looped back through its cluster?

(A) LOCAL_PREF (B) MULTI_EXIT_DISC
(C) ORIGIN (D) CLUSTER_LIST

19. What configuration do route-reflector clients need?

(A) The route-reflector-client command towards each other (B) A cluster ID matching the reflector's own ID
(C) None beyond normal iBGP peering with the reflectors (D) A confederation identifier for each client

20. Why must the IGP still carry eBGP next hops in a reflector design?

(A) Reflectors only accept routes with IGP origin (B) Clients rely on the IGP to elect the active reflector
(C) Reflectors drop routes without a MED value (D) Reflectors do not change the next hop by default

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MODULE 4 · BGP · LESSON 5 · 40 MIN

Multihoming to Two ISPs and BGP Troubleshooting

Control outbound and inbound traffic across two providers, and find out why a route is missing or not preferred.

- Design dual-ISP multihoming with clear primary/backup or load-sharing intent.
- Use local preference, AS-path prepending, MED, more-specific prefixes and communities appropriately.
- Troubleshoot missing, unpreferred or uninstalled BGP routes.

Multihoming to two providers protects an enterprise from a single provider outage and gives control over cost and performance. The design has two independent halves: **outbound** traffic engineering (which exit your users take) is fully under your control, while **inbound** traffic engineering (which link the world uses to reach you) can only be *influenced*, because other networks make the final decision.

Outbound versus inbound traffic engineering

Outbound (you decide)

- Raise LOCAL_PREF on routes from the preferred ISP
- Accept full, partial or default-only tables
- Per-prefix tuning for important destinations
- Effective immediately and predictably

Inbound (you influence)

- AS-path prepending towards the less-preferred ISP
- Provider communities that lower local preference inside that ISP
- Advertising more-specific prefixes on the preferred link
- MED only between links to the same ISP
- Results depend on other networks' policies

Inbound techniques compared

TECHNIQUE	STRENGTH	LIMITATION
AS-path prepending	Simple, works with any provider	Ignored by networks that prefer by local preference first; diminishing returns beyond 3 prepends
Provider communities	Acts inside the provider, where most decisions happen	Provider-specific; must be supported and documented
More-specific prefixes	Longest match always wins	Adds to global table size; IPv4 nothing longer than /24 propagates widely
MED	Precise between two links to one provider	Not meaningful between different providers

Dual-ISP reference design

Providers	`ISP-A (AS 64500): primary` `ISP-B (AS 64510): backup`
Edge	`E1 peers with ISP-A, E2 with ISP-B` `iBGP between E1 and E2 over loopbacks, next-hop-self`
Inside	`Firewalls and core learn a default from both edges` `Prefix 198.51.100.0/24 advertised from both edges`

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Backup edge router towards ISP-B

E2 · IOS-style CLI

```
route-map ISP-B-IN permit 10
 set local-preference 80
!
route-map ISP-B-OUT permit 10
 match ip address prefix-list OUR-NETS
 match as-path 1
 set as-path prepend 64505 64505 64505
!
router bgp 64505
 neighbor 192.0.2.9 remote-as 64510
 address-family ipv4 unicast
  neighbor 192.0.2.9 activate
  neighbor 192.0.2.9 prefix-list FROM-ISP in
  neighbor 192.0.2.9 route-map ISP-B-IN in
  neighbor 192.0.2.9 route-map ISP-B-OUT out
  neighbor 192.0.2.9 maximum-prefix 20000 90
```

Inbound local preference 80 (below E1's 200) keeps outbound traffic on ISP-A. Prepending makes the path through ISP-B longer for the rest of the internet, so inbound traffic prefers ISP-A while both links are up.

When a route is wrong, follow the route rather than guessing. First confirm the session is Established. Then ask four questions in order: **Was it received? Was it accepted? Is it best? Is it installed?** Each has a specific command and a specific set of causes, and the answer to one tells you where to look next.

Route troubleshooting ladder

QUESTION	HOW TO CHECK	IF THE ANSWER IS NO
Was it received?	Received-routes view (needs inbound soft reconfiguration) or the peer's advertised-routes	Peer's outbound policy, peer not originating it, session issue
Was it accepted?	`show ip bgp <prefix>` — present after inbound policy?	Inbound prefix list, route map, maximum-prefix, RPKI invalid
Is it best?	`show ip bgp <prefix>` — marked best (>)	Compare attributes in decision order; unreachable next hop
Is it installed?	`show ip route <prefix>`	A route with better administrative distance exists (RIB failure)
Is it advertised onwards?	`show ip bgp neighbors <peer> advertised-routes`	Outbound policy, iBGP split horizon, not best

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A classic iBGP next-hop problem

CORE1 · IOS-style CLI

```
CORE1# show ip bgp 0.0.0.0
BGP routing table entry for 0.0.0.0/0, version 3
Paths: (1 available, no best path)
  64500
    203.0.113.1 (inaccessible) from 10.255.0.1 (10.255.0.1)
      Origin IGP, metric 0, localpref 200, valid, internal
```

The next hop is the ISP's address, which CORE1 cannot reach because the edge link is not in the IGP. Configure next-hop-self on E1 towards its iBGP peers (or advertise the link into the IGP).

SCENARIO · DECIDE · A POLICY THAT DID NOT TAKE EFFECT

An engineer adds an inbound route map on E1 setting LOCAL_PREF 200 on routes from ISP-A, to make ISP-A primary. An hour later CORE1 still shows E1's paths with localpref 100 and traffic still leaves via E2. The session to ISP-A was not touched. What should be done?

- (A) Re-apply inbound policy with a soft reset or route refresh towards ISP-A.
- (B) Raise LOCAL_PREF further to 500 so the internal routers notice that the value has changed.
- (C) Set LOCAL_PREF on CORE1 as well, because LOCAL_PREF is never carried over iBGP sessions.
- (D) Prepend the AS path towards ISP-B so that internal routers start preferring the E1 exit path.

Circle your choice. Answer and feedback: module answer key.

CASE STUDY · GARMENT EXPORTER: DUAL-HOMING THE HEAD OFFICE

A garment exporter's head office hosts its order portal, supplier EDI and buyer video calls. It has a /24 of provider-independent space and its own public AS, and connects to ISP-A (fibre, 1 Gbit/s) and ISP-B (microwave backup, 300 Mbit/s). Today ISP-B carries almost no traffic and the portal is unreachable whenever ISP-A has an outage — even though the BGP session to ISP-B stays up. Investigation shows the exporter advertises its /24 only to ISP-A; towards ISP-B it advertises nothing, because the outbound prefix list on E2 still permits only the test prefix used when the link was commissioned. Management also wants ISP-B used for outbound bulk uploads to one cloud storage provider.

1. Why is the portal unreachable during ISP-A outages?
2. How would you design inbound traffic so ISP-A is primary and ISP-B is backup?
3. How would you send cloud storage uploads via ISP-B while keeping everything else on ISP-A?
4. What safeguards would you add to prevent a repeat?

Model answer: module answer key.

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TEMPLATE

BGP policy sheet

Define, per peer and direction, what is accepted or advertised and which attributes are set.

PEER (AS)	DIRECTION	MATCH	ACTION / ATTRIBUTES SET	INTENT	VERIFICATION COMMAND
ISP-A (64500)	*In*	*FROM-ISP prefix list*	*LOCAL_PREF 200*	*Primary exit for all traffic*	*show ip bgp 0.0.0.0*
ISP-B (64510)	*Out*	*OUR-NETS and AS-path ^$*	*Prepend 64505 ×3*	*Backup inbound path*	*show ip bgp neighbors 192.0.2.9 advertised-routes*

BGP edge readiness checklist

☐ Sessions authenticated and protected with TTL security where possible.

☐ Outbound policy permits only own prefixes and empty AS-path.

☐ Inbound policy filters bogons, own prefixes and over-specific routes.

☐ Maximum-prefix limits set on every external session.

☐ RPKI route origin validation enabled and ROAs published for own prefixes.

☐ iBGP between edges uses loopbacks, update-source and next-hop-self.

☐ Primary/backup intent documented and verified in both directions.

☐ Own prefixes monitored from outside the network.

☐ Policy changes applied with soft resets.

REFLECT

Draw your organisation's internet edge. Which link does inbound traffic actually use today, how do you know, and what would you change to make the behaviour match the business intent?

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Key terms

Autonomous system
A network under one routing policy, identified by an AS number.

eBGP
BGP between routers in different autonomous systems.

iBGP
BGP between routers in the same autonomous system.

Next-hop-self
Configuring a router to set itself as next hop for routes it sends to iBGP peers.

LOCAL_PREF
A well-known attribute shared inside an AS; higher values are preferred for outbound traffic.

AS_PATH
The list of AS numbers a route has crossed, used for loop prevention and path selection.

MED
Multi-exit discriminator; suggests the preferred entry point into an AS, lower is better.

AS-path prepending
Adding extra copies of your AS number to make a path look longer to others.

Community
A tag attached to routes to signal policy, standard (RFC 1997) or large (RFC 8092).

Route reflector
An iBGP speaker allowed to pass routes between iBGP peers, defined in RFC 4456.

CLUSTER_LIST
The list of reflector cluster IDs a route has passed, used to prevent loops.

Maximum-prefix
A limit on the number of prefixes accepted from a peer.

RPKI
Resource public key infrastructure used to validate that an AS may originate a prefix.

Soft reset
Re-applying BGP policy without tearing down the session.

KNOWLEDGE CHECK

21. Which technique gives an enterprise the most direct control over which ISP its users' outbound traffic uses?

(A) Prepending the AS path towards the backup ISP
(B) Setting LOCAL_PREF on routes received from each ISP
(C) Sending a lower MED value to the primary ISP
(D) Advertising more-specific prefixes to only one of the ISPs

22. An enterprise has links to two different ISPs. Why is MED a poor tool for inbound traffic engineering here?

(A) MED is normally only compared between links to the same AS
(B) MED is removed by every eBGP router on receipt
(C) MED applies only to IPv6 prefixes by default
(D) MED can only be set on iBGP sessions inside your own network

23. A route shows `(inaccessible)` next to its next hop on an internal router. What is the usual fix?

(A) Increase LOCAL_PREF for that route on the edge
(B) Configure next-hop-self on the edge router towards iBGP peers
(C) Add the route to an outbound prefix list
(D) Clear the eBGP session to the ISP with a full hard reset first

24. A BGP route is best in the BGP table but the routing table uses an OSPF route to the same prefix. Why?

(A) The OSPF route has a lower administrative distance than iBGP
(B) BGP routes are never installed alongside any OSPF routes at all
(C) The BGP route must have a higher LOCAL_PREF
(D) OSPF always overrides BGP for default routes

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25. Prepending your AS three times towards ISP-B has little effect on inbound traffic from some networks. Why?

(A) Prepending only works when a MED value is also set on each path

(B) Prepending removes your prefix from the table

(C) Those networks prefer by local preference before AS-path length

(D) Prepending is limited to iBGP sessions only

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 4 · BGP

Answer key

Knowledge checks

1. C Despite its name, Active means BGP is trying and failing to complete TCP. An Established session shows a prefix count instead of a state.

2. B The iBGP split-horizon rule prevents loops inside an AS but means each router must hear routes directly from the source unless reflection or confederations are used.

3. C Each AS adds its own number when advertising to eBGP peers; this is how loops between ASes are detected.

4. B OPEN message error subcode 2 means the AS in the peer's OPEN does not match what was configured.

5. A The network command requires an exact match in the routing table; a static route to Null0 is a common way to provide it.

6. D LOCAL_PREF is shared by all routers inside the AS and evaluated early, so it steers outbound traffic. MED and AS_PATH influence other ASes.

7. B ORIGIN is compared after AS_PATH: IGP is preferred over EGP, which is preferred over incomplete.

8. A Every AS number counts, including prepended duplicates; only an AS_SET counts as one.

9. D By default MED compares entry points into one neighbouring AS; comparing across ASes requires explicit configuration.

10. C eBGP over iBGP comes next, followed by the lowest IGP metric to the next hop and the final tie-breakers.

11. B An empty AS_PATH means the route has not crossed any AS boundary: it was originated locally.

12. D NO_EXPORT keeps the route inside the AS (or confederation). NO_ADVERTISE prevents advertising to any peer at all.

13. C Many implementations do not send communities by default; they must be enabled per neighbour.

14. A Soft reset re-applies policy and re-advertises without disrupting the TCP session or traffic.

15. B Common global practice is to accept nothing more specific than /24 for IPv4 (or /48 for IPv6); longer prefixes are likely leaks or errors.

16. A 12 × 11 ÷ 2 = 66.

17. C Routes from non-clients go to clients; routes from clients go to clients and non-clients.

18. D The reflector prepends its cluster ID; seeing it again means a loop. ORIGINATOR_ID is used for the same purpose at the originator level.

19. C Reflection is configured only on the reflector; clients are standard iBGP speakers.

20. D Reflected routes keep their original next hop (often the edge router's loopback with next-hop-self), so clients need an IGP route to it.

21. B Outbound path selection happens inside your AS, so LOCAL_PREF decides it. The other techniques influence how the world reaches you.

22. A Other networks compare MED only between paths from the same neighbouring AS, so it cannot steer between two providers.

23. B The internal router cannot reach the external next hop. Next-hop-self (or advertising the link in the IGP) makes the next hop reachable.

24. A The RIB installs the source with the lowest administrative distance; iBGP commonly has 200 and OSPF 110.

25. C Networks apply their own LOCAL_PREF first; if they prefer ISP-B for business reasons, AS-path length is never reached.

Scenarios & case studies

> **Scenario · Stuck in Active**
> **D** is best. Correct. Without it, TCP is sourced from the outgoing interface address, which does not match the peer's configured neighbour address, so the connection is refused.

> **Scenario · Unexpected exit**
> **A** is best. Correct. By default MED is compared only when paths come from the same neighbouring AS; here the first ASes differ.

> **Scenario · The accidental transit**
> **C** is best. Correct. A permit-all outbound policy turns the enterprise into a transit path between the two providers.

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Scenario · Missing alternative path
D is best. Correct. Clients only see one path — the reflector's choice — so they lose the closer exit. Placement or add-path addresses this.

Scenario · A policy that did not take effect
A is best. Correct. Inbound policy is applied as routes arrive; existing routes keep old attributes until they are re-received.

Case study · Garment exporter: dual-homing the head office
Because E2's outbound prefix list permits only the old test prefix, the /24 is never advertised to ISP-B, so when ISP-A fails the rest of the internet has no path to the exporter at all. Correct the prefix list, then **advertise the /24 to both ISPs**, with **AS-path prepending** (or ISP-B's backup community) towards ISP-B so inbound traffic prefers ISP-A while both are up. For outbound, set **higher LOCAL_PREF on ISP-A routes** (for example 200 against 80), then add a per-prefix exception: match the cloud provider's prefixes received from ISP-B and give them LOCAL_PREF 250, so only those uploads leave via ISP-B. Safeguards: a **BGP policy sheet** reviewed before change, `show ip bgp neighbors <peer> advertised-routes` checks after every change, monitoring that alerts if either ISP stops receiving the /24 (external looking-glass or route-monitoring service), maximum-prefix limits, inbound bogon and RPKI filtering, and soft resets for policy changes.

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05

MODULE 5 · OVERLAY

Overlay — Virtualisation & Overlays

Many logical networks on one underlay: VRFs, tunnels, IPsec, VXLAN EVPN and SD-WAN.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Segment an enterprise with VRFs and plan route distinguishers, route targets and route leaking.
- Build GRE and IPsec tunnels with correct MTU handling and avoid recursive routing.
- Explain how NHRP enables on-demand spoke-to-spoke tunnels in a dynamic multipoint VPN.
- Describe VXLAN encapsulation and the BGP EVPN control plane in a spine-leaf fabric.
- Design SD-WAN topology, application-aware routing and internet breakout policies.

LESSONS

1. VRFs and VRF-lite · 35 min
2. GRE Tunnels and Recursive Routing · 35 min
3. IPsec Site-to-Site and Dynamic Multipoint VPN · 40 min
4. VXLAN with an EVPN Control Plane · 40 min
5. SD-WAN Architecture and Policy · 40 min

MODULE 5 · OVERLAY · LESSON 1 · 35 MIN

VRFs and VRF-lite

Many routing tables on one box: isolation, route leaking and a route-target plan

- Explain how a VRF separates routing and forwarding tables on a single device.
- Configure and verify VRF-lite on a routed hop, including per-VRF routing.
- Distinguish route distinguishers from route targets and plan both for an enterprise.
- Choose a safe route-leaking method to reach shared services.

The Overlay stage of **The ZCNP Fabric** is about running several logical networks on one physical network. Modules 1 to 4 gave you a resilient underlay: a campus and spine-leaf design, a loop-free Layer 2 edge, a stable IGP and a disciplined BGP edge. Overlays reuse that investment. The first and simplest overlay tool is the **VRF**: a separate routing and forwarding instance that lets guests, IoT devices, corporate users and management traffic share links without ever seeing each other's routes.

> DEFINITION · VIRTUAL ROUTING AND FORWARDING (VRF)
> An independent routing table, forwarding table (FIB) and set of interfaces on one device. A packet arriving on an interface bound to a VRF is looked up **only** in that VRF's table, so identical prefixes can exist in different VRFs and traffic cannot cross between them unless you deliberately leak routes.

Where VRFs sit in an enterprise design

Shared services	`DNS and DHCP` `Identity and NAC servers` `Internet edge firewall`
Leaking and policy point	`Fusion firewall or router` `Selective import/export of routes` `Inter-VRF security policy`
VRF instances	`CORP` `GUEST` `IOT` `MGMT`
Physical underlay	`Routed uplinks and subinterfaces` `SVIs per VLAN` `Single IGP or per-VRF routing`

VRF-lite or a full VPN control plane?

VRF-lite

- VRFs configured hop by hop
- Each VRF needs its own subinterface or SVI on every transit link
- Per-VRF routing protocol instance or static routes
- Simple and fine for 2–4 VRFs across a few hops
- Operational effort grows with every VRF and every hop

MP-BGP VPN (L3VPN or EVPN)

- Only edge devices hold the VRFs
- Core carries labels or VXLAN, not per-VRF state
- One BGP session carries routes for all VRFs
- Route targets control who imports what
- Scales to hundreds of VRFs and sites

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VRF-lite on a transit link

R1 · IOS-style CLI

```
vrf definition GUEST
 rd 65010:20
 address-family ipv4
 exit-address-family
!
interface GigabitEthernet0/1.20
 encapsulation dot1Q 20
 vrf forwarding GUEST
 ip address 10.20.0.1 255.255.255.252
!
router ospf 20 vrf GUEST
 router-id 10.255.20.1
 network 10.20.0.0 0.0.255.255 area 0
```

The RD is optional for pure VRF-lite but harmless, and it keeps the configuration ready for a later MP-BGP migration.

WATCH OUT · ORDER OF OPERATIONS

Binding an interface to a VRF removes any IP address already configured on it. Apply `vrf forwarding` first, then the address. On a production device, stage the change in a maintenance window because the interface drops out of the global table immediately.

Verifying per-VRF routing

R1 · IOS-style CLI

```
R1# show ip route vrf GUEST
Routing Table: GUEST
      10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C        10.20.0.0/30 is directly connected, GigabitEthernet0/1.20
L        10.20.0.1/32 is directly connected, GigabitEthernet0/1.20
O        10.20.64.0/18 [110/2] via 10.20.0.2, 00:12:41, GigabitEthernet0/1.20
R1# ping vrf GUEST 10.20.64.1 source GigabitEthernet0/1.20
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
```

A plain `ping` uses the global table and would fail here. Most show and test commands need the `vrf` keyword.

RD, RT and VRF name: three different things

ITEM	WHAT IT IS	SCOPE	TYPICAL FORMAT
VRF name	Local label for the instance	Only the local device	GUEST, CORP, IOT
Route distinguisher (RD)	Prefix added to make overlapping routes unique inside MP-BGP	Per VRF per device (unique per PE is common)	ASN:nn or router-ID:nn
Route target (RT)	Extended community that controls import and export	Network-wide policy	ASN:nn, e.g. 65010:20

Ways to leak routes to shared services

1. **Fusion firewall** Each VRF peers with a firewall that holds the inter-VRF policy. Best visibility and control; the usual enterprise choice.

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2. **Fusion router** A router with all VRFs exchanges routes using per-VRF peering. Simple, but policy is limited to route filtering and ACLs.
3. **Route-target import** With MP-BGP, a VRF imports the shared-services RT and the shared-services VRF imports each client RT. Precise but easy to over-leak.
4. **Static inter-VRF routes** A static route in one VRF points to a next hop in another. Acceptable for one or two prefixes; hard to audit at scale.

TEMPLATE

VRF and route-target plan

Agree VRF names, identifiers and leaking rules before any device is touched, so every team configures the same values.

VRF	PURPOSE	RD SCHEME	EXPORT RT	IMPORT RT(S)	LEAKS TO / POLICY POINT
GUEST	*Visitor internet only*	*<loopback>:20*	*65010:20*	*65010:20, 65010:900 (DHCP/DNS)*	*Internet via fusion firewall only*
IOT	*Building systems*	*<loopback>:30*	*65010:30*	*65010:30, 65010:900*	*Vendor jump host via firewall; no CORP*

SCENARIO · DECIDE · GUESTS CANNOT GET AN ADDRESS

After moving guest SSIDs into a new GUEST VRF, clients associate but never receive an IP address. The DHCP server lives in a shared-services VRF. Relay is configured on the guest SVI. What is the most likely fix?

A. Move the DHCP server into the global routing table so that every VRF on the campus can reach it.
B. Configure the same route distinguisher on GUEST and the shared-services VRF so routes are shared.
C. Disable VRF-lite on the guest SVI until the DHCP issue is solved, then re-enable it afterwards.
D. Leak the DHCP server's prefix into GUEST and the guest subnets back, and use a VRF-aware relay.

Circle your choice. Answer and feedback: module answer key.

Key terms

VRF-lite
VRFs configured hop by hop without an MP-BGP control plane; each VRF needs its own logical interface on every transit link.

Route target (RT)
A BGP extended community attached on export and matched on import to decide which VRFs receive a route.

Route distinguisher (RD)
A 64-bit value prepended to an IPv4 or IPv6 prefix so overlapping customer routes stay unique inside MP-BGP.

KNOWLEDGE CHECK

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1. A router has the prefix 10.1.0.0/16 in both the CORP and IOT VRFs. What allows this?

   (A) The router silently prefers the VRF that was configured first on the device

   (B) Overlapping prefixes are only allowed when both VRFs share the same RD value

   (C) Each VRF has its own routing table and FIB, so the prefixes never collide

   (D) The prefixes are automatically summarised into one entry in the global table

2. In an MP-BGP VPN, which attribute decides whether a route is imported into a VRF?

   (A) The route target extended community

   (B) The route distinguisher prepended to the prefix

   (C) The VRF name configured on the receiving router

   (D) The advertising router's next hop

3. An engineer configures `ip address` and then `vrf forwarding CORP` on an interface. What happens?

   (A) The address moves into CORP automatically with no impact

   (B) The IP address is removed and must be configured again

   (C) The command is rejected until the address is removed by hand

   (D) The interface keeps its address in both the global table and CORP

4. Which route-leaking approach gives the most inspection and logging between VRFs?

   (A) Static inter-VRF routes pointing directly at a next hop in the other VRF

   (B) Importing every client route target into the shared-services VRF

   (C) Using one route distinguisher for all VRFs that need to communicate

   (D) Leaking through a fusion firewall that holds the inter-VRF policy

5. When does VRF-lite start to become hard to operate?

   (A) When many VRFs cross many routed hops, each needing its own subinterface

   (B) As soon as a second VRF is added to any single device in the campus network

   (C) When the VRFs use overlapping private address space in different buildings

   (D) When a management VRF is used for out-of-band access to every device

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MODULE 5 · OVERLAY · LESSON 2 · 35 MIN

GRE Tunnels and Recursive Routing

Point-to-point overlays, MTU budgets and the classic flapping-tunnel trap

- Describe GRE encapsulation and the overhead it adds.
- Configure a GRE tunnel with correct MTU and MSS settings.
- Recognise recursive routing and fix it with route separation or a front-door VRF.

Generic Routing Encapsulation (GRE) wraps one packet inside another so that it can cross a network that would not otherwise carry it. It is protocol-agnostic, supports multicast, and therefore lets you run an IGP between two sites over the internet or a provider core. GRE adds no confidentiality or integrity by itself; in production it is almost always protected by IPsec, which the next lesson covers.

GRE over IPv4: what is on the wire

Delivery IPv4 header (20 bytes)	`Source = local tunnel source` `Destination = remote tunnel endpoint` `IP protocol 47`
GRE header (4 bytes)	`Protocol type of the payload` `Optional key and sequence fields add 4 bytes each`
Passenger packet	`Original IP header` `Routing protocol hellos, multicast or user data`

Point-to-point GRE with an IGP over it

R1 · IOS-style CLI

```
interface Tunnel0
 description GRE to Branch-7
 ip address 172.31.7.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 keepalive 10 3
 tunnel source GigabitEthernet0/0
 tunnel destination 198.51.100.7
!
router ospf 1
 network 172.31.7.0 0.0.0.3 area 0
```

1400/1360 leaves room for IPsec later. For plain GRE, 1476 is the true maximum on a 1500-byte path.

Tunnel MTU arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
GRE IP MTU	`1500 − 20 (delivery IP) − 4 (GRE) = 1476`	The largest passenger IP packet that fits a 1500-byte transport without fragmentation.
TCP MSS	`IP MTU − 20 (IP) − 20 (TCP) = 1436 for GRE alone`	Clamping MSS on the tunnel stops TCP hosts from sending segments that will need fragmenting.
Safe IPsec-ready values	`ip mtu 1400, adjust-mss 1360`	Leaves headroom for ESP, NAT-T and padding so the same tunnel survives protection being added.

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DEFINITION · RECURSIVE ROUTING

A tunnel is recursively routed when the best route to its **own destination address** points back into the tunnel. The router cannot send a packet into a tunnel whose transport depends on that same tunnel, so it shuts the tunnel down, the routes learned over it disappear, the tunnel comes back, and the cycle repeats.

How the flap cycle forms

01 Tunnel up
Transport route to 198.51.100.7 uses the physical WAN.

→

02 IGP adjacency forms
OSPF runs over Tunnel0 and exchanges all prefixes.

→

03 Better route learned
The remote site advertises a route covering 198.51.100.7 through the tunnel.

→

04 Recursion detected
The route to the tunnel destination now points into the tunnel; the router brings Tunnel0 down.

→

05 Routes withdrawn
Adjacency drops, the physical route returns, and the tunnel comes up again.

The log signature of recursive routing

R1 · IOS-style CLI

```
%TUN-5-RECURDOWN: Tunnel0 temporarily disabled due to recursive routing
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
%OSPF-5-ADJCHG: Process 1, Nbr 10.255.0.7 on Tunnel0 from FULL to DOWN
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
%OSPF-5-ADJCHG: Process 1, Nbr 10.255.0.7 on Tunnel0 from LOADING to FULL
```

A tunnel that goes down and up on a regular rhythm, together with adjacency changes, is the tell-tale pattern.

Fixes that work and fixes that hide the problem

Durable fixes

- Never advertise transport (WAN) prefixes into the overlay IGP
- Pin the tunnel destination with a /32 static route via the physical next hop
- Put the transport interface in a front-door VRF so tunnel lookups never use the overlay table
- Filter the remote endpoint prefix inbound on the tunnel

Weak fixes

- Raising the IGP cost on the tunnel (a longer prefix still wins)
- Disabling keepalives so the flap is less visible
- Shortening timers so the tunnel recovers faster
- Rebooting the branch router

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Front-door VRF: transport and overlay in separate tables

R1 · IOS-style CLI

```
vrf definition INET
 address-family ipv4
 exit-address-family
!
interface GigabitEthernet0/0
 vrf forwarding INET
 ip address 203.0.113.2 255.255.255.252
!
ip route vrf INET 0.0.0.0 0.0.0.0 203.0.113.1
!
interface Tunnel0
 tunnel source GigabitEthernet0/0
 tunnel destination 198.51.100.7
 tunnel vrf INET
```

The tunnel endpoint is resolved only in INET, while the tunnel interface itself stays in the global or a corporate VRF. A default route from the internet can never collide with overlay routes.

GRE troubleshooting quick list

- Tunnel down/down: check the route to the tunnel destination and that the source interface is up.
- Up/up but no adjacency: verify that both sides use the same subnet and that multicast hellos pass.
- Small pings pass, large transfers stall: check `ip mtu`, MSS clamping and whether ICMP 'fragmentation needed' is filtered.
- Periodic flapping: suspect recursive routing before blaming the carrier.

SCENARIO · DECIDE · THE TUNNEL THAT FLAPS EVERY 70 SECONDS

A new GRE tunnel to a branch comes up, OSPF reaches FULL, and roughly a minute later both drop and return. The branch advertises a summary 198.51.100.0/24 into OSPF, which includes the head-end's tunnel destination. What do you change?

A. Stop advertising the transport range into OSPF, or put the WAN interface in a front-door VRF.

B. Set a higher OSPF cost on Tunnel0 so that the physical route to the branch is always preferred.

C. Increase the keepalive interval to 30 seconds so the tunnel has more time to stabilise itself.

D. Replace GRE with IP-in-IP encapsulation, which does not suffer from recursive routing at all.

Circle your choice. Answer and feedback: module answer key.

Key terms

GRE
A stateless encapsulation (IP protocol 47) that carries any network-layer packet, including multicast, inside a delivery IP packet.

Front-door VRF
A VRF that holds the transport interface and its routes, so tunnel endpoint lookups are isolated from the overlay routing table.

KNOWLEDGE CHECK

6. What is the IP MTU of a plain GRE tunnel over a 1500-byte IPv4 path with no key or sequence fields?

A. 1500 bytes
B. 1460 bytes
C. 1400 bytes
D. 1476 bytes

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7. Which log pattern most strongly suggests recursive routing?

   (A) The tunnel stays down/down after the tunnel source interface is administratively shut

   (B) The tunnel and its IGP adjacency go down and up again on a regular cycle

   (C) The tunnel stays up but small pings succeed while large ones fail

   (D) The tunnel is up/up but the IGP never progresses beyond the INIT state

8. Why does raising the IGP cost on a tunnel usually fail to fix recursive routing?

   (A) The IGP silently ignores manually configured cost values on all GRE tunnel interfaces

   (B) Tunnel interfaces always use a fixed administrative distance of zero in the routing table

   (C) Longest-prefix match is applied before metrics, so the more specific route still wins

   (D) Cost changes only take effect after the tunnel interface is reset

9. What does the `tunnel vrf INET` command achieve?

   (A) The tunnel interface itself is moved into the INET VRF for overlay routing

   (B) The tunnel endpoint is looked up in the INET VRF, not the overlay table

   (C) All traffic entering the tunnel is filtered against the INET routing table

   (D) The INET VRF routes are leaked into the global table through the tunnel

10. Users across a GRE tunnel can load small web pages but large file transfers hang. What is the most likely cause?

    (A) An IGP metric mismatch between the two ends of the tunnel interfaces

    (B) The GRE keepalive interval is too short for the tunnel round-trip time

    (C) The tunnel uses a /30 subnet instead of a /31 point-to-point subnet

    (D) A path MTU problem, with ICMP feedback filtered along the way

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MODULE 5 · OVERLAY · LESSON 3 · 40 MIN

IPsec Site-to-Site and Dynamic Multipoint VPN

IKEv2, route-based tunnels and on-demand spoke-to-spoke paths

- Explain the IKEv2 exchanges and the SAs they create.
- Configure a route-based IKEv2 tunnel protected by IPsec and interpret its counters.
- Describe how NHRP lets spokes build direct tunnels in a hub-and-spoke dynamic multipoint VPN.

IPsec adds confidentiality, integrity and peer authentication to an overlay. In modern designs it is built **route-based**: a virtual tunnel interface (VTI) carries traffic, routing decides what enters it, and IKEv2 negotiates the keys. This keeps security and routing cleanly separated, and it is the foundation for both classic site-to-site tunnels and hub-and-spoke dynamic multipoint VPNs.

IPsec building blocks

IKE

IKE SA

The authenticated, encrypted control channel between peers, built by IKEv2 (UDP 500, or 4500 when NAT is detected).

CHLD

Child SA

A pair of unidirectional IPsec SAs (one each way) that actually protect user traffic, identified by SPIs.

ESP

Encapsulating Security Payload

IP protocol 50. Provides encryption and integrity; AEAD ciphers such as AES-GCM do both in one step.

AUTH

Peer authentication

Pre-shared keys for small estates; certificates from an internal PKI for anything that must scale or rotate.

PFS

Perfect forward secrecy

A fresh Diffie-Hellman exchange per child SA rekey, so one compromised key does not expose earlier traffic.

IKEv2 in four exchanges

01 **IKE_SA_INIT**
Agree algorithms, exchange DH values and nonces, detect NAT.

→

02 **IKE_AUTH**
Prove identities with PSK or certificates; create the first child SA.

→

03 **CREATE_CHILD_SA**
Add more child SAs or rekey existing ones, optionally with PFS.

→

04 **INFORMATIONAL**
Liveness checks (dead peer detection), deletes and error notifications.

Policy-based or route-based?

Policy-based (crypto ACL)

- Interesting traffic defined by an ACL
- Each subnet pair can become its own SA
- No interface, so dynamic routing is awkward
- Mismatched ACLs between peers are a classic failure

Route-based (VTI)

- A tunnel interface with an IP address
- Routing (static, OSPF, BGP) decides what is protected
- One SA pair per tunnel, any traffic
- Interface counters, QoS and monitoring just work

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IKEv2 and IPsec profile

HQ-R1 · IOS-style CLI

```
crypto ikev2 proposal PROP-GCM
 encryption aes-gcm-256
 prf sha384
 group 20
crypto ikev2 policy POL-1
 proposal PROP-GCM
crypto ikev2 keyring KR-BR7
 peer BR7
  address 198.51.100.7
  pre-shared-key <from-secrets-vault>
crypto ikev2 profile PROF-BR7
 match identity remote address 198.51.100.7 255.255.255.255
 authentication remote pre-share
 authentication local pre-share
 keyring local KR-BR7
crypto ipsec transform-set TS-GCM esp-gcm 256
crypto ipsec profile IPSEC-BR7
 set transform-set TS-GCM
 set ikev2-profile PROF-BR7
```

AES-GCM is an AEAD cipher, so the proposal needs a PRF but no separate integrity algorithm. Group 20 is 384-bit ECP.

Static VTI and a one-way traffic symptom

HQ-R1 · IOS-style CLI

```
interface Tunnel7
 ip address 172.31.7.1 255.255.255.252
 ip mtu 1400
 ip tcp adjust-mss 1360
 tunnel source GigabitEthernet0/0
 tunnel mode ipsec ipv4
 tunnel destination 198.51.100.7
 tunnel protection ipsec profile IPSEC-BR7
!
HQ-R1# show crypto ipsec sa interface Tunnel7 | include pkts
    #pkts encaps: 15204, #pkts encrypt: 15204, #pkts digest: 15204
    #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
```

Encaps rising while decaps stays at zero means our packets leave protected but nothing comes back: check the peer's routing back into its tunnel, its firewall or NAT, not the crypto settings.

Reading an IPsec failure by phase

SYMPTOM	LIKELY CAUSE	FIRST CHECK
IKE_SA_INIT fails, 'no proposal chosen'	Algorithm or DH group mismatch	Compare proposals on both peers
IKE_AUTH fails	Wrong PSK, identity mismatch, expired or untrusted certificate	Peer identity type and certificate chain
Child SA up, encaps only	Return routing, ACL or NAT on the far side	Peer's route back via its tunnel
Small packets pass, large fail	MTU and fragmentation	Tunnel ip mtu and MSS clamp

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SYMPTOM	LIKELY CAUSE	FIRST CHECK
Tunnel drops behind NAT after idle	NAT mapping expires	NAT keepalives and DPD settings

A **dynamic multipoint VPN** scales the same idea to hundreds of sites. Every spoke has one multipoint GRE (mGRE) tunnel interface protected by IPsec. Spokes register their public (NBMA) address with the hub using the **Next Hop Resolution Protocol (NHRP)**. Traffic between spokes can start through the hub, and NHRP then lets the spokes discover each other's public addresses and build a direct tunnel on demand.

On-demand spoke-to-spoke tunnel

01 Register — Spoke sends an NHRP registration mapping its tunnel IP to its public address. →

02 First packets via hub — Spoke A sends traffic for Spoke B's subnet to the hub. →

03 Redirect — The hub forwards it and tells Spoke A a better path exists. →

04 Resolve — Spoke A asks for Spoke B's public address via NHRP. →

05 Direct tunnel — IKEv2 and IPsec build a spoke-to-spoke SA; a shortcut route is installed. →

06 Age out — Idle shortcuts and SAs time out, keeping state proportional to real traffic.

Minimal multipoint tunnel with NHRP redirect and shortcut

Hub and spoke · IOS-style CLI

```
! Hub
interface Tunnel100
 ip address 10.100.0.1 255.255.255.0
 ip nhrp network-id 100
 ip nhrp map multicast dynamic
 ip nhrp redirect
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile DMVPN-PROF
! Spoke
interface Tunnel100
 ip address 10.100.0.21 255.255.255.0
 ip nhrp network-id 100
 ip nhrp nhs 10.100.0.1 nbma 203.0.113.10 multicast
 ip nhrp shortcut
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 tunnel protection ipsec profile DMVPN-PROF
```

With redirect and shortcut, the hub can advertise only a summary to spokes; specific shortcut routes are created when traffic needs them.

PRO TIP · DESIGN NOTES FOR HUB-AND-SPOKE VPNS

Run two hubs on separate transports for resilience, summarise from the hubs, and choose a routing protocol that scales to the spoke count (BGP with route reflection on the hubs is common). Plan the IPsec SA count for the busiest spoke-to-spoke patterns, not the average.

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SCENARIO · DECIDE · BRANCH TUNNEL UP, BUT ONE-WAY TRAFFIC

A new branch VTI shows an established IKEv2 SA. On HQ, `#pkts encaps` increases steadily but `#pkts decaps` stays at zero. Branch users report no reply from HQ servers. What is the best next step?

(A) Change the IKEv2 proposal to a different Diffie-Hellman group on both of the peers.

(B) Check that the branch routes return traffic for HQ subnets into its own tunnel interface.

(C) Lower the tunnel IP MTU to 1300 bytes on both routers in case large packets are dropped.

(D) Replace the pre-shared key with a certificate in case peer authentication is failing silently.

Circle your choice. Answer and feedback: module answer key.

Key terms

IKEv2
The key-exchange protocol that authenticates IPsec peers and negotiates SAs in a small, fixed set of exchanges.

NHRP
Next Hop Resolution Protocol: maps tunnel addresses to public (NBMA) addresses so multipoint tunnels can find peers dynamically.

Virtual tunnel interface (VTI)
A routable tunnel interface whose traffic is protected by IPsec, making encryption route-based.

KNOWLEDGE CHECK

11. Which IKEv2 exchange authenticates the peers and creates the first child SA?

(A) IKE_AUTH
(B) IKE_SA_INIT
(C) CREATE_CHILD_SA
(D) INFORMATIONAL

12. Why does an AES-GCM IKEv2 proposal not include a separate integrity algorithm?

(A) AES-GCM disables integrity checking to reduce CPU load on the router
(B) Integrity checking is negotiated later in the IKE_AUTH exchange instead
(C) AES-GCM is an AEAD cipher: it encrypts and authenticates together
(D) GCM relies on the Diffie-Hellman group for integrity of each packet

13. What is the main operational advantage of route-based IPsec over policy-based IPsec?

(A) It removes the need to authenticate the two IPsec peers to each other
(B) It encrypts traffic faster because it uses a different cipher family
(C) It avoids the use of IKE for key exchange between the two peers
(D) Routing, interface counters and QoS work naturally on the tunnel

14. In a dynamic multipoint VPN, what does a spoke learn through NHRP resolution?

(A) The public (NBMA) address behind another spoke's tunnel address
(B) The pre-shared key that protects the hub's multipoint tunnel interface
(C) The routing table of the hub, used to replace its own IGP routes
(D) The MTU of the transport path between the hub and every spoke

15. An IPsec tunnel behind a home-office NAT drops after a few idle minutes and recovers when traffic restarts. What should you tune first?

(A) The IKEv2 encryption algorithm and Diffie-Hellman group on both peers
(B) The tunnel IP MTU and TCP MSS values on both of the tunnel interfaces
(C) NAT keepalives and dead peer detection to hold the NAT entry open
(D) The route distinguisher configured on the branch VRF for the tunnel

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MODULE 5 · OVERLAY · LESSON 4 · 40 MIN

VXLAN with an EVPN Control Plane

Layer 2 and Layer 3 segments over a routed fabric, learned through BGP

- Describe VXLAN encapsulation, VNIs and the MTU impact on the underlay.
- Explain how BGP EVPN route types distribute MAC, IP and tenant prefix information.
- Read and verify a basic leaf configuration with an L2VNI and an L3VNI.

Module 1 built a routed spine-leaf fabric. Applications still sometimes need the same Layer 2 segment on many racks, and tenants need isolation beyond 4,094 VLANs. **VXLAN** (RFC 7348) solves the data-plane part by tunnelling Ethernet frames in UDP between **VTEPs** on the leaves. **BGP EVPN** solves the control-plane part: instead of flooding to learn where hosts live, leaves advertise MAC and IP reachability to each other through BGP.

VXLAN encapsulation (outermost first)

Outer Ethernet (14 bytes)	`Next-hop MAC in the underlay`
Outer IP (20 bytes)	`Source = local VTEP loopback` `Destination = remote VTEP or multicast group`
Outer UDP (8 bytes)	`Destination port 4789` `Source port derived from a hash of the inner flow, giving ECMP entropy`
VXLAN header (8 bytes)	`24-bit VXLAN Network Identifier (VNI)` `Flags`
Original Ethernet frame	`Inner MACs, optional inner VLAN tag removed or mapped` `Tenant payload`

VXLAN numbers to remember

MEASURE	FORMULA	WHAT IT TELLS YOU
Segment space	`2^24 = 16,777,216 VNIs`	Versus 4,094 usable VLAN IDs; VNIs are global identifiers across the fabric.
Overhead	`14 + 20 + 8 + 8 = 50 bytes (54 with an outer 802.1Q tag)`	Added to every inner frame on the underlay.
Underlay MTU	`Inner MTU + 50 ≥ 1550; jumbo (9,000+) is the norm`	Without it, full-size host frames are dropped silently inside the fabric.

Flood-and-learn or EVPN?

Flood-and-learn VXLAN

- BUM traffic flooded via underlay multicast or head-end lists
- Remote MACs learned from data traffic
- No knowledge of host IPs, so ARP is flooded
- Simple, but noisy and slow to converge on moves

BGP EVPN control plane

- MAC and IP learned locally and advertised in BGP
- Remote VTEPs discovered automatically
- ARP suppression answers locally from EVPN data
- Supports multi-tenancy, mobility and multihoming

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EVPN route types you will meet

TYPE	NAME	CARRIES	USED FOR
1	Ethernet auto-discovery	Ethernet segment reachability	Multihoming fast convergence and aliasing
2	MAC/IP advertisement	Host MAC, optional IP, VNI(s)	Host reachability, ARP suppression, mobility
3	Inclusive multicast Ethernet tag	VTEP interest in a VNI	Building BUM replication lists
4	Ethernet segment	ES identifier	Designated forwarder election
5	IP prefix	IPv4/IPv6 prefix in a tenant VRF	External and subnet routes per tenant

How a host becomes reachable fabric-wide

01 Attach
A server sends traffic; leaf L3 learns its MAC and IP on VLAN 10.

→

02 Advertise
L3 sends a Type 2 route with VNI 10010, its VTEP address and route targets.

→

03 Reflect
Spines act as BGP route reflectors and pass the route to every leaf.

→

04 Import
Leaves with a matching RT install the MAC/IP pointing to L3's VTEP.

→

05 Forward
Traffic is VXLAN-encapsulated directly to L3; ARP is answered locally.

For routing between subnets, leaves use a **distributed anycast gateway**: the same gateway IP and MAC exist on every leaf, so a host always routes at its first hop, even after it moves. With **symmetric IRB**, the ingress leaf routes into a per-tenant **L3VNI**, and the egress leaf routes again into the destination subnet. Each leaf only needs the VNIs of locally attached subnets, which keeps state and configuration small.

One L2VNI, one L3VNI, iBGP EVPN to a spine reflector

Leaf-3 · data-centre CLI (illustrative)

```
nv overlay evpn
vlan 10
  vn-segment 10010
interface nve1
  no shutdown
  host-reachability protocol bgp
  source-interface loopback1
  member vni 10010
    ingress-replication protocol bgp
  member vni 50001 associate-vrf
router bgp 65001
  neighbor 10.0.0.1
    remote-as 65001
    update-source loopback0
    address-family l2vpn evpn
      send-community extended
evpn
  vni 10010 l2
    rd auto
    route-target both auto
```

Ingress replication via BGP uses Type 3 routes instead of underlay multicast. Route targets must carry extended communities, so `send-community extended` *is essential.*

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WATCH OUT · THE SILENT MTU FAILURE

If the underlay MTU is 1500, pings with default size work and the fabric looks healthy, yet databases and backups stall. Always test with full-size, do-not-fragment pings between VTEP loopbacks and between hosts before go-live.

Verification sequence on a new leaf

- Underlay: loopbacks reachable with ECMP; jumbo DF ping between VTEPs succeeds.
- BGP EVPN sessions to both spines established, prefixes received.
- Type 3 routes present for each local VNI, and remote VTEPs listed as peers.
- Type 2 routes appear for local hosts and are imported on remote leaves.
- Tenant VRF shows Type 5 or host routes; anycast gateway answers on every leaf.

SCENARIO · DECIDE · HOSTS IN THE SAME VNI CANNOT REACH EACH OTHER

Two servers in VNI 10010 on different leaves cannot ping each other. Both leaves show the EVPN session up. Leaf-3 advertises a Type 2 route for its server, but Leaf-5 does not install it. What should you check first?

(A) Whether the underlay MTU on the spine switches is set to 9216 bytes on every single fabric link.

(B) Whether spanning tree is blocking VLAN 10 on the spine uplinks of both of the leaf switches.

(C) Whether Leaf-3's exported route targets match Leaf-5's import route targets for VNI 10010.

(D) Whether the anycast gateway MAC address is identical on both leaves.

Circle your choice. Answer and feedback: module answer key.

Key terms

VTEP
VXLAN tunnel endpoint: the device (usually a leaf) that encapsulates and decapsulates VXLAN traffic.

VNI
The 24-bit VXLAN Network Identifier that names a Layer 2 segment (L2VNI) or a tenant routing instance (L3VNI).

EVPN
A BGP address family that advertises MAC, IP and prefix reachability for Layer 2 and Layer 3 overlay services.

Distributed anycast gateway
The same default-gateway IP and MAC configured on every leaf so hosts route at their first hop wherever they attach.

KNOWLEDGE CHECK

16. How many bytes does VXLAN add to an untagged inner frame on an IPv4 underlay?

(A) 24 bytes (B) 50 bytes
(C) 36 bytes (D) 64 bytes

17. Which EVPN route type advertises a host's MAC and IP address?

(A) Type 2 (B) Type 1
(C) Type 3 (D) Type 5

18. Why do VTEPs vary the outer UDP source port?

(A) To identify the tenant VRF that the inner frame belongs to
(B) Hash entropy, so underlay ECMP spreads flows across the spines
(C) To signal the VNI to devices that cannot read the VXLAN header
(D) To prevent the underlay from fragmenting the outer packets

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19. What is the main benefit of ARP suppression in an EVPN fabric?

(A) ARP packets are encrypted so hosts cannot spoof the gateway address

(B) Hosts no longer need a default gateway configured to reach other subnets

(C) The underlay MTU requirement drops back to a standard 1,500 bytes

(D) Leaves answer ARP requests locally from EVPN data, reducing flooding

20. In symmetric IRB, what does the L3VNI carry?

(A) Broadcast and unknown-unicast traffic for every bridged segment

(B) BGP EVPN control-plane updates between leaves and spines

(C) Routed traffic between subnets for one tenant VRF

(D) Management traffic from the leaves to the controller

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MODULE 5 · OVERLAY · LESSON 5 · 40 MIN

SD-WAN Architecture and Policy

Orchestrator, controllers and edges; application-aware routing across many transports

- Describe the planes of an SD-WAN solution and how an edge is onboarded.
- Design application-aware routing using measurable SLA classes.
- Choose between centralised and direct internet breakout and plan a phased rollout.

SD-WAN takes the overlay tools from this module (VRFs, IPsec, dynamic tunnels) and puts them under **central policy**. Instead of configuring each router, you describe intent once: which sites talk to which, which applications need which path quality, where internet traffic breaks out. Controllers distribute routes, keys and policy; edges measure every path continuously and steer each application to a path that meets its SLA.

SD-WAN planes

Orchestration and management	`Zero-touch provisioning` `Templates and device groups` `Dashboards, APIs and audit`
Control	`Controllers distribute overlay routes, keys and policy` `Edges never need full-mesh routing sessions`
Data	`Edge routers build IPsec tunnels over every transport` `Per-tunnel probes measure loss, latency and jitter`
Transport	`MPLS` `Broadband internet` `4G/5G` `Satellite`

Zero-touch onboarding of an edge

01 Power on
Edge gets an address on any transport and contacts the provisioning service.

→

02 Authenticate
Serial number and device certificate are checked against the allowed list.

→

03 Receive identity
Edge learns its controllers, site ID and configuration template.

→

04 Control up
Secure control connections to controllers are established.

→

05 Data up
IPsec tunnels form to permitted peers; policy and telemetry begin.

Policy types in a typical SD-WAN

TOP

Topology policy
Who builds tunnels to whom: full mesh, hub-and-spoke, regional mesh, per VRF.

AAR

Application-aware routing
Maps applications to SLA classes and preferred transports; moves flows when a path breaches its SLA.

DATA

Traffic policy
Per-application actions such as local breakout, service chaining or forward error correction.

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QOS

QoS policy

Queuing, shaping per transport and marking, consistent with the campus QoS model.

SEC

Security policy

Segmentation per VRF, edge firewalling and redirection to cloud security services.

LOC

Local policy

Device-specific items such as interface ACLs or transport preferences.

Example SLA classes (tune to your applications)

CLASS	LOSS	LATENCY (ONE-WAY)	JITTER	PREFERRED PATH
Voice	≤ 1%	≤ 150 ms	≤ 30 ms	MPLS, then broadband
Interactive video	≤ 1%	≤ 150 ms	≤ 30 ms	MPLS or best broadband
Business-critical apps	≤ 2%	≤ 250 ms	n/a	Any compliant path
Bulk and backup	≤ 5%	n/a	n/a	Cheapest path, off-peak

Path measurement basics

MEASURE	FORMULA	WHAT IT TELLS YOU
Loss %	`(probes sent − probes received) ÷ probes sent × 100`	Measured per tunnel over a sliding window, often several minutes.
Jitter	`mean of |delay(n) − delay(n−1)|`	Variation in delay between consecutive probes; drives voice quality.
SLA compliance	`loss ≤ L AND latency ≤ D AND jitter ≤ J`	A path is eligible for a class only when all thresholds are met; hysteresis prevents flapping.

Where should internet traffic leave?

Centralised breakout

- Branch internet hairpins to a data centre or hub
- One security stack to manage
- Adds latency to SaaS and cloud apps
- Consumes expensive WAN bandwidth

Direct internet access (DIA)

- Trusted SaaS breaks out locally at the branch
- Security delivered by edge firewall or cloud security service
- Lower latency, less hub load
- Requires consistent security policy at every site

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Intent expressed as data

Orchestrator · policy export (illustrative YAML)

```
app_route_policy: VOICE-FIRST
  vpn: CORP
  sequences:
    - match: { app_list: VOICE-APPS }
      sla_class: VOICE
      preferred: [mpls, internet]
      fallback: best-path
    - match: { app_list: SAAS-TRUSTED }
      action: local-internet-breakout
sla_classes:
  VOICE: { loss_pct: 1, latency_ms: 150, jitter_ms: 30 }
```

Field names differ between platforms; the shape (match, SLA class, preference, fallback) is common to most.

INSIGHT · CONTROLLERS ARE NOT IN THE DATA PATH

If edges lose contact with the controllers, existing tunnels and policies normally keep forwarding for a grace period. Design controller redundancy for change and convergence, but do not confuse a controller outage with a WAN outage.

CASE STUDY · A MINISTRY'S 140-SITE SD-WAN ROLLOUT

A national ministry runs 140 offices on a single MPLS service with central internet breakout. Video calls to cloud collaboration tools are poor, MPLS contracts are expensive, and three regional offices already have fibre broadband. Leadership wants SD-WAN within a year without disrupting tax-collection systems that run on a central data centre.

1. How would you phase the rollout?
2. Which applications would you allow to break out locally, and how would you secure them?
3. How would you protect the tax-collection systems during migration?
4. What would you measure to prove success?

Model answer: module answer key.

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SCENARIO · DECIDE · VOICE STILL POOR DURING BROADBAND BROWNOUTS

Branch voice calls break up every afternoon. Telemetry shows broadband loss rising to 4% for 20-minute periods while MPLS stays clean. The app-aware policy maps voice to the VOICE class with broadband preferred because it has more bandwidth. Calls are not moving. What is the most likely problem?

A. Order more bandwidth on the broadband circuit so that voice has more room during the afternoon.

B. Make MPLS preferred for voice, or make the policy move flows when the VOICE SLA is breached.

C. Scale up the SD-WAN controllers, which are overloaded and too slow to steer the voice flows.

D. Remove voice from the overlay and send it directly to the internet from each branch office.

Circle your choice. Answer and feedback: module answer key.

Overlay go-live checklist

☐ VRF names, RDs and route targets match the agreed plan on every device.

☐ Inter-VRF leaking happens only at the documented policy point.

☐ Tunnel endpoints are resolved via the transport (or a front-door VRF), never via the overlay.

☐ Tunnel IP MTU and TCP MSS are set and tested with full-size DF pings.

☐ IKEv2 proposals use AEAD ciphers and strong DH groups; keys or certificates come from a managed store.

☐ VXLAN underlay MTU is jumbo end to end; EVPN sessions and route imports are verified.

☐ SD-WAN SLA classes, preferred paths and fallback behaviour are tested by injecting loss and delay.

☐ Rollback steps and a monitoring dashboard exist before the first site migrates.

REFLECT

Think about the WAN you know best. Which three applications would you put into explicit SLA classes first, and what evidence would you collect before choosing their thresholds?

Key terms

Orchestrator
The SD-WAN component that onboards edges, assigns identity and distributes the list of controllers.

Application-aware routing
Steering each application to a transport path whose measured loss, latency and jitter meet the application's SLA class.

KNOWLEDGE CHECK

21. In an SD-WAN, which component typically validates a new edge and tells it which controllers to use?

A. The orchestrator or provisioning service

B. The nearest branch edge router acting as a hub

C. The MPLS provider's provider-edge router

D. The cloud security service used for internet breakout

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22. A voice SLA class requires loss ≤ 1%, latency ≤ 150 ms and jitter ≤ 30 ms. A path measures 0.4% loss, 90 ms latency and 45 ms jitter. Is it eligible?

    (A) Yes, because two out of three thresholds are met
    (B) Yes, because latency is well under the limit
    (C) No, because latency is too close to the threshold
    (D) No, because jitter exceeds the threshold

23. What happens to forwarding in most SD-WAN designs if all controllers become unreachable?

    (A) All tunnels drop immediately and the branch falls back to static routes
    (B) Edges switch to flooding traffic over every transport until recovery
    (C) Existing tunnels keep forwarding on the last policy for a while
    (D) Traffic is sent to the orchestrator for forwarding instead of the edges

24. Which is the strongest reason to adopt direct internet access at branches?

    (A) It removes the need for any security inspection of internet-bound traffic
    (B) Lower SaaS latency and less backhaul, with consistent security enforced
    (C) It allows every application to bypass the SD-WAN policy completely
    (D) It makes the MPLS transport unnecessary for all internal applications

25. Why do SD-WAN policies apply hysteresis to path switching?

    (A) To encrypt the probes that measure loss, latency and jitter on each overlay tunnel
    (B) To stop flows bouncing between paths when measurements hover near a limit
    (C) To make sure the controllers approve every individual path change first
    (D) To delay onboarding of new edges until all transports are stable

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 5 · OVERLAY

Answer key

Knowledge checks

1. C VRFs are separate tables. Overlap is normal; collisions only become a concern when routes are leaked between VRFs.

2. A Route targets are policy. The RD only makes prefixes unique; the VRF name is local.

3. B Binding an interface to a VRF clears its IP configuration, so apply the VRF first.

4. D A firewall at the leaking point adds stateful policy and logs. RT import and static leaks only control reachability.

5. A VRF-lite effort grows with VRFs multiplied by hops. MP-BGP VPNs keep that state only at the edge.

6. D 1500 minus 20 bytes of delivery IP header and 4 bytes of GRE header gives 1476.

7. B Recursion causes a rhythmic flap: the adjacency brings in the bad route, the tunnel drops, the route goes away and the tunnel returns.

8. C Routing chooses the most specific prefix first. Cost matters only between equal-length routes from the same protocol.

9. B It sets the VRF used for the transport lookup only. The tunnel interface stays in whatever VRF its `vrf forwarding` command places it.

10. D This is the classic MTU black hole. Lower `ip mtu`, clamp MSS and allow ICMP type 3 code 4.

11. A IKE_SA_INIT negotiates algorithms and DH; IKE_AUTH proves identities and installs the first child SA.

12. C AEAD ciphers authenticate the data they encrypt, so only a PRF is configured for key derivation.

13. D A VTI behaves like any interface, so dynamic routing and monitoring are straightforward.

14. A NHRP maps overlay addresses to underlay addresses so spokes can build direct tunnels.

15. C Idle NAT entries expire; periodic keepalives keep the UDP 4500 mapping open.

16. B Outer Ethernet 14, outer IPv4 20, UDP 8 and VXLAN 8 make 50.

17. A Type 2 is the MAC/IP advertisement route; Type 3 announces VTEP interest for BUM replication; Type 5 carries IP prefixes.

18. B The source port is derived from a hash of inner headers, giving per-flow load sharing.

19. D Because EVPN already knows IP-to-MAC bindings, the leaf can reply on the remote host's behalf.

20. C Ingress and egress leaves both route; the transit VNI between them is the tenant's L3VNI.

21. A Onboarding and identity are orchestration functions; controllers then distribute routes and policy.

22. D All thresholds must be met; one breach makes the path non-compliant.

23. C The data plane is on the edges; controllers are needed for changes and new state.

24. B DIA improves SaaS experience but only works safely with consistent, centrally managed security.

25. B Hysteresis requires sustained improvement before switching back, preventing oscillation.

Scenarios & case studies

> **Scenario · Guests cannot get an address**
> **D** is best. Correct. The relay forwards requests from the GUEST VRF, so the server's route must exist there, and replies need a return route. Both directions must be leaked through the policy point.

> **Scenario · The tunnel that flaps every 70 seconds**
> **A** is best. Correct. Removing the overlap between transport and overlay routing removes the recursion permanently.

> **Scenario · Branch tunnel up, but one-way traffic**
> **B** is best. Correct. Crypto negotiation already succeeded; zero decaps means nothing protected is coming back, which is usually a routing, NAT or ACL issue on the far side.

> **Scenario · Hosts in the same VNI cannot reach each other**
> **C** is best. Correct. A route that is received but not installed is almost always an import policy problem, and RTs are the import policy.

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Case study · A ministry's 140-site SD-WAN rollout

Start with a design and pilot phase: two hub edges in the data centre, controllers in redundant locations, and a pilot of three to five sites with both MPLS and broadband. Keep separate VRFs for tax systems, general corporate traffic and guest access from day one. Allow DIA only for an approved SaaS list (collaboration and productivity suites), secured by a cloud security service or edge firewall with consistent policy; everything else continues to the data centre. Tax-collection traffic stays in its own VRF, pinned to MPLS with broadband as SLA-checked fallback, and sites migrate in waves with a rollback plan and a frozen change window around tax deadlines. Success measures: SLA compliance per class, MOS or call-quality scores for collaboration, mean time to provision a new site, WAN cost per site and zero incidents on tax systems.

Scenario · Voice still poor during broadband brownouts

B is best. Correct. Voice needs a compliant path more than bandwidth, and the policy must act on SLA breaches rather than a static preference.

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06

MODULE 6 · WIRELESS

Wireless — Enterprise Wireless

Design, secure and operate WLANs that carry voice, video and critical devices.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Design WLAN coverage and capacity using RF arithmetic, surveys and channel and power plans.
- Choose a WLAN architecture and enable fast, reliable roaming with 802.11k, v and r.
- Secure enterprise and guest wireless with WPA3, certificate-based EAP and OWE.
- Map voice and video QoS over Wi-Fi and validate it with real clients.
- Evaluate location services, analytics and Wi-Fi 7 features for business value.

LESSONS

1. RF Design, Site Surveys and Channel Planning · 40 min
2. WLAN Architectures and Seamless Roaming · 40 min
3. Wireless Security, Guest Access and QoS · 40 min
4. Location, Analytics and Wi-Fi 7 · 35 min

MODULE 6 · WIRELESS · LESSON 1 · 40 MIN

RF Design, Site Surveys and Channel Planning

Designing coverage and capacity you can prove, band by band

- Apply decibel arithmetic, free-space path loss and SNR to WLAN design decisions.
- Choose survey types and design targets for data, voice and location use cases.
- Plan channels, widths and transmit power across 2.4, 5 and 6 GHz.

Wireless is the Fabric stage where the physical medium is shared, invisible and constantly changing. Good enterprise WLANs are designed like any other network: requirements first, then a design with measurable targets, then validation. This lesson gives you the RF arithmetic and the survey discipline that every later wireless decision rests on.

RF arithmetic you will use every day

MEASURE	FORMULA	WHAT IT TELLS YOU
Decibel rules	`+3 dB ≈ double power, +10 dB = ten times power; −3 dB ≈ half`	Gains and losses in dB simply add and subtract.
EIRP	`Tx power (dBm) + antenna gain (dBi) − cable/connector loss (dB)`	What actually leaves the antenna; regulators limit EIRP, not just radio power.
Free-space path loss	`FSPL (dB) = 20·log10(d km) + 20·log10(f MHz) + 32.44`	At 20 m and 5,500 MHz this is about 73 dB; walls and bodies add more.
SNR	`SNR (dB) = RSSI (dBm) − noise floor (dBm)`	−65 dBm signal over a −92 dBm noise floor gives 27 dB SNR.

The three enterprise bands

BAND	NON-OVERLAPPING 20 MHZ CHANNELS	STRENGTHS	WATCH-OUTS
2.4 GHz	3 in most regions (1, 6, 11)	Range, legacy and IoT support	Crowded, interference-prone, too few channels for density
5 GHz	Up to about 25, depending on region and DFS use	Capacity, mature client support	DFS channels can be vacated when radar is detected
6 GHz	Up to 59 where the full band is allowed; 24 where only the lower 500 MHz is	Clean spectrum, wide channels, modern clients	Requires WPA3 or OWE; shorter range through walls

Common design targets by use case

USE CASE	MINIMUM SIGNAL (RSSI)	MINIMUM SNR	OTHER TARGETS
General data	−70 dBm	20 dB	Coverage everywhere users sit
Voice and real-time video	−67 dBm	25 dB	Every spot covered by a second AP for roaming; low retry rates
Location services	−75 dBm from at least 3 APs	—	APs around the perimeter as well as the interior

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USE CASE	MINIMUM SIGNAL (RSSI)	MINIMUM SNR	OTHER TARGETS
High density (halls, lecture rooms)	−67 dBm	25 dB	Capacity-driven AP count, narrower channels, minimum data rates raised

Survey types and when to use them

Predictive

- Floor plans and wall materials modelled in software
- Fast, cheap, good for budgets and first AP placement
- Only as accurate as the wall data you enter

On-site (AP-on-a-stick, passive or active)

- Measures real attenuation with a temporary AP
- Essential for unusual buildings: hospitals, warehouses, heritage sites
- Active mode also measures throughput and retries

Validation

- Walks the finished network
- Proves targets for signal, SNR, overlap and roaming
- Becomes the baseline for later troubleshooting

A disciplined survey process

1. **Gather requirements** Use cases, client types, densities per area, applications and their SLAs, security and compliance needs.
2. **Model predictively** Calibrate floor plans, assign wall materials and place APs for the strictest use case in each zone.
3. **Measure on site** Validate attenuation of unusual materials and check the existing noise floor and interferers with a spectrum analyser.
4. **Adjust the design** Revise AP count, mounting, antennas and channel plan; document every assumption.
5. **Validate after install** Run a post-installation survey against the agreed targets and fix gaps before handover.

Channel and power planning rules

- Use 20 MHz channels in 2.4 GHz only, and consider turning surplus 2.4 GHz radios off or into a second 5 GHz/monitor role.
- In dense 5 GHz deployments, 20 or 40 MHz usually beats 80 MHz because more channels mean less co-channel contention.
- 6 GHz has room for 80 MHz or wider channels in most enterprises, but still plan reuse.
- Set AP transmit power close to typical client power to avoid one-way links where the client hears the AP but the AP cannot hear the client.
- Raise minimum basic data rates to shrink effective cell size and save airtime spent on slow management frames.
- Let automatic radio resource management do day-to-day tuning, but give it sensible minimum and maximum power and allowed channel lists.

WATCH OUT · MORE POWER IS NOT MORE COVERAGE

Turning every AP to maximum makes the client think it has a strong link while its own, weaker transmissions fail to reach the AP. The result is retries, sticky clients and co-channel interference. Coverage problems are usually fixed by AP placement, not power.

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Reading a client's view of the link

Linux laptop · terminal

```
$ iw dev wlan0 link
Connected to 3c:fd:fe:a1:22:41 (on wlan0)
        SSID: CORP-SECURE
        freq: 5500
        signal: -63 dBm
        rx bitrate: 866.7 MBit/s VHT-MCS 9 80MHz short GI VHT-NSS 2
        tx bitrate: 650.0 MBit/s VHT-MCS 7 80MHz short GI VHT-NSS 2
```

5,500 MHz is channel 100, a DFS channel. −63 dBm meets a voice-grade target; the lower transmit MCS hints that the AP hears the client less well than the client hears the AP.

TEMPLATE WLAN design sheet

Capture per-area requirements and design decisions so that survey, installation and validation teams work to the same targets.

AREA	USE CASE AND DENSITY	BANDS AND CHANNEL WIDTH	RSSI / SNR TARGET	SSIDS AND SECURITY	AP MODEL, MOUNT AND COUNT
Ward 3 East	*Voice badges, tablets, telemetry; 60 clients*	*5 GHz 20 MHz, 6 GHz 40 MHz*	*−67 dBm / 25 dB, 2 APs heard*	*CLIN-SECURE (WPA3-Ent), IOT-MED*	*Internal antennas, ceiling, 9 APs*
Lecture hall B	*300 students, laptops and phones*	*5 GHz 20 MHz, 6 GHz 80 MHz*	*−67 dBm / 25 dB*	*CAMPUS (WPA3-Ent), GUEST (OWE)*	*Directional under-seat, 12 APs*

SCENARIO · DECIDE · WAREHOUSE SCANNERS DROP OUT IN THE HIGH RACKS

A warehouse with metal racking was designed predictively. Handheld scanners disconnect in the aisles between the tallest racks, although the coverage map showed −65 dBm everywhere. What is the best next step?

- (A) Survey on site with an AP-on-a-stick in the aisles, then redesign placement and antennas.
- (B) Raise every AP in the warehouse to maximum transmit power so the signal reaches the aisles.
- (C) Move the scanners to 2.4 GHz with 40 MHz channels to gain range between the metal racks.
- (D) Enable 802.11r on the scanner SSID so the scanners roam faster between the aisle APs.

Circle your choice. Answer and feedback: module answer key.

Key terms

EIRP
Effective isotropic radiated power: transmit power plus antenna gain minus losses; the value regulators limit.

SNR
Signal-to-noise ratio: the difference in dB between received signal strength and the noise floor.

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DFS

Dynamic Frequency Selection: rules that require APs on certain 5 GHz channels to detect radar and move away from it.

KNOWLEDGE CHECK

1. A radio transmits at 17 dBm through a cable with 2 dB loss into a 5 dBi antenna. What is the EIRP?

   (A) 24 dBm
   (B) 22 dBm
   (C) 15 dBm
   (D) 20 dBm

2. A client sees the AP at −68 dBm with a noise floor of −95 dBm. What is the SNR?

   (A) 163 dB
   (B) −27 dB
   (C) 27 dB
   (D) 68 dB

3. Why are narrower 5 GHz channels often preferred in a dense office?

   (A) More channels allow better reuse, so nearby APs contend less
   (B) Narrow channels give each individual client a higher peak data rate
   (C) Narrow channels remove the need to use DFS channels in the 5 GHz band
   (D) Narrow channels allow WPA2 and WPA3 clients to share the same SSID

4. Which survey type proves that the installed network meets its targets?

   (A) A predictive survey from floor plans
   (B) A spectrum scan performed before the APs are ordered
   (C) A post-installation validation survey
   (D) An AP-on-a-stick survey performed during the design phase

5. What is the main risk of setting AP transmit power far above typical client power?

   (A) Clients hear the AP, but the AP cannot reliably hear the clients
   (B) Clients refuse to associate with APs that transmit above 14 dBm
   (C) DFS radar detection is disabled when power exceeds the client level
   (D) Channel width is automatically reduced to 20 MHz on all radios

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · WIRELESS · LESSON 2 · 40 MIN

WLAN Architectures and Seamless Roaming

Controller, distributed and cloud models; 802.11k, v and r in practice

- Compare controller-based, distributed and cloud-managed WLAN architectures.
- Explain how 802.11k, 802.11v and 802.11r cooperate to speed up roaming.
- Design for Layer 3 roaming and controller high availability.

Every enterprise WLAN separates three jobs: managing configuration, controlling RF and client state, and forwarding client data. Architectures differ mainly in **where** each job runs. Your choice affects roaming behaviour, failure domains, WAN dependence and how you segment traffic.

WLAN functional planes

Management	`Configuration templates` `Firmware and inventory` `Dashboards and APIs`
Control	`Radio resource management` `Client authentication state and key caching` `Roaming coordination`
Data	`Tunnelled to a controller, or switched locally at the AP`
Access	`APs, radios and antennas` `Clients and their roaming decisions`

Architecture options

MODEL	WHERE DATA IS SWITCHED	STRENGTHS	TRADE-OFFS
Controller with central tunnelling	Controller (APs tunnel client traffic, e.g. CAPWAP, RFC 5415)	Central policy, easy Layer 3 roaming, one place to segment	Controller capacity and placement matter; hairpinning for remote sites
Controller with local switching	At the AP onto local VLANs	Branch survivability, no hairpin over the WAN	Roaming across VLANs needs careful design
Cloud-managed	At the AP; only management in the cloud	Fast rollout, simple multi-site operations	Features depend on the platform; plan for cloud reachability loss
Fabric-integrated	At the access switch into an overlay	Same segmentation for wired and wireless	Requires a matching campus fabric

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Access ports for the two data-plane models

Switch SW-ACC3 · IOS-style CLI

```
interface GigabitEthernet1/0/24
 description AP-3F-07 (local switching)
 switchport mode trunk
 switchport trunk native vlan 30
 switchport trunk allowed vlan 30,110,120,130
 spanning-tree portfast trunk
!
interface GigabitEthernet1/0/25
 description AP-3F-08 (central tunnelling)
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
```

With local switching the AP needs a trunk carrying every client VLAN it maps SSIDs to (VLAN 30 untagged for AP management). With central tunnelling the AP only needs its management VLAN; client VLANs live at the controller.

INSIGHT · CLIENTS DECIDE WHEN TO ROAM

The infrastructure can advise, but the client's driver chooses when and where to roam. Your job is to give clients good information (802.11k), clear nudges (802.11v) and a fast way in (802.11r), then validate with the actual devices your users carry.

The roaming amendments

AMENDMENT	WHAT IT ADDS	BENEFIT
802.11k	Neighbour reports and radio measurements	Client scans a short list of channels instead of all of them
802.11v	BSS transition management requests	Network can suggest a better AP or warn of disassociation
802.11r	Fast BSS Transition (FT) key hierarchy	Re-keys in the authentication/reassociation frames, skipping full 802.1X

A fast roam for a voice handset

01 Signal drops — Handset's roaming threshold is crossed as the user walks. →

02 Neighbour list — 802.11k report tells it which channels to scan. →

03 Nudge — 802.11v request may suggest a target AP with better load. →

04 Fast transition — 802.11r derives new keys during reassociation, no full EAP exchange. →

05 Resume — Voice resumes typically within tens of milliseconds.

Roam time budget

MEASURE	FORMULA	WHAT IT TELLS YOU
Roam time	`scan + authentication + reassociation + key exchange`	Each term is a place to save time; FT removes the 802.1X exchange from the critical path.
Voice target	`roam < 50 ms (commonly used goal)`	Above roughly 150 ms the user typically hears a gap.

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MEASURE	FORMULA	WHAT IT TELLS YOU
Full 802.1X re-auth	`EAP round trips × RADIUS latency + 4-way handshake`	Can reach hundreds of milliseconds across a WAN to the RADIUS server.

WATCH OUT · TEST 802.11R WITH YOUR REAL CLIENTS

Some older or specialised clients (scanners, medical devices, printers) misbehave on FT-enabled SSIDs. Use a mixed or adaptive FT mode where supported, or a separate SSID, and run a pilot with every critical device type.

Layer 3 roaming and high availability

- Keep a client's VLAN and IP when it roams: span the client VLAN across the roaming domain, use central tunnelling, or anchor the session at a home controller.
- Make controllers redundant (active/standby with state sync or N+1) and test failover during the working day.
- Place controllers near the data they serve; avoid hairpinning branch traffic across the WAN.
- For cloud-managed estates, confirm what keeps working if the cloud is unreachable: SSIDs, authentication, new clients.

CASE STUDY · A HOSPITAL WIRELESS REDESIGN

A 600-bed hospital has a ten-year-old WLAN with one flat SSID for staff, guests and devices. Nurses' voice badges drop calls between wards, infusion pumps lose connectivity during upgrades, and the infection-control team now forbids ceiling access in some wards without a permit. The board has approved a redesign over 18 months.

1. How would you segment users and devices on the new WLAN?
2. What roaming design would you choose for voice badges?
3. How would you work within the ceiling-access restrictions?
4. What would you validate before declaring each ward live?

Model answer: module answer key.

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SCENARIO · DECIDE · CALLS DROP BETWEEN FLOORS

Voice handsets drop calls when users walk from the third floor stairwell into the fourth floor corridor. Each floor has its own client VLAN and IP subnet, switched locally at the APs. Survey data shows good overlap in the stairwell. What should you change first?

- (A) Increase AP transmit power on both floors to strengthen the signal in the stairwell area.
- (B) Disable 802.11k so that the handsets scan every channel before choosing the next AP.
- (C) Keep the handset's subnet when it roams: one voice VLAN, central tunnelling or anchoring.
- (D) Move the voice SSID to 2.4 GHz so the handsets get better range through the stairwell.

Circle your choice. Answer and feedback: module answer key.

Key terms

BSS transition management
The 802.11v mechanism that lets the network suggest a better AP to a client or warn it before disassociation.

CAPWAP
An IETF protocol (RFC 5415) for managing APs from a controller and optionally tunnelling client data to it.

Fast BSS Transition (FT)
802.11r roaming that derives new keys during reassociation so the client avoids a full 802.1X exchange.

KNOWLEDGE CHECK

6. Which amendment provides clients with a list of neighbouring APs and their channels?
   - (A) 802.11r
   - (B) 802.11v
   - (C) 802.11mc
   - (D) 802.11k

7. What does 802.11r remove from the critical path of a roam on an 802.1X SSID?
   - (A) The probe requests used to find candidate APs
   - (B) The full EAP exchange with the RADIUS server
   - (C) The reassociation frames exchanged with the new AP
   - (D) The WMM admission control for the voice queue

8. Who makes the final decision to roam in an 802.11 WLAN?
   - (A) The controller
   - (B) The client device
   - (C) The RADIUS server
   - (D) The AP the client is leaving

9. Branch APs switch traffic locally onto per-floor VLANs, and voice calls drop between floors. What is the root design issue?
   - (A) Clients change IP subnet during the roam, breaking active sessions
   - (B) 802.11r keys cannot be cached on locally switched access points
   - (C) Locally switched SSIDs cannot apply WMM priority to the voice traffic
   - (D) Branch APs cannot send 802.11k neighbour reports to their clients

10. What should you do before enabling 802.11r on an SSID shared with legacy scanners?
    - (A) Lower AP transmit power so that the scanners roam more often
    - (B) Enable 40 MHz channels in 2.4 GHz to improve scanner throughput
    - (C) Remove the WPA3 configuration and fall back to an open SSID
    - (D) Pilot every critical client type, or use an adaptive FT mode

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 6 · WIRELESS · LESSON 3 · 40 MIN

Wireless Security, Guest Access and QoS

WPA3-Enterprise, certificate-based EAP, safe guest networks and voice-grade WMM

- Compare WPA2- and WPA3-Enterprise and choose an EAP method.
- Configure a supplicant for EAP-TLS with proper server validation.
- Design a guest WLAN and map voice and video QoS end to end.

Wireless security is identity plus encryption plus management-frame protection. **WPA3** raises the floor: it mandates **Protected Management Frames (PMF)**, removes legacy ciphers on WPA3-only SSIDs, and brings stronger options such as a 192-bit mode. The method used to prove identity, the EAP method, matters as much as the WPA version.

Enterprise security modes

WPA2-Enterprise

- 802.1X/EAP with AES-CCMP
- PMF optional
- Still widely supported by legacy clients
- Not permitted in 6 GHz

WPA3-Enterprise

- 802.1X/EAP with PMF required
- SHA-256-based key management
- Transition mode exists but keeps some WPA2 weaknesses
- Required for 6 GHz enterprise SSIDs

WPA3-Enterprise 192-bit

- GCMP-256 and SHA-384 suites
- EAP-TLS with strong certificates
- For high-assurance environments
- Needs client and RADIUS support end to end

802.1X on a WLAN, end to end

01 **Associate** — Client joins the SSID; the AP blocks all but EAP. →

02 **EAP exchange** — AP or controller relays EAP to RADIUS; a TLS tunnel forms. →

03 **Mutual proof** — Client validates the server certificate; server validates the client certificate or credentials. →

04 **Access-Accept** — RADIUS returns keying material plus policy: VLAN, role or ACL. →

05 **4-way handshake** — Session keys are derived and the client's traffic is encrypted.

Choosing an EAP method

METHOD	CLIENT PROVES IDENTITY WITH	STRENGTH	TYPICAL USE
EAP-TLS	Client certificate	Strongest; no passwords to phish	Managed laptops, phones and IoT with device PKI
TEAP	Certificate and/or password inside a tunnel; can chain machine and user	Strong and flexible	Modern managed fleets needing machine plus user auth
PEAP-MSCHAPv2	Username and password inside TLS	Weak if server validation is skipped	Legacy; migrate away
EAP-TTLS	Inner password or other methods	Depends on inner method	Mixed or non-domain clients

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EAP-TLS for a WPA3-Enterprise SSID

Linux laptop · wpa_supplicant.conf

```
network={
    ssid="CORP-SECURE"
    key_mgmt=WPA-EAP-SHA256
    ieee80211w=2
    eap=TLS
    identity="laptop042@corp.example"
    ca_cert="/etc/wifi/corp-root-ca.pem"
    client_cert="/etc/wifi/laptop042.pem"
    private_key="/etc/wifi/laptop042.key"
    domain_suffix_match="radius.corp.example"
}
```

`ieee80211w=2` *makes PMF mandatory;* `ca_cert` *plus* `domain_suffix_match` *ensure the laptop only trusts your RADIUS server.*

WATCH OUT · SKIPPING SERVER VALIDATION DEFEATS THE DESIGN

If clients do not validate the RADIUS server's certificate and name, an attacker's look-alike SSID can harvest credentials. Push supplicant profiles through device management; never rely on users clicking 'trust'.

Designing guest access

1. **Isolate** Put guests in their own VRF or VLAN with internet-only access, or tunnel them to a DMZ anchor.
2. **Encrypt** Use OWE (Enhanced Open) so even open guest traffic is encrypted over the air.
3. **Onboard** Captive portal for terms, sponsor approval or self-registration where policy requires it.
4. **Contain** Client isolation, per-client rate limits and session timeouts.
5. **Log** Retain association and portal logs as legal requirements demand, and no longer.

Mapping voice and video QoS onto Wi-Fi (RFC 8325 guidance)

TRAFFIC	DSCP	802.11 UP	WMM ACCESS CATEGORY
Voice	EF (46)	6	AC_VO
Interactive video	AF41 (34)	4	AC_VI
Low-latency data	AF21 (18)	3	AC_BE
Default	DF (0)	0	AC_BE
Scavenger / bulk	CS1 (8)	1	AC_BK

Voice and video over WLAN practices

- Trust DSCP from managed clients and map it to UP on the AP; do not rely on UP set by clients.
- Re-mark or police traffic from untrusted SSIDs such as guest.
- Use admission control on the voice access category where clients support it.
- Raise minimum data rates and keep cells small enough that voice frames go out at high rates.
- Measure with real calls: jitter, loss, retries and MOS while walking the roaming paths.

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SCENARIO · DECIDE · GUEST NETWORK DESIGN FOR A CONFERENCE CENTRE

A conference centre needs guest Wi-Fi for 2,000 visitors. Legal requires acceptance of terms; security requires that guests can never reach internal systems; visitors complain that open networks feel unsafe. Which design is best?

(A) A WPA2-Personal SSID with the password printed on badges, on the corporate VLAN with an ACL.

(B) A WPA3-Enterprise SSID with an individual RADIUS account created for every single visitor each day.

(C) An open SSID with no portal at all, relying on HTTPS on each website for visitors' privacy.

(D) An OWE SSID in a guest VRF with internet-only access, a terms portal, isolation and rate limits.

Circle your choice. Answer and feedback: module answer key.

WLAN security and QoS go-live checklist

☐ Enterprise SSIDs use WPA3-Enterprise (or a time-boxed transition plan) with PMF required.

☐ EAP-TLS or TEAP is the default method; supplicant profiles enforce server name validation.

☐ RADIUS returns roles or VLANs per identity; unknown devices land in a restricted role.

☐ Guest SSID uses OWE, a separate VRF, client isolation and rate limits.

☐ Voice SSID maps EF to UP 6 and AF41 to UP 4; untrusted SSIDs are re-marked.

☐ 6 GHz radios carry only WPA3 or OWE SSIDs.

☐ A rogue and wireless intrusion detection policy is enabled and alerts reach the NOC.

☐ A validation survey and roaming test walk with real devices are signed off.

Key terms

WPA3
The current Wi-Fi security generation; it mandates PMF and offers SAE for personal and stronger suites for enterprise networks.

Protected Management Frames (PMF)
802.11w protection that prevents forged deauthentication and disassociation frames.

OWE
Opportunistic Wireless Encryption (Enhanced Open): encrypts traffic on an open SSID without a shared password.

WMM
Wi-Fi Multimedia: the four access categories (voice, video, best effort, background) that prioritise traffic over the air.

KNOWLEDGE CHECK

11. Which security option is required for any enterprise SSID broadcast in the 6 GHz band?

(A) WPA2-Enterprise only
(B) WPA2-Personal with a long passphrase
(C) WPA3-Enterprise with PMF
(D) An open SSID protected by a captive portal

12. Why is EAP-TLS considered stronger than PEAP-MSCHAPv2?

(A) It uses a faster cipher suite inside the outer TLS tunnel for each session
(B) It does not require a RADIUS server to authenticate the clients
(C) It works without the client validating any server certificate
(D) Client certificates replace passwords, so there is nothing to phish

13. Following RFC 8325 guidance, voice marked EF should map to which 802.11 user priority?

(A) UP 6 (AC_VO)
(B) UP 4 (AC_VI)
(C) UP 7 (AC_VO)
(D) UP 5 (AC_VI)

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14. What does OWE add to a guest SSID?

(A) Mandatory user authentication against a RADIUS server

(B) Automatic acceptance of the captive portal terms

(C) Over-the-air encryption without a shared password

(D) Isolation of guest traffic into a separate VRF

15. A supplicant profile omits server certificate validation. What is the main risk?

(A) Clients will roam more slowly between APs on the same floor

(B) A rogue SSID can pose as the network and harvest credentials

(C) The RADIUS server will reject all clients that use the profile

(D) PMF will be disabled for every client on the secure SSID

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MODULE 6 · WIRELESS · LESSON 4 · 35 MIN

Location, Analytics and Wi-Fi 7

Turning RF into insight, and planning for multi-link, wide-channel WLANs

- Compare Wi-Fi and complementary location techniques and their typical accuracy.
- Handle analytics data with privacy and MAC randomisation in mind.
- Explain the main Wi-Fi 7 features and plan a realistic adoption path.

A WLAN constantly measures its environment: who is connected, where, how well and for how long. Location services and analytics turn that data into wayfinding, asset tracking, space planning and faster troubleshooting. At the same time, Wi-Fi 7 changes design assumptions with multi-link operation and much wider channels. This lesson covers both at the level a professional needs to make decisions.

Location techniques

TECHNIQUE	HOW IT WORKS	INDICATIVE ACCURACY	NOTES
RSSI trilateration	Signal strength heard by several APs	5–10 m	Works with any client; affected by bodies and walls
RF fingerprinting	Compares readings to a calibrated map	3–5 m	Needs calibration and re-calibration after changes
Angle of arrival	Antenna arrays measure signal direction	1–3 m	Needs specialised APs
Fine Timing Measurement	Round-trip time between client and APs	1–2 m	Needs client and AP support
BLE beacons and tags	Bluetooth Low Energy radios in APs or tags	1–5 m	Good for asset tags with long battery life
Ultra-wideband	Precise time-of-flight	Sub-metre	Separate infrastructure; high-value assets

From radio to insight

01 Sense — APs, BLE radios and sensors collect signal and timing data. →

02 Collect — Controller or cloud aggregates readings per device. →

03 Locate — Location engine computes positions on calibrated maps. →

04 Analyse — Presence, dwell time, occupancy and asset movement. →

05 Act — Dashboards, APIs, alerts and integration with other systems.

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Privacy and data quality

- Modern phones use randomised, often per-SSID, MAC addresses; counts of unassociated devices are estimates, not people.
- Prefer aggregated, anonymised analytics; store identifiable location data only with a clear purpose and consent.
- Set retention limits and access controls for location history, especially for staff and patients.
- Document what is collected on public signage or privacy notices where the law requires it.

Wi-Fi generations at a glance

GENERATION	STANDARD	HEADLINE FEATURES
Wi-Fi 6	802.11ax	OFDMA, uplink and downlink MU-MIMO, BSS colouring, target wake time, 1024-QAM
Wi-Fi 6E	802.11ax in 6 GHz	Wi-Fi 6 features plus clean 6 GHz spectrum and WPA3-only operation
Wi-Fi 7	802.11be	320 MHz channels in 6 GHz, 4096-QAM, multi-link operation, multi-RU and preamble puncturing

Wi-Fi 7 features that change design

MLO

Multi-link operation

A client and AP use links in two or more bands at once or switch between them quickly, improving latency and resilience.

320

320 MHz channels

Only in 6 GHz; very high peak rates but few channels, so suited to low-density, high-throughput spaces.

4K

4096-QAM

12 bits per symbol versus 10, about 20% more peak throughput, but only at very high SNR close to the AP.

PUNC

Preamble puncturing

Uses a wide channel while skipping a sub-channel that has interference, instead of falling back to a narrower width.

MRU

Multi-RU

Lets one client receive several resource units, improving spectral efficiency in OFDMA scheduling.

Why 4096-QAM is not a capacity plan

MEASURE	FORMULA	WHAT IT TELLS YOU
Bits per symbol	`log2(4096) = 12 vs log2(1024) = 10`	A 20% gain in peak rate per spatial stream.
Where it works	`Very high SNR, roughly 40 dB or more`	Only clients close to the AP benefit; the cell edge does not.

INSIGHT · WI-FI 7 AND SECURITY

Wi-Fi 7 features such as multi-link operation expect WPA3 with protected management frames. Treat legacy WPA2-only SSIDs as a blocker for Wi-Fi 7 benefits and plan their retirement early.

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A realistic Wi-Fi 7 adoption path

1. **Assess clients** Inventory client capabilities; benefits arrive only as Wi-Fi 7 clients arrive.
2. **Prepare the wire** Multi-gigabit access ports and adequate PoE budgets for new APs.
3. **Clean up security** Move SSIDs to WPA3 and consolidate SSIDs across bands for MLO.
4. **Pilot** Deploy in high-value areas and measure latency, roaming and throughput.
5. **Refresh by lifecycle** Replace APs as they age; avoid forklift upgrades with no client benefit.

SCENARIO · DECIDE · 320 MHZ EVERYWHERE?

A director has read about Wi-Fi 7 and wants 320 MHz channels on every AP in a dense open-plan office with 900 users. What do you recommend?

- (A) Agree, because wider channels always give every user more capacity, even in a dense open-plan office.
- (B) Use 320 MHz channels in the 5 GHz band instead, so that 6 GHz stays free for future expansion.
- (C) Use 80 MHz or narrower in open-plan areas, and keep 320 MHz for low-density, high-throughput rooms.
- (D) Disable 6 GHz entirely and keep the office on 5 GHz with 160 MHz channels for all of the APs.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Pick one building you know. Which wireless use case there would benefit most from location services or Wi-Fi 7, and what would you need to change in the network first?

Key terms

Multi-link operation (MLO)
A Wi-Fi 7 capability that lets a device use links in several bands simultaneously or switch rapidly between them.

Fine Timing Measurement (FTM)
An 802.11 round-trip-time measurement that enables metre-level Wi-Fi positioning when clients and APs support it.

Preamble puncturing
Transmitting on a wide channel while excluding an interfered sub-channel rather than shrinking the whole channel.

KNOWLEDGE CHECK

16. Which location technique typically needs specialised antenna-array APs?
    - (A) RSSI trilateration
    - (B) Angle of arrival
    - (C) Fingerprinting
    - (D) Fine Timing Measurement
17. Why are counts of nearby phones in Wi-Fi analytics only estimates?
    - (A) Phones are only detected when they are associated to an enterprise SSID
    - (B) Analytics engines discard all probe requests sent on 6 GHz channels
    - (C) Randomised MAC addresses can change over time and differ per SSID
    - (D) Every phone uses the same MAC address on every network that it sees

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18. In which band are Wi-Fi 7's 320 MHz channels available?

(A) 6 GHz only
(B) 5 GHz only
(C) Both 5 and 6 GHz
(D) All three bands

19. What does 4096-QAM improve compared with 1024-QAM?

(A) About double the range at the cell edge for every client
(B) Better penetration through walls and floors for all bands
(C) Faster roaming between APs on the same controller
(D) About 20% higher peak rate, only at very high SNR

20. What is the main benefit of multi-link operation for a latency-sensitive client?

(A) It doubles the number of SSIDs the client can see on each AP
(B) It removes the need for the client to support WPA3 at all
(C) It lets 2.4 GHz use 320 MHz channels when the band is idle
(D) Frames can use whichever band's link is free, cutting delay

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MODULE 6 · WIRELESS

Answer key

Knowledge checks

1. D 17 − 2 + 5 = 20 dBm. Gains and losses in dB add and subtract.

2. C SNR = −68 − (−95) = 27 dB, which meets a 25 dB voice target.

3. A Capacity in dense spaces is limited by contention; more channels mean better reuse.

4. C Validation surveys measure the finished network and become the operational baseline.

5. A Link asymmetry causes retries and sticky clients; match AP power to the client population.

6. D 802.11k neighbour reports shorten scanning. 802.11r speeds up re-keying, 802.11v lets the network suggest a target AP, and 802.11mc adds fine timing measurement for location.

7. B FT derives keys from a cached hierarchy, avoiding the slow EAP round trips.

8. B The network can advise with 802.11k/v, but the client's driver decides.

9. A A Layer 3 change mid-call kills the session; keep the client's subnet stable.

10. D Some legacy clients fail on FT SSIDs; validate first and use compatibility modes.

11. C 6 GHz permits only WPA3 or OWE; WPA2 and open SSIDs are not allowed there.

12. D Certificate-based identity removes the password attack surface entirely.

13. A EF maps to UP 6 in the voice access category.

14. C OWE encrypts each client's traffic; isolation and portals are separate controls.

15. B Without server validation, clients will talk to any RADIUS server, including an attacker's.

16. B Angle of arrival measures signal direction using antenna arrays built into specific APs.

17. C MAC randomisation means one device can appear as several, and identities are not stable.

18. A Only 6 GHz has enough contiguous spectrum for 320 MHz channels.

19. D 12 versus 10 bits per symbol; the gain applies only close to the AP.

20. D MLO spreads or switches traffic across links in different bands.

Scenarios & case studies

Scenario · Warehouse scanners drop out in the high racks
A is best. Correct. Metal racking and stock change attenuation in ways predictive models rarely capture. Measure, then redesign.

Case study · A hospital wireless redesign
Segment by SSID only where security or client capability truly differs: a WPA3-Enterprise clinical SSID with role-based policy from RADIUS, a separate SSID for legacy medical devices that cannot do modern security (isolated VRF, strict ACLs), and an OWE guest SSID with captive-portal terms. Voice badges get 802.11k/v/r on the clinical SSID after vendor-certified testing, a −67 dBm / 25 dB design with two-AP coverage and QoS mapped from EF to the voice queue. Ceiling restrictions push the plan towards a predictive design calibrated with targeted on-site measurements, staged ward-by-ward installs scheduled with infection control, and APs mounted to avoid repeated access. Before each ward goes live: validation survey, roaming test walks with real badges, pump connectivity during controlled failover, and a sign-off by clinical engineering.

Scenario · Calls drop between floors
C is best. Correct. Coverage is fine; the handset changes subnet mid-call. Preserving its Layer 3 identity fixes the drop.

Scenario · Guest network design for a conference centre
D is best. Correct. OWE encrypts over the air without a shared password, the VRF enforces isolation, and the portal meets the legal requirement.

Scenario · 320 MHz everywhere?
C is best. Correct. In dense areas, channel reuse matters more than peak width; there are very few 320 MHz channels even in the full 6 GHz band.

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07

MODULE 7 · ASSURE

Assure — Assurance, Security & Troubleshooting

See everything, prove service levels, find root cause fast and protect the network itself.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Collect SNMPv3, syslog and IPFIX data securely and interpret it correctly.
- Design streaming telemetry subscriptions and synthetic probes that reveal user-impacting events.
- Apply a structured troubleshooting method across VRFs, overlays and wireless.
- Deploy 802.1X and MAB with dynamic segmentation through a phased rollout.
- Protect devices with control-plane policing, hardening baselines and infrastructure ACLs.

LESSONS

1. Monitoring Foundations: SNMPv3, Syslog and Flow Data · 35 min
2. Streaming Telemetry and Synthetic Testing · 40 min
3. Structured Troubleshooting · 35 min
4. Network Access Control: 802.1X, MAB and Dynamic Segmentation · 40 min
5. Infrastructure Protection: CoPP, Hardening and iACLs · 35 min

MODULE 7 · ASSURE · LESSON 1 · 35 MIN

Monitoring Foundations: SNMPv3, Syslog and Flow Data

The three classic signals, configured securely and read correctly

- Configure SNMPv3 with authentication, privacy and source restrictions.
- Build a reliable syslog pipeline with accurate time.
- Export flow records with IPFIX and estimate traffic from sampled data.

The Assure stage of **The ZCNP Fabric** asks one question continuously: is the network doing what we designed it to do? You cannot answer it without trustworthy data. This lesson covers the three signals every network already produces (counters and state through **SNMP**, events through **syslog**, and conversations through **flow records**) and how to collect them without opening new security holes.

SNMP versions and security levels

SNMPv2c

- Community string sent in clear text
- No real authentication or encryption
- Still common on legacy kit; restrict tightly or retire

SNMPv3 security levels

- noAuthNoPriv: username only, avoid
- authNoPriv: messages authenticated, not encrypted
- authPriv: authenticated and encrypted, the target state

SNMPv3 with authPriv, a read-only view and a source ACL

R1 · IOS-style CLI

```
ip access-list standard NMS-HOSTS
 permit 10.10.50.0 0.0.0.255
!
snmp-server view NMS-VIEW iso included
snmp-server group NMS-GRP v3 priv read NMS-VIEW access NMS-HOSTS
snmp-server user nms-poller NMS-GRP v3 auth sha <auth-secret> priv aes 256 <priv-secret>
snmp-server host 10.10.50.20 version 3 priv nms-poller
snmp-server enable traps snmp linkdown linkup
snmp-server enable traps bgp
```

Use the strongest authentication and privacy algorithms your platforms support. SNMPv3 users are not shown in the running configuration, so record them in your secrets store.

Syslog severity levels

LEVEL	KEYWORD	TYPICAL EXAMPLE
0	Emergency	System unusable
1	Alert	Immediate action needed, e.g. temperature critical
2	Critical	Hardware component failure
3	Error	Interface or process error
4	Warning	Threshold approaching, configuration conflict
5	Notice	Interface up/down, neighbour changes

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LEVEL	KEYWORD	TYPICAL EXAMPLE
6	Informational	Configuration change, ACL hit log
7	Debug	Troubleshooting output; never send continuously to a server

Syslog to a collector with trustworthy timestamps

R1 · IOS-style CLI

```
ntp server 10.10.50.5 prefer
ntp server 10.10.50.6
service timestamps log datetime msec localtime show-timezone
logging source-interface Loopback0
logging host 10.10.50.30
logging trap informational
logging buffered 64000 informational
```

A consistent source interface makes events from one device easy to correlate. Where supported, prefer TCP or TLS transport so events are not silently lost.

PRO TIP · TIME IS THE BACKBONE OF ASSURANCE

Correlation across devices, flows, logs and tickets only works if every clock agrees. Use at least two internal NTP sources, authenticate them where possible, alarm on unsynchronised devices, and log in one agreed time zone or UTC.

Flow export formats

FORMAT	STRUCTURE	NOTES
NetFlow v5	Fixed fields, IPv4 only	Legacy; no IPv6, no extensibility
NetFlow v9	Template-based (RFC 3954)	Flexible fields, IPv6 support
IPFIX	Template-based IETF standard (RFC 7011), default port 4739	Variable-length and enterprise-specific fields; the modern default

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Flexible flow export using IPFIX

R1 · IOS-style CLI

```
flow record FR-APP
 match ipv4 source address
 match ipv4 destination address
 match ipv4 protocol
 match transport source-port
 match transport destination-port
 match interface input
 collect counter bytes long
 collect counter packets long
 collect timestamp absolute first
 collect timestamp absolute last
flow exporter FE-COLLECTOR
 destination 10.10.50.40
 source Loopback0
 transport udp 4739
 export-protocol ipfix
flow monitor FM-APP
 record FR-APP
 exporter FE-COLLECTOR
 cache timeout active 60
```

Apply with `ip flow monitor FM-APP input` *on the interfaces you want to observe. Match fields define a flow key; collect fields are the measurements stored per flow.*

Working with flow data

MEASURE	FORMULA	WHAT IT TELLS YOU
Sampled estimate	`estimated bytes ≈ sampled bytes × N (for 1-in-N sampling)`	Accurate for large aggregates, unreliable for small flows.
Active timeout effect	`long flows are reported every active-timeout seconds`	A 60 s timeout gives near-real-time visibility of long transfers.
Average rate	`bits/s = (bytes × 8) ÷ (last − first timestamp)`	Useful for finding the real top talkers on a congested link.

SCENARIO · DECIDE · THE MYSTERY CONGESTION

A branch WAN link runs at 95% every morning from 09:00 to 10:30. SNMP interface graphs confirm the load but nobody knows the cause. Which data source answers the question fastest?

- (A) Syslog messages from the branch router during the morning, filtered for interface-related events.
- (B) IPFIX flow records from the branch router, grouped by source, destination and port for that window.
- (C) A faster SNMP polling interval on the WAN interface so that the morning peak is graphed in more detail.
- (D) A packet capture on the WAN link for the whole morning, analysed afterwards packet by packet.

Circle your choice. Answer and feedback: module answer key.

Key terms

SNMPv3 authPriv
The SNMPv3 security level that both authenticates and encrypts messages using per-user keys.

IPFIX
IP Flow Information Export (RFC 7011): the IETF standard, template-based protocol for exporting flow records.

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Flow key
The set of match fields (for example the 5-tuple and input interface) that defines which packets belong to one flow.

KNOWLEDGE CHECK

1. Which SNMPv3 security level should be the target for production polling?
   - (A) authPriv
   - (B) noAuthNoPriv
   - (C) authNoPriv
   - (D) community-based v2c
2. An engineer sends debug-level syslog continuously to the central server from 400 devices. What is the main problem?
   - (A) Severity 7 messages are never accepted by any syslog collector
   - (B) Severity 7 floods the collector and buries important events
   - (C) Debug messages are sent without timestamps to remote servers
   - (D) Debug level disables the local logging buffer on each device
3. What does a template-based flow format such as IPFIX allow that NetFlow v5 does not?
   - (A) Encryption of every flow record sent between the exporter and the collector
   - (B) Real-time packet payload inspection for every exported flow record
   - (C) Flexible, extensible fields, including IPv6 and enterprise elements
   - (D) Guaranteed delivery of flow records using its own retransmission
4. With 1-in-1,000 sampling, a collector sees 2 MB for a flow aggregate. Roughly how much traffic was there?
   - (A) About 2 MB
   - (B) About 200 MB
   - (C) About 2 GB
   - (D) About 20 GB
5. Why configure a consistent logging source interface such as a loopback?
   - (A) A stable source address for every event, which eases correlation
   - (B) It encrypts syslog messages sent to the collector by default
   - (C) It raises the severity of all interface events to the critical level
   - (D) It allows debug messages to bypass the logging rate limiter

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MODULE 7 · ASSURE · LESSON 2 · 40 MIN

Streaming Telemetry and Synthetic Testing

Push-based, model-driven data and probes that measure what users feel

- Contrast polled and streamed telemetry and choose cadences.
- Subscribe to model-driven telemetry with an open-source collector.
- Use synthetic probes and tracking to measure and react to path quality.

SNMP polling asks each device for counters every few minutes. At modern scale that is both too slow and too expensive: problems shorter than the poll interval are invisible, and devices waste CPU answering. **Streaming telemetry** reverses the direction. Devices push structured, model-based data at a chosen cadence or on change, and collectors store it in time-series databases built for this load.

Poll or push?

Polling (SNMP)

- Collector asks, device answers
- Typical interval 1–5 minutes
- Microbursts and short flaps are missed
- MIBs vary between platforms

Streaming telemetry

- Device pushes on a schedule or on change
- Intervals of seconds, or immediate for state changes
- Efficient encodings over gRPC
- Uses YANG models, including vendor-neutral ones

A telemetry stack

Use	`Dashboards` `Alerting and SLO reports` `Capacity planning`
Store	`Time-series database` `Retention and down-sampling policies`
Collect	`Collectors that terminate subscriptions` `Normalisation and tagging`
Transport	`gNMI or other gRPC-based dial-in/dial-out` `Protobuf or JSON encoding over TLS`
Model	`YANG paths on the device` `OpenConfig and native models`

Dial-in subscriptions to OpenConfig paths

Collector host · gnmic

```
# credentials supplied via environment variables, TLS verified against the internal CA
$ gnmic -a leaf1.dc.example:57400 --tls-ca /etc/pki/ca.pem -u telemetry \
    subscribe --path "/interfaces/interface/state/counters" \
    --stream-mode sample --sample-interval 10s
# on-change subscription for routing state
$ gnmic -a leaf1.dc.example:57400 --tls-ca /etc/pki/ca.pem -u telemetry \
    subscribe --path "/network-instances/network-
instance/protocols/protocol/bgp/neighbors/neighbor/state/session-state" \
    --stream-mode on-change
```

Counters suit periodic sampling; state such as BGP session status suits on-change, so you learn of a drop within seconds without streaming unchanged values.

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Choosing cadence

DATA	MODE	TYPICAL CADENCE	WHY
Interface counters	Sample	10–30 s	Catches bursts that 5-minute polling averages away
Optical levels, CPU, memory	Sample	30–60 s	Slow-moving, trend-oriented
BGP/IGP neighbour state	On change	Immediate	Every change matters; no point repeating unchanged state
Interface oper status	On change	Immediate	Drives fast alerting
Queue drops on critical links	Sample	5–10 s	Short congestion events are the target

Telemetry tells you what devices see. **Synthetic probes** tell you what users experience. A probe sends test traffic at a fixed rate between two points and measures reachability, latency, jitter and loss continuously, even when no real user is active. The same idea powers SD-WAN path measurement, SLA reporting to the business and automatic failover.

Reachability tracking and a voice-quality probe

BR7 · IOS-style CLI

```
ip sla 10
 icmp-echo 198.51.100.1 source-interface GigabitEthernet0/0
 frequency 10
ip sla schedule 10 life forever start-time now
track 10 ip sla 10 reachability
ip route 198.51.100.1 255.255.255.255 203.0.113.1
ip route 0.0.0.0 0.0.0.0 203.0.113.1 track 10
ip route 0.0.0.0 0.0.0.0 192.0.2.1 250
!
ip sla 20
 udp-jitter 10.50.255.1 16384 codec g711alaw
 tos 184
 frequency 60
ip sla schedule 20 life forever start-time now
```

The tracked default route is withdrawn when the probe fails, and the floating static route (AD 250) takes over. The jitter probe needs a responder on the far end; `tos 184` *marks probes as EF so they share the voice queue. The /32 host route pins the probe target to the primary ISP, so the probe cannot falsely succeed through the backup path.*

Service-level arithmetic

MEASURE	FORMULA	WHAT IT TELLS YOU
Availability	`uptime ÷ total time × 100`	99.9% allows about 8.8 hours of downtime per year; 99.99% about 53 minutes.
From MTBF and MTTR	`availability = MTBF ÷ (MTBF + MTTR)`	Shortening repair time improves availability as much as reducing failures.
Probe loss	`lost probes ÷ sent probes × 100`	Use enough probes per window for the percentage to be meaningful.

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TEMPLATE

Telemetry KPI sheet

Define every KPI once, with its source, cadence and alert rule, so dashboards and alerts are consistent across teams.

KPI	SOURCE AND PATH OR PROBE	METHOD AND CADENCE	ALERT THRESHOLD	OWNER
WAN link utilisation	*/interfaces/interface/state/counters on edge routers*	*Stream, sample 10 s*	*> 85% for 5 min*	*WAN team*
Branch-to-DC voice quality	*UDP jitter probe BR→DC, EF marked*	*Probe every 60 s*	*Jitter > 30 ms or loss > 1% for 3 intervals*	*UC team*

CASE STUDY · A TELECOM NOC MOVES FROM POLLING TO STREAMING

A mobile operator's NOC polls 6,000 routers by SNMP every five minutes. Customers report brief outages that the NOC never sees, the polling servers are overloaded, and every vendor platform needs different MIB handling. The operator wants near-real-time visibility within a year.

1. Which data would you move to streaming first, and at what cadence?

2. How would you handle platforms that cannot stream yet?

3. What changes in storage and alerting are needed?

4. How would you prove the new system detects more incidents?

Model answer: module answer key.

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SCENARIO · DECIDE · BRIEF OUTAGES NOBODY SEES

Customers report one-minute outages on a metro ring several times a week. Five-minute SNMP graphs look flat and no tickets are raised automatically. What is the best improvement?

(A) On-change telemetry for interface and neighbour state, plus ring counters streamed every 10–30 s.

(B) Reduce the SNMP polling interval to 30 seconds for all 6,000 devices to catch the short outages.

(C) Ask customers to log the exact time of each outage so that the NOC can correlate it afterwards.

(D) Increase syslog severity to debug on all ring routers so that every short event is fully recorded.

Circle your choice. Answer and feedback: module answer key.

Key terms

Streaming telemetry
Model-driven data that devices push to collectors periodically or on change, instead of being polled.

On-change subscription
A telemetry subscription that sends an update only when the subscribed value changes.

Synthetic probe
Test traffic generated at a fixed rate to measure reachability, latency, jitter and loss between two points.

KNOWLEDGE CHECK

6. Which subscription mode best suits BGP neighbour session state?

(A) Sample every 10 seconds
(B) Sample every 5 minutes
(C) Once
(D) On change

7. A default route tracks an ICMP probe to the ISP. What happens when the probe fails?

(A) The router reloads its WAN interface to try to recover the ISP link quickly
(B) The tracked route is withdrawn and the floating static route takes over
(C) The probe automatically switches to probing the backup ISP instead
(D) Both default routes stay installed and traffic is load-balanced

8. Why is a UDP jitter probe marked with ToS 184?

(A) So the probe bypasses every QoS policy along the path to the far end
(B) So it is queued like EF voice and measures what voice experiences
(C) So the far-end responder can authenticate the probe packets it receives
(D) So routers treat the probe as control-plane traffic and police it

9. A service has MTBF of 999 hours and MTTR of 1 hour. What is its availability?

(A) 99.9%
(B) 99%
(C) 99.99%
(D) 99.5%

10. What is the main limitation of five-minute SNMP polling for interface utilisation?

(A) SNMP cannot report interface byte counters on modern devices
(B) SNMPv3 encryption makes interface counters inaccurate over time
(C) Short bursts and brief outages are averaged away between polls
(D) Polling requires every device to support streaming telemetry first

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MODULE 7 · ASSURE · LESSON 3 · 35 MIN

Structured Troubleshooting

A repeatable method that finds root cause faster and leaves evidence behind

- Apply a structured troubleshooting process from problem definition to documentation.
- Select an approach (top-down, bottom-up, divide-and-conquer, follow-the-path) for a given symptom.
- Use per-VRF and overlay-aware commands to isolate a fault across the Fabric.

Senior engineers are not faster because they know more commands. They are faster because they follow a method: they define the problem precisely, gather facts before forming opinions, test one hypothesis at a time and leave a trail others can follow. At ZCNP level, the network spans VRFs, tunnels, fabrics and wireless, so a method matters even more.

The structured troubleshooting cycle

01 Define Who is affected, what exactly fails, since when, what changed? →

02 Gather Collect facts: telemetry, logs, configs, traces, baselines. →

03 Analyse Compare with the baseline and with working cases. →

04 Hypothesise Rank likely causes; pick the most likely and cheapest to test. →

05 Test and fix Change one thing, with a rollback ready. →

06 Verify Confirm the original symptom is gone and nothing else broke. →

07 Document Record cause, fix and prevention; update runbooks and monitoring.

Choosing an approach

APPROACH	HOW IT WORKS	BEST WHEN
Top-down	Start at the application and work down the layers	One application fails while others work
Bottom-up	Start at physical and work up	Many services fail on one link or device
Divide and conquer	Start in the middle (e.g. can you ping the gateway?) and go up or down	Symptoms are vague; quickest to halve the search space
Follow the path	Trace hop by hop, both directions	Multi-hop, overlay or asymmetric routing problems
Spot the difference	Compare a failing case with a working one	Similar sites or devices behave differently

Worked example: a branch cannot reach an application

1. **Define** Branch 7 users (10.70.7.0/24, VRF CORP) cannot reach the HR application at 10.50.8.20 since last night's change window. Other branches are fine.
2. **Gather** Change records show a DC routing-policy update. Branch telemetry shows the SD-WAN tunnel up.
3. **Follow the path** From the branch, a VRF-aware traceroute reaches the hub and then times out.
4. **Check the return path** On the DC core, the branch prefix is missing from VRF CORP.
5. **Fix and verify** The new import policy omitted Branch 7's route target. Correct it, confirm the route, test the application, then document.

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Following the path in both directions

Branch and DC · IOS-style CLI

```
BR7# traceroute vrf CORP 10.50.8.20 source Loopback10
  1 172.31.7.1 2 msec 1 msec 2 msec
  2 10.50.255.1 3 msec 3 msec 4 msec
  3  *  *  *
  4  *  *  *
!
DC-CORE1# show ip route vrf CORP 10.70.7.0
% Subnet not in table
DC-CORE1# show bgp vpnv4 unicast vrf CORP 10.70.7.0/24
% Network not in table
```

Forward traffic reaches the DC; replies have no route back. Always check the return path; most 'one-way' failures live there.

Sources of truth while troubleshooting

- Baselines: normal utilisation, latency, routing table sizes and neighbour counts.
- Change records: what changed in the last 24–72 hours, including automated changes.
- Telemetry and logs, aligned to the same clock.
- Configuration history from version control, to diff against the last known good state.
- Topology and design documents, including VRF and route-target plans.

WATCH OUT · ONE CHANGE AT A TIME, WITH A WAY BACK

Changing several things at once may make the symptom disappear without telling you why, and may hide a second fault. Record each change, test, and roll back if it did not help.

Symptom to first place to look

SYMPTOM	FIRST SUSPECT
Everything at one site down	Physical, power, transport, tunnel endpoint routing
One application unreachable from many sites	Application, DNS, firewall or DC routing policy
Works for small packets, fails for large	MTU in tunnels, VXLAN underlay or IPsec
Intermittent, at regular intervals	Recursive routing, timers, flapping links, scheduled jobs
Only wireless users affected	RF, roaming domain, RADIUS, DHCP scope for the WLAN
Only after authentication	RADIUS policy: wrong VLAN, ACL or group assigned

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Troubleshooting discipline checklist

☐ Problem statement written: who, what, where, since when.

☐ Recent changes checked, including automated pipeline runs.

☐ Scope confirmed with a working comparison case.

☐ Path tested in both directions, in the correct VRF.

☐ Each hypothesis tested with one change and a rollback ready.

☐ Fix verified against the original symptom and by the affected users.

☐ Root cause, fix and prevention recorded; monitoring updated to catch it next time.

SCENARIO · DECIDE · WHERE DO YOU START?

After a planned change, users at all 30 branches can browse the internet but cannot reach one internal ERP application in the data centre. Other internal applications work. What approach and first step make the most sense?

(A) Bottom-up: check the cabling and interface error counters at every one of the 30 branches first.

(B) Reload the WAN hub routers first to clear any stale state left behind by the planned change.

(C) Open a ticket with the internet provider, because a change in their network may be the cause.

(D) Top-down: check the ERP service, its DNS record and the DC firewall and routing changes first.

Circle your choice. Answer and feedback: module answer key.

REFLECT

Recall the last difficult incident you worked on. Which step of the structured cycle did you skip or rush, and what would you do differently with a written method?

Key terms

Baseline
Recorded normal behaviour (utilisation, latency, table sizes) used to judge whether current data is abnormal.

MTTR
Mean time to repair: the average time from detecting a failure to restoring service.

Follow-the-path
A troubleshooting approach that checks forwarding hop by hop in both directions along the traffic's route.

KNOWLEDGE CHECK

11. Many services fail for every user behind one access switch. Which approach is usually fastest?

(A) Top-down, starting with each of the affected applications in turn

(B) Spot the difference against an unrelated data-centre switch

(C) Follow the path from the internet edge to the user subnet

(D) Bottom-up, starting with physical and link checks on that switch

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12. A traceroute in the correct VRF reaches the DC core and then times out. What should you check next?

(A) Whether the source branch router has a working default route

(B) Whether the traceroute was run using the global routing table

(C) Whether the branch Wi-Fi uses WPA3-Enterprise with PMF enabled

(D) Whether the DC has a route back to the source in the same VRF

13. Why change only one variable at a time when testing a hypothesis?

(A) Because devices reject more than one configuration change per minute

(B) So you know which change fixed it and can roll back the rest

(C) Because change records only allow one line per incident ticket

(D) Because monitoring tools cannot display several changes together

14. Which data source best supports a 'what changed?' question during an incident?

(A) The current running configuration of the device that is being investigated

(B) Five-minute SNMP utilisation graphs for the affected interfaces

(C) Version-controlled config history plus change and pipeline records

(D) Customer descriptions of when they first noticed the problem

15. Small pings across an overlay succeed but file transfers stall. Which suspect comes first?

(A) An MTU or MSS problem in the tunnel or VXLAN underlay

(B) A routing loop between the two sites that drops large packets

(C) An SNMPv3 authentication mismatch on the edge routers

(D) A spanning-tree topology change on the access layer

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MODULE 7 · ASSURE · LESSON 4 · 40 MIN

Network Access Control: 802.1X, MAB and Dynamic Segmentation

Identity at the edge port, policy from RADIUS, and a safe rollout path

- Explain the 802.1X roles and the fallback to MAC Authentication Bypass.
- Configure a switch port for 802.1X with MAB and change of authorisation.
- Design dynamic segmentation with RADIUS-assigned VLANs, ACLs or group tags, and roll it out in phases.

Every network port is a door. **802.1X** makes the device prove who it is before the door opens, and **RADIUS** returns not just yes or no but **policy**: which segment, which ACL, which group. Devices that cannot run a supplicant, such as printers or cameras, use **MAC Authentication Bypass (MAB)**, which must be paired with profiling and tight policy because MAC addresses are easy to spoof.

802.1X roles

Policy and identity	`Directory and certificate authority` `Device profiling and posture data`
Authentication server	`RADIUS server evaluates identity and returns policy`
Authenticator	`Switch port or WLAN; relays EAP over RADIUS and enforces the result`
Supplicant	`Software on the endpoint that speaks EAP over LAN (EAPOL)`

What happens when a device plugs in

01 Link up — Port allows only EAPOL (and anything a pre-auth ACL permits).

→

02 Try 802.1X — Switch sends EAP identity requests.

→

03 Fallback — No supplicant answers in time: switch tries MAB with the MAC as identity.

→

04 Decide — RADIUS matches identity, profile and posture to a policy.

→

05 Enforce — Switch applies the returned VLAN, ACL or group tag.

→

06 Adapt — Change of authorisation (CoA) can re-assign policy later.

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Global AAA, RADIUS and CoA

SW-ACC1 · IOS-style CLI

```
aaa new-model
radius server NAC-1
 address ipv4 10.10.60.10 auth-port 1812 acct-port 1813
 key <from-secrets-vault>
aaa group server radius NAC-GRP
 server name NAC-1
aaa authentication dot1x default group NAC-GRP
aaa authorization network default group NAC-GRP
aaa accounting dot1x default start-stop group NAC-GRP
aaa server radius dynamic-author
 client 10.10.60.10 server-key <from-secrets-vault>
dot1x system-auth-control
```

Add a second RADIUS server in the group for resilience. `dynamic-author` *lets the policy server send CoA messages (RFC 5176) to re-authorise or disconnect sessions.*

Access port with 802.1X, MAB fallback and a phone

SW-ACC1 · IOS-style CLI

```
interface GigabitEthernet1/0/5
 switchport mode access
 switchport access vlan 100
 switchport voice vlan 200
 authentication host-mode multi-domain
 authentication order dot1x mab
 authentication priority dot1x mab
 authentication port-control auto
 authentication periodic
 authentication timer reauthenticate server
 mab
 dot1x pae authenticator
 dot1x timeout tx-period 10
 spanning-tree portfast
```

Multi-domain allows one voice device and one data device. The shorter tx-period speeds up fallback to MAB for devices without a supplicant.

Host modes

MODE	BEHAVIOUR	USE
Single-host	One authenticated MAC per port	Highest assurance desktops
Multi-domain	One voice and one data device, each authenticated	Phone with a PC behind it
Multi-auth	Every MAC authenticates separately (one voice VLAN device)	Unmanaged hubs or virtual machines on a port
Multi-host	First device authenticates; others ride along	Rarely acceptable; weakest option

Rolling out access control without outages

1. **Monitor mode** Authenticate everything but allow all traffic (open authentication). Learn which devices fail and why.
2. **Fix the gaps** Deploy supplicant profiles and certificates; profile and register MAB devices.

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3. **Low-impact mode** Keep the port open with a pre-authentication ACL allowing only essentials (DHCP, DNS, PXE) until authentication succeeds.
4. **Closed mode** Nothing but EAPOL until authorised, for areas where risk justifies it.
5. **Operate** Watch failure dashboards, keep the endpoint inventory current and review policies quarterly.

RADIUS authorisation results

RESULT	HOW IT IS SIGNALLED	EFFECT
VLAN assignment	Tunnel-Type = VLAN (13), Tunnel-Medium-Type = 802 (6), Tunnel-Private-Group-ID = VLAN	Port placed in the returned VLAN
Named ACL	Filter-Id names an ACL defined on the switch	Traffic filtered per session
Downloadable ACL	Platform-specific attributes carry the ACL entries	Central ACL management
Group tag	Platform-specific attribute assigns a group identifier	Group-based policy enforced across the fabric
Re-authentication timer	Session-Timeout with Termination-Action	Forces periodic re-authentication

Two ways to segment dynamically

VLAN and ACL based

- Identity mapped to a VLAN and an IP subnet
- Policy written as IP ACLs
- Works on almost any hardware
- Rules multiply as subnets and groups grow

Group-based policy

- Identity mapped to a group tag carried with the traffic
- Policy is a matrix: group to group, allow or deny
- Independent of IP addressing
- Needs end-to-end support for carrying or mapping tags

WATCH OUT · MAB IS NOT AUTHENTICATION

A MAC address is printed on the label and easy to clone. Treat MAB devices as low trust: profile them, give them only the access they need (a printer reaches print servers, not the finance VLAN), and alert when a known MAC appears with an unexpected profile.

SCENARIO · DECIDE · PRINTERS VANISHED AFTER ENFORCEMENT

After moving a floor from monitor to closed mode, every network printer is offline. 802.1X laptops work. Logs show printers time out on 802.1X and never try MAB within the switch's limits. What is the best fix?

(A) Return the whole floor to monitor mode permanently so that printers and laptops both work again.

(B) Put the printers on multi-host ports behind the nearest laptop so they share its session.

(C) Install a supplicant and certificate on every printer immediately so they can do 802.1X.

(D) Enable MAB with a short tx-period for fast fallback, and map printer MACs to a printer policy.

Circle your choice. Answer and feedback: module answer key.

Key terms

Supplicant
The endpoint software that performs 802.1X authentication using EAP over LAN.

MAB
MAC Authentication Bypass: using a device's MAC address as its identity when it cannot run 802.1X.

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Change of Authorisation (CoA)

A RADIUS message (RFC 5176) from the policy server that changes or ends an existing session's authorisation.

KNOWLEDGE CHECK

16. In 802.1X, which component enforces the policy returned by RADIUS?

    (A) The supplicant software running on the endpoint
    (B) The directory that stores the user's account
    (C) The certificate authority that issued the client certificate
    (D) The authenticator, such as the switch port or WLAN

17. A desk has an IP phone with a PC behind it, each authenticating separately. Which host mode fits?

    (A) Single-host
    (B) Multi-domain
    (C) Multi-host
    (D) Open authentication only

18. What is the purpose of monitor mode in an access-control rollout?

    (A) To block every device until administrators approve each MAC address
    (B) To disable RADIUS accounting temporarily so the policy server is not overloaded
    (C) Authenticate everything without blocking, to reveal failures first
    (D) To mirror all user traffic to a monitoring appliance for inspection

19. Which RADIUS attribute carries the VLAN identifier in a dynamic VLAN assignment?

    (A) Tunnel-Private-Group-ID
    (B) Filter-Id
    (C) Session-Timeout
    (D) Chargeable-User-Identity

20. Why should MAB-authenticated devices get narrowly scoped access?

    (A) MAB sessions cannot be re-authenticated after the first login
    (B) MAB uses an encryption method that is too weak for most traffic
    (C) MAB devices can only reach the RADIUS server and nothing else
    (D) MAC addresses are easy to spoof, so identity confidence is low

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MODULE 7 · ASSURE · LESSON 5 · 35 MIN

Infrastructure Protection: CoPP, Hardening and iACLs

Protect the control and management planes before an attacker or a storm tests them

- Identify management, control and data plane threats and controls.
- Build and roll out a control-plane policing policy safely.
- Apply a hardening baseline and an infrastructure ACL at the network edge.

Routers and switches are targets in their own right. A flood of packets aimed at a device's CPU can drop routing adjacencies and take a site offline even when links are healthy. Infrastructure protection treats each **plane** separately: lock down how people and tools manage the device, protect the protocols that keep it in the network, and filter traffic that has no business reaching the device at all.

Planes and their protection

Management plane	`SSH only, AAA with TACACS+ or RADIUS` `Management VRF or out-of-band network` `Source-restricted access`
Control plane	`Control-plane policing (CoPP)` `Routing protocol authentication` `Maximum-prefix and neighbour limits`
Data plane	`Infrastructure ACLs at the edge` `Unicast RPF and anti-spoofing` `Storm control and DHCP/ARP protection at access`

CoPP classification

R1 · IOS-style CLI

```
ip access-list extended COPP-ACL-ROUTING
 permit tcp any any eq bgp
 permit tcp any eq bgp any
 permit ospf any any
ip access-list extended COPP-ACL-MGMT
 permit tcp 10.10.50.0 0.0.0.255 any eq 22
 permit udp 10.10.50.0 0.0.0.255 any eq snmp
!
class-map match-any COPP-ROUTING
 match access-group name COPP-ACL-ROUTING
class-map match-any COPP-MGMT
 match access-group name COPP-ACL-MGMT
```

Classes should reflect how important the traffic is to keeping the device in the network, from routing protocols down to everything else.

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CoPP policy applied to the control plane

R1 · IOS-style CLI

```
policy-map COPP-POLICY
 class COPP-ROUTING
  police 2000000 conform-action transmit exceed-action transmit
 class COPP-MGMT
  police 512000 conform-action transmit exceed-action drop
 class class-default
  police 256000 conform-action transmit exceed-action drop
!
control-plane
 service-policy input COPP-POLICY
!
R1# show policy-map control-plane input class class-default
```

Routing is set to transmit on exceed while you baseline it; never drop routing traffic until you know its real peak. Counters on `class-default` *reveal traffic you did not classify.*

Rolling out CoPP safely

1. **Inventory** List every protocol the device must process: routing, management, monitoring, first-hop redundancy, DHCP relay.
2. **Classify** Build ACLs and classes from that list, with a catch-all class.
3. **Baseline** Apply with exceed-action transmit and collect conform/exceed counters through busy periods and failovers.
4. **Enforce** Set rates above observed peaks with headroom; switch exceed to drop for lower classes first.
5. **Monitor** Alert on exceed counters; review after topology or protocol changes.

Infrastructure ACL on the upstream interface

Internet edge R-EDGE1 · IOS-style CLI

```
ip access-list extended IACL-IN
 remark anti-spoofing
 deny ip 10.0.0.0 0.255.255.255 any
 deny ip 172.16.0.0 0.15.255.255 any
 deny ip 192.168.0.0 0.0.255.255 any
 deny ip 198.51.100.0 0.0.0.255 any
 remark required protocols to infrastructure
 permit tcp host 203.0.113.1 host 203.0.113.2 eq bgp
 permit tcp host 203.0.113.1 eq bgp host 203.0.113.2
 permit icmp any 198.51.100.0 0.0.0.255 packet-too-big
 remark protect infrastructure, permit transit
 deny ip any 198.51.100.0 0.0.0.255
 deny ip any host 203.0.113.2
 permit ip any any
!
interface GigabitEthernet0/0
 ip access-group IACL-IN in
```

198.51.100.0/24 is the infrastructure block (loopbacks and links); 203.0.113.0/30 is the ISP peering link. Customer and user traffic transits freely; only the devices themselves are shielded.

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A hardening baseline excerpt

R1 · IOS-style CLI

```
no ip http server
no ip http secure-server
ip ssh version 2
enable algorithm-type scrypt secret <from-secrets-vault>
login block-for 120 attempts 5 within 60
aaa authentication login default group TACACS-GRP local
line vty 0 15
 access-class VTY-MGMT in
 transport input ssh
 exec-timeout 10 0
interface GigabitEthernet0/0
 no ip proxy-arp
 no ip redirects
 no lldp transmit
```

Disable the web server only if no tool depends on it. Keep a local break-glass account for when AAA servers are unreachable, stored in the secrets vault.

Hardening baseline themes

THEME	EXAMPLES
Access	SSH only, AAA with per-user accounts, source-restricted VTY, session timeouts, login rate limits
Services	Disable unused services (HTTP, discovery protocols on external interfaces, proxy ARP, redirects)
Integrity	Signed software images, secure boot where available, configuration archive in version control
Protocols	Routing protocol authentication, BGP maximum-prefix, TTL security for external peers
Visibility	Accurate NTP, central syslog, SNMPv3, accounting of commands via AAA

SCENARIO · DECIDE · ROUTING FLAPS DURING AN ATTACK

During a traffic flood aimed at a core router's interface addresses, OSPF and BGP sessions drop and recover repeatedly. Links are not saturated. The router has no CoPP or infrastructure ACL. What is the best long-term protection?

- (A) Increase OSPF and BGP hold timers on all core routers so that sessions survive much longer during floods.
- (B) An edge infrastructure ACL to stop traffic to device addresses, plus CoPP protecting routing protocols.
- (C) Upgrade the core router to a model with a faster CPU so that it can process the flood traffic.
- (D) Disable ICMP on all interfaces of the core router so that the flood traffic is ignored entirely.

Circle your choice. Answer and feedback: module answer key.

Key terms

Control-plane policing (CoPP)
A QoS policy applied to traffic destined for the device's CPU, rate-limiting each class to protect critical protocols.

Infrastructure ACL (iACL)
An edge ACL that permits only required protocols to the network's own device addresses while letting transit traffic pass.

KNOWLEDGE CHECK

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21. Why baseline CoPP with exceed-action transmit before enforcing?

(A) Because policing cannot drop traffic until after a reload of the device

(B) To learn real peaks without dropping legitimate control traffic

(C) Because transmit mode encrypts control-plane traffic during the baseline

(D) Because CoPP counters are only populated when exceed-action is transmit

22. What is the main purpose of an infrastructure ACL at the internet edge?

(A) Block traffic to the network's own devices while letting transit pass

(B) Replace the firewall that protects internal users and data-centre servers

(C) Encrypt management traffic sent between routers and the NOC

(D) Enforce QoS markings for voice traffic entering the network

23. Which item belongs to management-plane protection?

(A) Rate-limiting BGP and OSPF packets destined for the route processor

(B) Blocking spoofed private-address sources on the internet uplink

(C) Restricting VTY access to SSH from a management subnet with AAA

(D) Enabling storm control on access ports facing user devices

24. Why permit ICMP packet-too-big messages towards infrastructure in an iACL?

(A) They carry the routing updates that BGP exchanges between the edge routers

(B) They are required for SSH sessions to complete their handshake

(C) Path MTU discovery relies on them; blocking them causes black holes

(D) They prove the source address of every packet is not spoofed

25. What should happen to traffic matching CoPP's class-default?

(A) Police it at a modest rate and review it, since it was not expected

(B) It should always be dropped completely, including routing protocol traffic

(C) It should be allowed without any limit because it is mostly user traffic

(D) It should be redirected to the management VRF for later inspection

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MODULE 7 · ASSURE

Answer key

Knowledge checks

1. A authPriv authenticates and encrypts. The others leave data or identity unprotected.

2. B Debug output is for short, targeted troubleshooting, not continuous export.

3. C Templates describe the fields, so new information elements can be added.

4. C Multiply by the sampling rate: 2 MB × 1,000 ≈ 2 GB.

5. A A stable source address makes filtering and correlation reliable regardless of which interface is used.

6. D State changes are rare but important; on-change reports them immediately without repeating unchanged data.

7. B Tracking ties route installation to probe state; the higher-AD floating route becomes active.

8. B ToS 184 equals DSCP 46 (EF); probes should experience the same treatment as the traffic they represent.

9. A 999 ÷ (999 + 1) = 0.999.

10. C Averages over five minutes hide microbursts and short events.

11. D Widespread failure on one device suggests lower-layer issues; start there.

12. D Forward progress to the core means the problem is likely on the return path.

13. B Multiple simultaneous changes hide cause and effect.

14. C Diffs against the last known good state show exactly what changed and when.

15. A Size-dependent failure is the signature of MTU problems.

16. D The authenticator relays EAP and then enforces the result.

17. B Multi-domain allows one voice and one data session on the port.

18. C Monitor mode gives visibility with no user impact, so gaps can be fixed first.

19. A Tunnel-Type and Tunnel-Medium-Type set the VLAN context; Tunnel-Private-Group-ID carries the VLAN.

20. D MAB proves little; limit the blast radius with tight policy and profiling.

21. B Enforcing rates you have not measured can drop routing traffic and cause outages.

22. A iACLs shield device addresses; they are not a replacement for user-facing firewalls.

23. C Management-plane controls govern how people and tools access the device.

24. C Filtering 'fragmentation needed' messages breaks PMTUD for tunnels and TCP sessions.

25. A Class-default catches unexpected traffic; police it and investigate what appears there.

Scenarios & case studies

Scenario · The mystery congestion
B is best. Correct. Flow data shows who talked to whom and how much, which is exactly what interface counters cannot tell you.

Case study · A telecom NOC moves from polling to streaming
Start with high-value, fast-changing data: interface counters and queue drops on aggregation and core links (10–30 s sampling), and on-change subscriptions for interface, BGP and IGP neighbour state. Prefer OpenConfig paths to reduce per-platform differences, falling back to native models where needed. Keep SNMP for legacy platforms but tighten it to SNMPv3 and reduce the polled set. Collectors scale horizontally behind a load balancer and write to a time-series database with tiered retention (raw for days, down-sampled for months). Alerting moves from single thresholds to rules over windows, deduplicated per service. Prove value by running both systems in parallel for a quarter and comparing incidents detected, time to detect and false alarms, and by replaying known customer-reported outages.

Scenario · Brief outages nobody sees
A is best. Correct. Short events are invisible to five-minute polling; on-change state and fine-grained sampling expose them.

Scenario · Where do you start?
D is best. Correct. One application failing from everywhere points to something shared near the application, not to branch links.

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Scenario · Printers vanished after enforcement
D is best. Correct. Printers have no supplicant; they depend on fast MAB fallback and a matching policy.

Scenario · Routing flaps during an attack
B is best. Correct. The iACL stops most hostile traffic at the border; CoPP ensures what remains cannot starve the routing processes.

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08

MODULE 8 · PROGRAM

Program — Network Automation & NetDevOps

Model-driven APIs, automation code, pipelines and closed-loop operations, with guardrails.

BY THE END OF THIS MODULE YOU WILL BE ABLE TO

- Read YANG models and use NETCONF, RESTCONF and gNMI to read and change devices.
- Write idempotent Python and Ansible automation with secrets handled safely.
- Build a NetDevOps pipeline with intent tests, lab validation, canary rollout and pre/post checks.
- Design event-driven workflows and apply guardrails to AI-assisted operations.
- Integrate all eight stages of The ZCNP Fabric into a capstone design and automation project.

LESSONS

1. Data Models and Programmable Interfaces · 45 min
2. Python and Ansible for Network Automation · 45 min
3. NetDevOps: Source Control, CI/CD and Testing · 45 min
4. Event-Driven Automation and AI-Assisted Operations · 40 min
5. The ZCNP Fabric End to End and Your Capstone · 45 min

MODULE 8 · PROGRAM · LESSON 1 · 45 MIN

Data Models and Programmable Interfaces

YANG, NETCONF, RESTCONF and gNMI: structured data instead of screen-scraping

- Read a YANG module and its tree representation.
- Compare NETCONF, RESTCONF and gNMI and choose one for a task.
- Build NETCONF and RESTCONF requests and subscribe or set values with gNMI.

The Program stage of **The ZCNP Fabric** turns everything you have designed into code-driven operations. Its foundation is **structured data**. Instead of sending CLI text and parsing human-readable output, automation talks to devices through APIs whose data is described by **YANG** models. The device validates requests against the model, and tools can generate code, documentation and tests from it.

The model-driven stack

Automation and applications	`Python scripts, Ansible, controllers, CI/CD pipelines`
Protocols	`NETCONF (RFC 6241) over SSH` `RESTCONF (RFC 8040) over HTTPS` `gNMI over gRPC`
Encodings	`XML` `JSON` `Protobuf`
Data models	`YANG (RFC 7950)` `IETF and OpenConfig models` `Native platform models`
Device datastores	`Running, candidate, startup` `Operational state`

A small YANG module

zz-vlans.yang · YANG 1.1

```
module zz-vlans {
  yang-version 1.1;
  namespace "urn:example:zz-vlans";
  prefix zzv;
  container vlans {
    list vlan {
      key "id";
      leaf id {
        type uint16 { range "1..4094"; }
      }
      leaf name {
        type string { length "1..32"; }
      }
      leaf enabled { type boolean; default true; }
    }
  }
}
```

A container groups data, a list holds keyed entries, and leaves carry typed values with constraints the device enforces before accepting a change.

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Reading a model as a tree (excerpt)

Automation host · pyang

```
$ pyang -f tree ietf-interfaces.yang
module: ietf-interfaces
  +--rw interfaces
     +--rw interface* [name]
        +--rw name            string
        +--rw description?    string
        +--rw type            identityref
        +--rw enabled?        boolean
        +--ro oper-status     enumeration
        +--ro statistics
           +--ro in-octets?   yang:counter64
```

`rw` *nodes are configuration,* `ro` *nodes are operational state,* `*` *marks a list with its key in brackets and* `?` *marks an optional leaf.*

Three programmable interfaces

NETCONF

- SSH, port 830
- XML encoding
- Datastores: running, candidate, startup
- Locks, validate, confirmed commit
- Best for transactional configuration

RESTCONF

- HTTPS with REST methods
- JSON or XML
- Resource URLs built from the YANG tree
- Easy for web tools and quick integrations
- No candidate datastore in the protocol itself

gNMI

- gRPC over HTTP/2 with TLS
- Protobuf or JSON encoding
- Capabilities, Get, Set, Subscribe
- Excellent for streaming telemetry
- Common with OpenConfig models

A safe NETCONF transaction

01 Hello — Exchange capabilities; confirm candidate and confirmed-commit support.

→

02 Lock — Lock the candidate datastore so no one else changes it.

→

03 Edit — edit-config into candidate with the desired data.

→

04 Validate — Ask the device to validate the candidate.

→

05 Confirmed commit — Commit with a timeout; it rolls back unless confirmed.

→

06 Confirm and unlock — Run post-checks, confirm, then release the lock.

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Edit the candidate, then a confirmed commit

NETCONF RPCs · XML

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <edit-config>
    <target><candidate/></target>
    <config>
      <interfaces xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces">
        <interface>
          <name>GigabitEthernet2</name>
          <description>WAN to ISP-B</description>
        </interface>
      </interfaces>
    </config>
  </edit-config>
</rpc>
<rpc message-id="102" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <commit><confirmed/><confirm-timeout>120</confirm-timeout></commit>
</rpc>
```

If no confirming commit arrives within 120 seconds, the device reverts automatically: a safety net for changes that might cut off your own access.

Reading one interface as JSON

Automation host · RESTCONF with curl

```
$ curl -s -u "$NET_USER:$NET_PASS" -H "Accept: application/yang-data+json" \
    https://r1.lab.example/restconf/data/ietf-interfaces:interfaces/interface=GigabitEthernet2
{
  "ietf-interfaces:interface": [
    {
      "name": "GigabitEthernet2",
      "description": "WAN to ISP-B",
      "type": "iana-if-type:ethernetCsmacd",
      "enabled": true
    }
  ]
}
```

Key values containing `/` *must be percent-encoded in the URL: GigabitEthernet0/0/1 becomes* `interface=GigabitEthernet0%2F0%2F1`.

RESTCONF methods

METHOD	EFFECT	TYPICAL SUCCESS CODE
GET	Read configuration and state	200
POST	Create a child resource or invoke an operation	201 (create) or 204/200
PUT	Create or fully replace the target resource	201 or 204
PATCH	Merge changes into the target resource	204 or 200
DELETE	Remove the target resource	204

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gNMI Capabilities, Get and Set with OpenConfig paths

Automation host · gnmic

```
$ gnmic -a leaf1.dc.example:57400 --tls-ca /etc/pki/ca.pem -u automation capabilities
$ gnmic -a leaf1.dc.example:57400 --tls-ca /etc/pki/ca.pem -u automation \
    get --path "/interfaces/interface[name=Ethernet1]/state/oper-status"
$ gnmic -a leaf1.dc.example:57400 --tls-ca /etc/pki/ca.pem -u automation \
    set --update-path "/interfaces/interface[name=Ethernet1]/config/description" \
        --update-value "uplink to spine1"
```

OpenConfig separates intended configuration (`config`) from operational values (`state`) under each node.

SCENARIO · DECIDE · CHOOSING THE INTERFACE

Your team must push a multi-part routing change to 40 edge routers so that either all parts apply on a router or none do, with automatic rollback if management access is lost. All routers support NETCONF, RESTCONF and gNMI. Which approach fits best?

- (A) RESTCONF PATCH requests, one per configuration section, sent to each router in turn.
- (B) gNMI Subscribe in on-change mode, applied to the BGP configuration paths on each router.
- (C) NETCONF with the candidate datastore, validate and a confirmed commit on each router.
- (D) CLI commands sent over SSH to each router and verified afterwards with regular expressions.

Circle your choice. Answer and feedback: module answer key.

Key terms

YANG
The data-modelling language (RFC 7950) that defines the structure, types and constraints of configuration and state data.

OpenConfig
Vendor-neutral YANG models written by network operators, widely used with gNMI.

Confirmed commit
A NETCONF commit that reverts automatically unless a confirming commit is received within a timeout.

KNOWLEDGE CHECK

1. In a YANG tree, what does `+--ro oper-status` indicate?
   - (A) A required configuration leaf with a default
   - (B) A list key used for indexing
   - (C) A deprecated configuration node
   - (D) A read-only operational state leaf
2. Which NETCONF feature lets a change revert automatically if you lose access to the device?
   - (A) Locking the running datastore
   - (B) Confirmed commit with a confirm timeout
   - (C) The get-config operation with a subtree filter
   - (D) The hello exchange that lists the device capabilities
3. A RESTCONF URL must address interface GigabitEthernet0/0/1. How is the key written?
   - (A) interface=GigabitEthernet0%2F0%2F1
   - (B) interface/GigabitEthernet0/0/1
   - (C) interface[name=GigabitEthernet0/0/1]
   - (D) interface=GigabitEthernet0-0-1
4. Which gNMI RPC is used for streaming telemetry?
   - (A) Get
   - (B) Set
   - (C) Capabilities
   - (D) Subscribe

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5. Which RESTCONF method merges changes into an existing resource rather than replacing it?

(A) PUT
(B) POST
(C) PATCH
(D) DELETE

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MODULE 8 · PROGRAM · LESSON 2 · 45 MIN

Python and Ansible for Network Automation

From one-off scripts to idempotent, reviewed playbooks

- Use Python libraries to collect structured data and apply configuration safely.
- Write an Ansible inventory, Jinja2 template and playbook with verification.
- Apply idempotency, secrets handling and dry runs to every automation task.

Python and Ansible are the everyday tools of network automation. Python gives you full control for data gathering, analysis and custom logic. Ansible gives you a declarative, readable way to apply the same change to many devices. Both work best when they handle **data**: inventories, variables and templates, rather than hard-coded commands.

Choosing a tool

Netmiko-style SSH libraries

- Send CLI commands over SSH
- Parse output with templates into structured data
- Works on almost any device
- You own error handling and idempotency

NAPALM-style abstraction

- Same functions across platforms (facts, BGP neighbours, config)
- Load, compare, commit and rollback workflow
- Shows a diff before committing

Ansible

- Inventory, variables and playbooks in YAML
- Modules for CLI and API-based platforms
- Check and diff modes where modules support them
- Readable by people who do not write Python

Parallel, parsed data collection

collect_bgp.py · Python

```
import os
from concurrent.futures import ThreadPoolExecutor
from netmiko import ConnectHandler
DRIVER = os.environ["NET_DRIVER"]      # Netmiko platform driver name
HOSTS = ["edge1.site7.example", "edge2.site7.example"]
def bgp_summary(host):
    params = {"device_type": DRIVER, "host": host,
              "username": os.environ["NET_USER"], "password": os.environ["NET_PASS"]}
    with ConnectHandler(**params) as conn:
        return host, conn.send_command("show ip bgp summary", use_textfsm=True)
with ThreadPoolExecutor(max_workers=10) as pool:
    for host, peers in pool.map(bgp_summary, HOSTS):
        print(host, peers)
```

Credentials come from the environment (or a vault), never from the script. `use_textfsm=True` *returns a list of dictionaries when a parsing template exists for the command.*

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Diff before commit

ntp_change.py · Python

```
from napalm import get_network_driver
driver = get_network_driver(DRIVER)
with driver(host, user, password) as device:
    device.load_merge_candidate(filename="ntp_baseline.cfg")
    diff = device.compare_config()
    if diff:
        print(diff)
        device.commit_config()
    else:
        device.discard_config()
```

Committing only when a diff exists makes the script idempotent: running it twice changes nothing the second time.

Habits of reliable automation code

- Keep secrets in environment variables or a vault, never in Git.
- Prefer APIs and structured output over parsing free text.
- Make every task idempotent: check current state, change only what differs.
- Set timeouts and handle failures per device so one bad host does not stop the run.
- Log what was changed, where and by whom; keep the output as evidence.

Inventory and data

inventory.yml and group_vars/all.yml · YAML

```
all:
  children:
    access_switches:
      hosts:
        sw-acc1.site7.example:
        sw-acc2.site7.example:
      vars:
        ansible_connection: ansible.netcommon.network_cli
        ansible_network_os: <your platform's network OS name>
# group_vars/all.yml
ntp_servers:
  - 10.10.50.5
  - 10.10.50.6
```

The NTP servers live in data, not in the playbook, so the same playbook serves every site.

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Template, push and verify

templates/ntp.j2 and ntp.yml · Jinja2 and Ansible

```
{% for server in ntp_servers %}
ntp server {{ server }}
{% endfor %}
---
- name: Enforce NTP baseline
  hosts: access_switches
  gather_facts: false
  tasks:
    - name: Render and push NTP configuration
      ansible.netcommon.cli_config:
        config: "{{ lookup('ansible.builtin.template', 'ntp.j2') }}"
    - name: Read NTP associations
      ansible.netcommon.cli_command:
        command: show ntp associations
      register: ntp
    - name: Verify both servers are present
      ansible.builtin.assert:
        that: "'10.10.50.5' in ntp.stdout and '10.10.50.6' in ntp.stdout"
```

Run first with `ansible-playbook -i inventory.yml ntp.yml --check --diff --limit sw-acc1.site7.example` *to preview the change on one device.*

A playbook run, done properly

01 Select — Limit to a canary device or site. →

02 Preview — Check and diff mode shows intended changes. →

03 Apply — Push the rendered configuration. →

04 Verify — Assert the resulting state, not just the command's success. →

05 Expand — Widen the limit in waves, stopping on any failure.

WATCH OUT · IDEMPOTENT IS NOT THE SAME AS SAFE

A playbook can be perfectly idempotent and still apply the wrong intent to 500 devices. Safety comes from review, testing in a lab, canary rollout and post-checks, which the next lesson builds into a pipeline.

SCENARIO · DECIDE · THE SCRIPT THAT CHANGED TOO MUCH

A colleague's Python script adds an ACL line to every edge router. Run twice by mistake, it added duplicate entries and broke the ACL's sequence. What is the best improvement?

- (A) Make it idempotent: compare the current ACL with the intended state and apply only the difference.
- (B) Add a comment at the top of the script warning colleagues never to run it twice in a row.
- (C) Rewrite the script in a different programming language with stronger error handling built in.
- (D) Run the script only on Sunday mornings, when traffic is low and fewer users would notice.

Circle your choice. Answer and feedback: module answer key.

Key terms

Idempotency
The property of an automation task that produces the same end state whether it runs once or many times.

Inventory
The structured list of devices, groups and variables that automation tools target.

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Jinja2 template
A text template with variables and loops used to render device configuration from data.

KNOWLEDGE CHECK

6. What makes the NAPALM-style workflow idempotent in the example?

   (A) It reconnects to every device twice before committing a change
   (B) It commits only when compare_config returns a difference
   (C) It removes the existing configuration before loading the file
   (D) It runs the commit inside a background thread for each device

7. Where should device credentials for automation live?

   (A) In the playbook header so every engineer can read them easily
   (B) In a comment inside the Jinja2 template for each platform
   (C) In the Git repository alongside the inventory and variables
   (D) In environment variables or a vault, injected at run time

8. What does Ansible's `--check --diff` combination provide?

   (A) A permanent rollback point stored on every target device
   (B) An automatic peer review of the playbook by another engineer
   (C) A preview of the changes, where modules support check mode
   (D) A parallel run against both lab and production inventories

9. Why does the playbook end with an assert task?

   (A) To prove the resulting device state, not just accepted commands
   (B) To make the playbook run faster on large inventories of network devices
   (C) To encrypt the output of the show command in the logs
   (D) To prevent the playbook from running in check mode

10. Why are NTP server addresses kept in group_vars instead of in the playbook?

    (A) Ansible cannot read IP addresses written directly inside playbooks
    (B) Separating data from logic lets one playbook serve many sites
    (C) Group variables are encrypted automatically by Ansible at run time
    (D) Playbooks may only contain module names and never any values

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MODULE 8 · PROGRAM · LESSON 3 · 45 MIN

NetDevOps: Source Control, CI/CD and Testing

Treat network changes like software: versioned, reviewed, tested and reversible

- Use Git branches and reviews for network intent and templates.
- Design a CI/CD pipeline with linting, rendering, testing and staged deployment.
- Define pre-checks and post-checks that gate every change.

NetDevOps applies software delivery practice to network change. The **source of truth** is data in version control, not the running configuration of a device. Every change is a commit, reviewed by a colleague, tested automatically and deployed by a pipeline that checks the network before and after. The goal is not speed for its own sake; it is fewer failed changes and faster, safer recovery when something does go wrong.

From intent to device

Intent data	`Sites, VLANs, VRFs, RDs/RTs, BGP peers in YAML or an IPAM/inventory system`
Templates and logic	`Jinja2 templates, playbooks, Python modules`
Rendered artefacts	`Per-device configurations or API payloads, built by the pipeline`
Network	`Devices receive changes only through the pipeline; drift is detected and reported`

A change starts as a branch

Engineer workstation · Git

```
$ git switch -c change/CHG-2417-ntp-baseline
$ git add group_vars/all.yml templates/ntp.j2
$ git commit -m "CHG-2417: add second NTP server to baseline"
$ git push -u origin change/CHG-2417-ntp-baseline
# open a merge request: the pipeline lints, renders, tests and deploys to the lab
$ git log --oneline -3
9c1e4ab CHG-2417: add second NTP server to baseline
51d07f2 CHG-2399: GUEST VRF route-target update
e8a3310 CHG-2391: CoPP management class rate increase
```

The change ID in the branch and commit message links code, ticket and pipeline evidence together.

A network CI/CD pipeline

01 Lint — YAML, templates and playbooks checked for syntax and style. →

02 Render and test — Configurations rendered; unit tests check intent rules. →

03 Model and lab — Offline analysis of rendered configs, then deployment to a virtual lab. →

04 Review and approve — Peer review plus change approval for production. →

05 Pre-check and canary — Verify current state, deploy to a small set of devices. →

06 Post-check and expand — Verify, then roll out in waves; roll back automatically on failure.

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Testing intent before it reaches a device

tests/test_intent.py · pytest

```
import yaml
def load(path):
    with open(path) as f:
        return yaml.safe_load(f)
def test_route_distinguishers_unique():
    vrfs = load("intent/vrfs.yml")["vrfs"]
    rds = [v["rd"] for v in vrfs]
    assert len(rds) == len(set(rds)), "duplicate RD in VRF plan"
def test_guest_never_imports_corp():
    guest = next(v for v in load("intent/vrfs.yml")["vrfs"] if v["name"] == "GUEST")
    assert "65010:10" not in guest["import_rt"], "GUEST must not import CORP routes"
```

Tests encode design rules from earlier modules, such as the VRF and route-target plan, so a pull request that breaks them fails before any device is touched.

Pipeline stages expressed as code

pipeline.yml · CI definition (illustrative)

```
stages: [lint, build, test, lab, approve, deploy]
lint:    { script: ["yamllint .", "ansible-lint playbooks/"] }
build:   { script: ["python tools/render.py --out build/"] }
test:    { script: ["pytest tests/ --junitxml=reports/unit.xml",
                    "python tools/model_checks.py build/"] }
lab:     { script: ["ansible-playbook -i inventory/lab.yml deploy.yml",
                    "python tools/postchecks.py --inventory inventory/lab.yml"] }
approve: { when: manual }
deploy:  { script: ["python tools/prechecks.py --site $SITE",
                    "ansible-playbook -i inventory/prod.yml deploy.yml --limit $CANARY",
                    "python tools/postchecks.py --site $SITE"] }
```

Syntax differs between CI systems; the structure (stages, gates, manual approval, canary and checks) is what matters.

Tests at each level

LEVEL	QUESTION IT ANSWERS	EXAMPLES
Static	Is the code well-formed?	YAML lint, template render, playbook lint
Intent (unit)	Does the data obey our design rules?	Unique RDs, no overlapping subnets, MTU ≥ 9,000 in the fabric
Model	Would the rendered configs behave correctly?	Offline analysis of reachability and ACLs, e.g. with open-source config analysis tools
Lab	Does it work on real software?	Deploy to a virtual lab, run post-checks
Production checks	Is the live network healthy before and after?	Neighbour counts, route counts, probes, interface errors

TEMPLATE

Change pre-check and post-check list

Define, per change type, what must be true before a change starts and what must be proven after, so pipelines and engineers gate changes the same way.

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CHECK	COMMAND / API / PROBE	EXPECTED VALUE	PRE	POST	STOP IF
BGP neighbours on edge	*gNMI get neighbor session-state*	*All ESTABLISHED*	*Record count*	*Same count*	*Any neighbour not ESTABLISHED after 3 min*
Branch-to-DC voice probe	*UDP jitter probe results*	*Jitter < 30 ms, loss < 1%*	*Baseline*	*Within 10% of baseline*	*SLA breach for 2 intervals*

CASE STUDY · AN ISP BUILDS AN AUTOMATION PIPELINE

A regional ISP provisions business internet and L3VPN services by hand. Each order takes five days, one in twelve changes causes a ticket, and configurations differ between engineers. The ISP wants self-service ordering within two years and asks you to design the pipeline that will sit behind it.

1. What would be your source of truth, and how would you migrate to it?

2. Which tests would you require before a service change reaches a PE router?

3. How would you roll out changes and recover from failures?

4. Which measures would show the pipeline is working?

Model answer: module answer key.

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NetDevOps readiness checklist

☐ All intent data and templates are in Git with protected main branch and mandatory review.

☐ No secrets in the repository; the pipeline injects them from a vault.

☐ Every merge request runs lint, render and intent tests automatically.

☐ A virtual lab mirrors production topology closely enough for meaningful tests.

☐ Pre-checks and post-checks exist for every common change type.

☐ Deployments use canary waves and have an automated rollback path.

☐ Drift between intent and live configuration is reported at least daily.

☐ Change failure rate and lead time are measured and reviewed monthly.

SCENARIO · DECIDE · A RISKY MERGE REQUEST

A merge request changes the GUEST VRF import list. Lint and rendering pass, but the intent test `test_guest_never_imports_corp` fails. A senior engineer asks you to skip tests just this once because the change is urgent. What do you do?

(A) Skip the test just this once because the change is urgent, and add a note to the change ticket.

(B) Delete the test so that future urgent changes are not blocked by the same rule again.

(C) Keep the gate: inspect the failure, fix the data or formally update the rule, then re-run.

(D) Apply the change by hand on the devices to avoid the pipeline, and update Git afterwards.

Circle your choice. Answer and feedback: module answer key.

Key terms

Source of truth
The authoritative, version-controlled data describing how the network should be configured.

Configuration drift
Differences between the intended configuration in the source of truth and what is running on devices.

CI/CD
Continuous integration and delivery: automatically testing every change and deploying it through controlled stages.

KNOWLEDGE CHECK

11. In NetDevOps, what should be the source of truth for intended configuration?

(A) The running configuration on each production device

(B) The most recent backup taken by the monitoring system

(C) Version-controlled intent data and templates

(D) The design documents stored on a shared drive

12. Which test catches a duplicate route distinguisher before any device is touched?

(A) A post-check on BGP neighbour counts after deployment

(B) A YAML lint check of the playbook formatting

(C) A synthetic branch probe

(D) An intent unit test over the VRF data

13. What is the purpose of a canary stage in network deployment?

(A) To run linting and template rendering on the change

(B) To limit the blast radius by changing a few devices first

(C) To obtain formal approval from the change advisory board for the rollout

(D) To encrypt configuration files before they are pushed

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14. A post-check fails after the canary deploy. What should the pipeline do?

(A) Stop the rollout and roll back the canary devices

(B) Continue with the next wave and raise a low-priority ticket

(C) Re-run the post-check repeatedly until it eventually passes

(D) Delete the failing post-check so the rollout can continue

15. Why is configuration drift reported regularly?

(A) To measure how quickly the pipeline renders configurations

(B) To decide which engineers should approve merge requests

(C) To compress old configuration backups and save storage

(D) To find out-of-band changes that diverge from intent

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MODULE 8 · PROGRAM · LESSON 4 · 40 MIN

Event-Driven Automation and AI-Assisted Operations

Closing the loop from signal to action, with guardrails

- Design event-driven workflows that enrich, decide and act on network events.
- Decide which actions may run automatically and which need human approval.
- Apply guardrails when using AI assistance for network operations.

Assurance tells you something happened; automation can respond. **Event-driven automation** connects the two: a telemetry alert, syslog message or ticket triggers a workflow that gathers context, decides what to do and acts or asks for approval. AI assistance is increasingly part of that loop, summarising evidence and suggesting actions. Both multiply the speed of operations, and both multiply mistakes unless they are bounded by clear guardrails.

The closed loop

01 Detect — Telemetry, syslog, probes or tickets raise an event. →

02 Enrich — Add context: device role, recent changes, topology, open incidents. →

03 Decide — Rules (and optionally AI-generated suggestions) choose an action. →

04 Act — Run diagnostics automatically; run changes per approval policy. →

05 Verify — Confirm the outcome with fresh data. →

06 Record — Update the ticket with evidence and actions taken.

Events mapped to safe, read-only first actions

rulebooks/bgp.yml · event-driven Ansible rulebook

```
- name: React to BGP neighbour loss
  hosts: all
  sources:
    - ansible.eda.webhook:
        host: 0.0.0.0
        port: 5000
  rules:
    - name: Collect diagnostics when a BGP session drops
      condition: event.payload.alert == "bgp_session_down"
      action:
        run_playbook:
          name: playbooks/bgp_diagnostics.yml
    - name: Notify on repeated interface flaps
      condition: event.payload.alert == "interface_flap" and event.payload.count > 5
      action:
        run_playbook:
          name: playbooks/open_ticket.yml
```

The alerting system posts JSON to the webhook. The first rule only gathers evidence; changing the network is left to a human or to a separately approved workflow.

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What may run automatically?

EVENT	AUTOMATIC ACTION	NEEDS APPROVAL
BGP session down	Collect neighbour, route and log data; attach to ticket	Clearing sessions, changing policy
Interface flapping	Gather optics and error counters; open ticket	Shutting the interface
Config drift detected	Report the diff to the owning team	Overwriting the device from intent
WAN SLA breach	Confirm with extra probes; notify	Permanent path-preference changes
Known, low-risk fix (e.g. re-enable a port after an approved maintenance)	Run if pre-checks pass	Anything outside the approved runbook

Guardrails for AI-assisted operations

READ
Read-only by default
Assistants query telemetry, logs and configs; write access is granted per approved workflow only.

GRND
Grounded in your data
Answers must cite the source of truth, telemetry and runbooks, not guesses.

HUM
Human approval for change
Proposed changes become merge requests that pass the same pipeline and review as any other change.

SEC
No secrets or personal data in prompts
Redact credentials, keys and customer data before anything leaves the controlled environment.

LOG
Full audit trail
Prompts, evidence, suggestions and decisions are logged with the change or incident.

EVAL
Measured, not assumed
Track accuracy of suggestions against outcomes and withdraw uses that do not earn trust.

Where AI assistance helps and where it hurts

Good early uses

- Summarising a noisy incident timeline
- Explaining an unfamiliar log message or config section
- Drafting a runbook or a test from a description
- Natural-language queries over telemetry, with links to the data
- Suggesting a likely cause for an engineer to verify

Risky uses

- Pushing generated configuration directly to devices
- Trusting answers that cite no evidence
- Pasting full configs with secrets into external tools
- Letting suggestions bypass change control
- Using it to replace understanding of the network

Adopting AI assistance safely

1. **Pick a read-only use case** For example, incident summaries attached to tickets.
2. **Connect approved data only** Telemetry, logs, the source of truth and runbooks, with access controls.
3. **Define success** Time saved per incident and accuracy judged by engineers.
4. **Pilot with review** Engineers rate every output; failures are analysed.
5. **Expand deliberately** Only after evidence; any write action goes through the pipeline.

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WATCH OUT · AUTOMATION AMPLIFIES

A wrong manual change hits one device. A wrong automated or AI-suggested change can hit every device in minutes. Rate-limit automated actions, require pre-checks, and keep a kill switch that stops all event-driven workflows at once.

SCENARIO · DECIDE · AUTO-REMEDIATION PROPOSAL

An engineer proposes that whenever an access-switch uplink flaps more than five times in ten minutes, automation should shut the uplink and force traffic to the redundant one, with no human involved. Uplinks are dual-homed. What do you recommend?

(A) Collect evidence and ticket first; add shutdown later with health pre-checks and rate limits.

(B) Approve the proposal as written, because the uplinks are dual-homed and therefore redundant.

(C) Reject automation entirely, because interface flaps must always be handled manually by people.

(D) Let an AI assistant decide case by case whether to shut each uplink, and apply its choice.

Circle your choice. Answer and feedback: module answer key.

Key terms

Event-driven automation
Workflows triggered automatically by events such as alerts, logs or tickets, rather than run by hand on a schedule.

Kill switch
A single control that halts all automated actions immediately when something goes wrong.

Enrichment
Adding context (device role, recent changes, topology) to an event before deciding how to respond.

KNOWLEDGE CHECK

16. What is the safest first action for most event-driven network workflows?

(A) Shutting down the interface that raised the event

(B) Reloading the device to clear transient state

(C) Collecting diagnostics and attaching them to a ticket

(D) Reverting the device to the configuration from last week

17. An AI assistant suggests a BGP policy change during an incident. How should it be applied?

(A) Directly on the device, because the incident is urgent and time matters

(B) As a merge request through the normal pipeline and review

(C) Only on the core routers where the impact is highest

(D) By pasting it into the console of every affected router

18. Why redact data before using external AI tools during troubleshooting?

(A) Configs and logs may hold secrets and personal or customer data

(B) External tools cannot read any text that includes IP addresses or hostnames

(C) Redaction makes the suggestions more accurate in every case

(D) Logs must be shortened to fit within the tool's display

19. What does 'enrichment' add in an event-driven workflow?

(A) Context: device role, recent changes and topology

(B) Encryption of the event payload as it moves between systems

(C) Compression of telemetry data before it is stored

(D) A second copy of the event for disaster recovery

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20. Which guardrail limits damage if an automated workflow misbehaves?

(A) A larger webhook port range on the automation server

(B) More detailed log messages from every network device

(C) Faster telemetry sampling for every interface in the whole network

(D) Rate limits on actions plus a kill switch for all workflows

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MODULE 8 · PROGRAM · LESSON 5 · 45 MIN

The ZCNP Fabric End to End and Your Capstone

One network, eight stages, and the project that proves you can run them all

- Trace a user's traffic and an engineer's change through all eight Fabric stages.
- Connect design artefacts from every module into one coherent package.
- Plan your capstone deliverables and milestones.

You have now worked through all eight stages of **The ZCNP Fabric**. Real networks do not respect module boundaries: a voice call from a ward tablet crosses wireless, the campus, an overlay, the WAN, a data-centre fabric and back, and every hop is designed, assured and increasingly programmed. This lesson stitches the stages together and introduces your capstone.

The ZCNP Fabric, stage by stage

ARC
Architect
Campus, spine-leaf and high-availability design set the shape of everything else.

SW
Switch
A loop-free, protected Layer 2 edge with RSTP/MST and edge protections.

IGP
IGP
OSPF and IS-IS design, redistribution and policy routing give stable reachability.

BGP
BGP
Policy, scale and resilience at the edges and in the core.

OVL
Overlay
VRFs, tunnels, IPsec, VXLAN EVPN and SD-WAN create logical networks on the underlay.

WLS
Wireless
RF design, roaming, security and QoS extend the network to every mobile user and device.

ASR
Assure
Telemetry, probes, troubleshooting, access control and device protection prove and defend service.

PRG
Program
Models, APIs, pipelines and event-driven workflows make changes repeatable and safe.

A voice call through the Fabric

01
Wireless
A clinician's handset joins a WPA3-Enterprise SSID with EAP-TLS; RADIUS assigns the CLINICAL role.

→

02
Switch and Architect
The AP's traffic enters the campus access layer and a resilient distribution pair.

→

03
Overlay
Traffic sits in VRF CORP and leaves the branch over an SD-WAN tunnel chosen by the VOICE SLA class.

→

04
IGP and BGP
Hub routing and the DC's BGP EVPN fabric deliver it to the call-control servers.

→

05
Assure
Jitter probes and streaming telemetry confirm the SLA every minute.

→

06
Program
The SSID, VRF and SLA policy were all deployed from Git through the pipeline.

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Artefacts that connect the stages

STAGE	KEY ARTEFACT	VERIFIED BY
Architect	High-level and low-level design	Design review and failure-mode analysis
Switch / IGP / BGP	Protocol design and routing policy	Lab tests and convergence measurements
Overlay	VRF and route-target plan, tunnel and SD-WAN policy	Intent tests and path-steering tests
Wireless	WLAN design sheet and validation survey	Survey report and roaming test walks
Assure	Telemetry KPI sheet, NAC policy, hardening baseline	Dashboards, audits and incident drills
Program	Repository, pipeline and pre/post checks	Pipeline runs and change failure rate

Your **capstone** asks you to take one realistic organisation through the whole Fabric. You will design a multi-site network with a campus, at least two branches, a data-centre fabric and enterprise wireless; segment it with VRFs; connect it with an overlay WAN; define how it is assured and protected; and automate at least one meaningful change end to end with tests and pre/post checks. The organisation can be real (anonymised) or realistic, such as a hospital group, a university, a ministry or a regional bank.

Capstone milestones

1. **Requirements and constraints** Users, sites, applications and their SLAs, security and compliance needs, budget and growth.
2. **Architecture** Topology, high availability, IGP and BGP design, with rationale.
3. **Overlay and segmentation** VRF and route-target plan, WAN overlay and SD-WAN policy, data-centre fabric.
4. **Wireless** WLAN design sheet for at least two contrasting areas, security and QoS.
5. **Assurance and protection** Telemetry KPI sheet, NAC rollout plan, CoPP and hardening baseline, troubleshooting runbook.
6. **Automation** A Git repository with intent data, templates, tests and a pipeline that deploys one change with pre/post checks.
7. **Defence** A short presentation of key decisions, trade-offs and risks.

TEMPLATE

Capstone design decision log

Record each significant decision with its alternatives and evidence, so reviewers can follow your reasoning across all eight stages.

STAGE	DECISION	ALTERNATIVES CONSIDERED	WHY THIS CHOICE	RISK AND MITIGATION	HOW IT IS VERIFIED
Overlay	*SD-WAN with DIA for approved SaaS; tax systems pinned to MPLS*	*Central breakout; full DIA*	*Lower SaaS latency without exposing critical apps*	*Inconsistent security at branches: cloud security service with one policy*	*Path-steering tests and monthly SLA report*
Program	*NETCONF confirmed commits from the pipeline*	*Ansible CLI push; manual change*	*Atomic per device with automatic rollback*	*Pipeline outage: documented manual fallback*	*Canary run with post-checks*

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STAGE	DECISION	ALTERNATIVES CONSIDERED	WHY THIS CHOICE	RISK AND MITIGATION	HOW IT IS VERIFIED

Capstone readiness checklist

☐ Requirements include measurable SLAs for at least three applications.

☐ Every design choice is logged with alternatives and evidence.

☐ The VRF and route-target plan, WLAN design sheet and telemetry KPI sheet are complete.

☐ NAC and hardening plans include phased rollout and rollback.

☐ The repository contains intent data, templates, tests and a pipeline definition.

☐ One change has run through the pipeline with pre-checks, canary and post-checks.

☐ A troubleshooting runbook covers the three most likely failure scenarios.

☐ No secrets, real customer data or proprietary configurations are included.

SCENARIO · DECIDE · WHERE TO START THE CAPSTONE

You have six weeks for the capstone. A colleague suggests starting by writing the automation pipeline because it is the most impressive part. What is the best starting point?

(A) Build the pipeline first and fit the design around whatever the pipeline is able to deploy.

(B) Spend the six weeks perfecting the wireless design, because it is the most visible part.

(C) Copy a published reference design, change the names, and focus the remaining time on automation.

(D) Start with requirements and architecture, then automate a change that the design actually needs.

Circle your choice. Answer and feedback: module answer key.

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REFLECT

Look back over all eight stages. Which stage is strongest in the network you work on today, which is weakest, and what single change would most improve the weakest one in the next three months?

Key terms

The ZCNP Fabric
The eight-stage professional framework of this programme: Architect, Switch, IGP, BGP, Overlay, Wireless, Assure and Program.

Design decision log
A record of significant design choices, the alternatives considered, the evidence and the verification method.

KNOWLEDGE CHECK

21. A branch voice call depends on an SD-WAN SLA class. Which Fabric stage proves the SLA is being met?

(A) Architect, through the high-level design
(B) Assure, through probes and telemetry
(C) Switch, through spanning-tree protection
(D) Wireless, through the channel plan

22. Which artefact links the Overlay stage to automated testing in the Program stage?

(A) The radio channel plan for each floor of every building
(B) The syslog severity table on each device
(C) The VRF and route-target plan held as intent data
(D) The spanning-tree root bridge priorities

23. Why should the capstone automate a change that the design actually needs?

(A) Automated changes do not need to be tested in a lab beforehand
(B) Pipelines can only deploy changes that were chosen after the design was finished
(C) Reviewers only assess automation that changes wireless settings
(D) Automation should encode real intent, proving design and pipeline together

24. What is the main purpose of a design decision log?

(A) To show the alternatives, evidence and checks behind each choice
(B) To list every command entered on every device in the network over time
(C) To replace the low-level design with a shorter document
(D) To record the names of engineers who approved changes

25. In the voice-call walk-through, how did the SSID, VRF and SLA policy reach the network?

(A) By manual configuration during a weekend window
(B) From version control through the CI/CD pipeline
(C) By the wireless clients requesting them at join time
(D) By the RADIUS server writing device configuration

☐ Lesson complete · Date _____ · Knowledge check score ____ / 5

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MODULE 8 · PROGRAM

Answer key

Knowledge checks

1. D `ro` marks state data that cannot be configured.

2. B Without a confirming commit before the timeout, the device rolls back.

3. A Reserved characters in key values are percent-encoded; `/` becomes %2F.

4. D Subscribe supports stream, once and poll modes.

5. C PUT replaces the target, DELETE removes it, and PATCH merges into it.

6. B No diff means no change; re-running is harmless.

7. D Secrets in code or Git leak; inject them at run time from a protected store.

8. C Check mode simulates; diff shows the change. Module support varies, so verify behaviour first.

9. A Successful command execution does not prove the intended state was reached.

10. B Data-driven automation reuses the same logic with different inputs.

11. C Devices hold the actual state; intent lives in version control and is compared against it.

12. D Intent tests validate design rules on the data itself.

13. B Problems surface on a small set before the change reaches everything.

14. A Failed checks must halt and reverse the change.

15. D Drift reveals manual or unauthorised changes that the pipeline does not know about.

16. C Read-only actions add value immediately and cannot make the outage worse.

17. B AI-generated changes follow the same controls as human ones.

18. A Credentials and personal data must never leave controlled systems.

19. A Context turns a raw alert into something a rule or person can act on correctly.

20. D Bounding how fast automation acts, and being able to stop it, contains failures.

21. B Assurance measures service continuously; design stages set it up.

22. C Intent tests in the pipeline enforce the VRF and RT plan.

23. D Automation without design intent demonstrates tooling, not engineering.

24. A Reviewers need the reasoning, not only the result.

25. B The Program stage makes every other stage's configuration repeatable and reviewable.

Scenarios & case studies

Scenario · Choosing the interface
C is best. Correct. Candidate, validate and confirmed commit give an atomic change per device with automatic rollback.

Scenario · The script that changed too much
A is best. Correct. Comparing intended and actual state is the core of safe, repeatable automation.

Case study · An ISP builds an automation pipeline
Make service intent the source of truth: a Git repository (or an inventory/IPAM system with Git-backed templates) holding customers, VRFs, RDs/RTs, interfaces and BGP parameters as data. Migrate by discovering existing services, normalising them into that data model and reconciling until rendered configs match live configs, with drift reports along the way. Required tests: schema and lint checks, intent rules (unique RDs, no RT that leaks one customer into another, valid ASNs and prefixes), rendered-config analysis, and a virtual-lab deployment with post-checks. Rollout: NETCONF confirmed commits per PE, canary customers first, automated pre/post checks on BGP sessions, routes and probes, with automatic rollback when a check fails. Measures: lead time per order, change failure rate, mean time to restore, percentage of services under pipeline control and configuration drift count.

Scenario · A risky merge request
C is best. Correct. The test protects a security boundary. Either the change is wrong or the rule must be formally updated, never silently bypassed.

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Scenario · Auto-remediation proposal
A is best. Correct. Automate diagnostics first, then the action, with conditions that prevent it from isolating a switch.

Scenario · Where to start the capstone
D is best. Correct. Automation encodes intent; without a clear design there is nothing trustworthy to automate.

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REFERENCE

Glossary

A

ABR Area Border Router, connecting area 0 to one or more other areas. (IGP)

Alternate port An RSTP port that receives a better BPDU from another bridge and stands by to replace the root port. (Switch)

Application-aware routing Steering each application to a transport path whose measured loss, latency and jitter meet the application's SLA class. (Overlay)

AS_PATH The list of AS numbers a route has crossed, used for loop prevention and path selection. (BGP)

AS-path prepending Adding extra copies of your AS number to make a path look longer to others. (BGP)

ASBR Autonomous System Boundary Router, injecting external routes into OSPF. (IGP)

Attached bit Set by an IS-IS L1/L2 router so Level 1 routers install a default route towards it. (IGP)

Autonomous system A network under one routing policy, identified by an AS number. (BGP)

B

Baseline Recorded normal behaviour (utilisation, latency, table sizes) used to judge whether current data is abnormal. (Assure)

BFD Bidirectional Forwarding Detection (RFC 5880), a lightweight protocol that detects path failure in milliseconds. (Architect)

Border node A fabric node that connects the overlay to external networks. (Architect)

BPDU guard Err-disables an edge port that receives any BPDU. (Switch)

BSS transition management The 802.11v mechanism that lets the network suggest a better AP to a client or warn it before disassociation. (Wireless)

C

CAPWAP An IETF protocol (RFC 5415) for managing APs from a controller and optionally tunnelling client data to it. (Wireless)

Change of Authorisation (CoA) A RADIUS message (RFC 5176) from the policy server that changes or ends an existing session's authorisation. (Assure)

CI/CD Continuous integration and delivery: automatically testing every change and deploying it through controlled stages. (Program)

CLUSTER_LIST The list of reflector cluster IDs a route has passed, used to prevent loops. (BGP)

Collapsed core A design where the distribution and core functions are combined into one redundant pair of switches. (Architect)

Community A tag attached to routes to signal policy, standard (RFC 1997) or large (RFC 8092). (BGP)

Configuration drift Differences between the intended configuration in the source of truth and what is running on devices. (Program)

Confirmed commit A NETCONF commit that reverts automatically unless a confirming commit is received within a timeout. (Program)

Control-plane policing (CoPP) A QoS policy applied to traffic destined for the device's CPU, rate-limiting each class to protect critical protocols. (Assure)

D

Design decision log A record of significant design choices, the alternatives considered, the evidence and the verification method. (Program)

DFS Dynamic Frequency Selection: rules that require APs on certain 5 GHz channels to detect radar and move away from it. (Wireless)

Distributed anycast gateway The same default-gateway IP and MAC configured on every leaf so hosts route at their first hop wherever they attach. (Overlay)

Distribution block A distribution pair and the access switches beneath it, treated as one failure and summarisation domain. (Architect)

E

eBGP BGP between routers in different autonomous systems. (BGP)

ECMP Equal-cost multipath: forwarding traffic over several routes with the same cost, usually hashed per flow. (Architect)

Edge port A port expected to connect only to end hosts; it forwards immediately and does not cause topology changes. (Switch)

EIRP Effective isotropic radiated power: transmit power plus antenna gain minus losses; the value regulators limit. (Wireless)

Enrichment Adding context (device role, recent changes, topology) to an event before deciding how to respond. (Program)

Event-driven automation Workflows triggered automatically by events such as alerts, logs or tickets, rather than run by hand on a schedule. (Program)

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EVPN A BGP address family that advertises MAC, IP and prefix reachability for Layer 2 and Layer 3 overlay services. (Overlay)

F

Fast BSS Transition (FT) 802.11r roaming that derives new keys during reassociation so the client avoids a full 802.1X exchange. (Wireless)

Fine Timing Measurement (FTM) An 802.11 round-trip-time measurement that enables metre-level Wi-Fi positioning when clients and APs support it. (Wireless)

Flow key The set of match fields (for example the 5-tuple and input interface) that defines which packets belong to one flow. (Assure)

Follow-the-path A troubleshooting approach that checks forwarding hop by hop in both directions along the traffic's route. (Assure)

Front-door VRF A VRF that holds the transport interface and its routes, so tunnel endpoint lookups are isolated from the overlay routing table. (Overlay)

G

GRE A stateless encapsulation (IP protocol 47) that carries any network-layer packet, including multicast, inside a delivery IP packet. (Overlay)

Group-based policy Access rules written between groups of users or devices rather than between IP addresses. (Architect)

I

iBGP BGP between routers in the same autonomous system. (BGP)

Idempotency The property of an automation task that produces the same end state whether it runs once or many times. (Program)

IKEv2 The key-exchange protocol that authenticates IPsec peers and negotiates SAs in a small, fixed set of exchanges. (Overlay)

Infrastructure ACL (iACL) An edge ACL that permits only required protocols to the network's own device addresses while letting transit traffic pass. (Assure)

Inventory The structured list of devices, groups and variables that automation tools target. (Program)

IPFIX IP Flow Information Export (RFC 7011): the IETF standard, template-based protocol for exporting flow records. (Assure)

IST Internal spanning tree, MST instance 0; the only instance that sends BPDUs and talks outside the region. (Switch)

J

Jinja2 template A text template with variables and loops used to render device configuration from data. (Program)

K

Kill switch A single control that halts all automated actions immediately when something goes wrong. (Program)

L

LACP Link Aggregation Control Protocol (IEEE 802.1AX) for negotiating link bundles. (Switch)

LOCAL_PREF A well-known attribute shared inside an AS; higher values are preferred for outbound traffic. (BGP)

Loop guard Blocks a root or alternate port whose BPDUs stop arriving, protecting against unidirectional links. (Switch)

LSDB Link-state database; identical on every router in an OSPF area. (IGP)

M

MAB MAC Authentication Bypass: using a device's MAC address as its identity when it cannot run 802.1X. (Assure)

MAC flapping A MAC address repeatedly learned on different ports, a common sign of a loop. (Switch)

Maximum-prefix A limit on the number of prefixes accepted from a peer. (BGP)

MED Multi-exit discriminator; suggests the preferred entry point into an AS, lower is better. (BGP)

Min-links The minimum number of active members a bundle needs to stay up. (Switch)

MST region Switches sharing the same MST name, revision and VLAN-to-instance mapping. (Switch)

MTTR Mean time to repair: the average time from detecting a failure to restoring service. (Assure)

Multi-chassis link aggregation Two switches presenting one LACP identity so a device can bundle links to both. (Architect)

Multi-link operation (MLO) A Wi-Fi 7 capability that lets a device use links in several bands simultaneously or switch rapidly between them. (Wireless)

N

NET Network entity title: the IS-IS address of a router, encoding area and system ID. (IGP)

Next-hop-self Configuring a router to set itself as next hop for routes it sends to iBGP peers. (BGP)

NHRP Next Hop Resolution Protocol: maps tunnel addresses to public (NBMA) addresses so multipoint tunnels can find peers dynamically. (Overlay)

NSSA Not-so-stubby area: blocks type 5 LSAs but allows local redistribution as type 7. (IGP)

O

On-change subscription A telemetry subscription that sends an update only when the subscribed value changes. (Assure)

OpenConfig Vendor-neutral YANG models written by network operators, widely used with gNMI. (Program)

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Orchestrator The SD-WAN component that onboards edges, assigns identity and distributes the list of controllers. (Overlay)

Overlay A virtual network carried over the underlay, usually by encapsulation such as VXLAN. (Architect)

Oversubscription The ratio of downstream (host-facing) bandwidth to upstream (uplink) bandwidth at a switch. (Architect)

OWE Opportunistic Wireless Encryption (Enhanced Open): encrypts traffic on an open SSID without a shared password. (Wireless)

P

Policy-based routing Forwarding selected packets according to policy rather than the routing table. (IGP)

Preamble puncturing Transmitting on a wide channel while excluding an interfered sub-channel rather than shrinking the whole channel. (Wireless)

Prefix list An ordered list matching prefixes by address and length, with ge and le options. (IGP)

Private VLAN A VLAN subdivided into isolated and community secondary VLANs sharing one subnet. (Switch)

Promiscuous port A private VLAN port that can communicate with all isolated and community ports. (Switch)

Proposal/agreement The RSTP handshake that moves a point-to-point designated port to forwarding without timers. (Switch)

Protected Management Frames (PMF) 802.11w protection that prevents forged deauthentication and disassociation frames. (Wireless)

R

Reference bandwidth The value divided by interface bandwidth to calculate OSPF cost. (IGP)

Root guard Blocks a port that receives a superior BPDU, keeping the root where it was designed. (Switch)

Route distinguisher (RD) A 64-bit value prepended to an IPv4 or IPv6 prefix so overlapping customer routes stay unique inside MP-BGP. (Overlay)

Route reflector An iBGP speaker allowed to pass routes between iBGP peers, defined in RFC 4456. (BGP)

Route tag A numeric label attached to a route, used to make policy decisions during redistribution. (IGP)

Route target (RT) A BGP extended community attached on export and matched on import to decide which VRFs receive a route. (Overlay)

Routed access An access design where the access switch is the default gateway and uplinks are Layer 3 links. (Architect)

RPKI Resource public key infrastructure used to validate that an AS may originate a prefix. (BGP)

RSTP Rapid Spanning Tree Protocol, the handshake-based spanning tree now defined within IEEE 802.1Q. (Switch)

S

SNMPv3 authPriv The SNMPv3 security level that both authenticates and encrypts messages using per-user keys. (Assure)

SNR Signal-to-noise ratio: the difference in dB between received signal strength and the noise floor. (Wireless)

Soft reset Re-applying BGP policy without tearing down the session. (BGP)

Source of truth The authoritative, version-controlled data describing how the network should be configured. (Program)

Spine-leaf A two-tier Clos topology in which every leaf connects to every spine, giving equal-cost two-hop paths. (Architect)

Streaming telemetry Model-driven data that devices push to collectors periodically or on change, instead of being polled. (Assure)

Supplicant The endpoint software that performs 802.1X authentication using EAP over LAN. (Assure)

Switch stack Several physical switches operating with one shared control plane and management identity. (Architect)

Synthetic probe Test traffic generated at a fixed rate to measure reachability, latency, jitter and loss between two points. (Assure)

T

The ZCNP Fabric The eight-stage professional framework of this programme: Architect, Switch, IGP, BGP, Overlay, Wireless, Assure and Program. (Program)

Totally stubby area An area receiving only intra-area routes and a default route from the ABR. (IGP)

Type 3 LSA Summary LSA generated by an ABR to advertise inter-area prefixes. (IGP)

Type 5 LSA AS external LSA flooded throughout the domain except stub and NSSA areas. (IGP)

Type 7 LSA NSSA external LSA, translated into type 5 by an ABR. (IGP)

U

Underlay The physical routed network that provides reachability between fabric nodes. (Architect)

V

Virtual tunnel interface (VTI) A routable tunnel interface whose traffic is protected by IPsec, making encryption route-based. (Overlay)

VNI The 24-bit VXLAN Network Identifier that names a Layer 2 segment (L2VNI) or a tenant routing instance (L3VNI). (Overlay)

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VRF-lite VRFs configured hop by hop without an MP-BGP control plane; each VRF needs its own logical interface on every transit link. (Overlay)

VTEP VXLAN tunnel endpoint: the device (usually a leaf) that encapsulates and decapsulates VXLAN traffic. (Overlay)

W

WMM Wi-Fi Multimedia: the four access categories (voice, video, best effort, background) that prioritise traffic over the air. (Wireless)

WPA3 The current Wi-Fi security generation; it mandates PMF and offers SAE for personal and stronger suites for enterprise networks. (Wireless)

Y

YANG The data-modelling language (RFC 7950) that defines the structure, types and constraints of configuration and state data. (Program)

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CAPSTONE

Capstone: Modernise a multi-site enterprise network

Take a realistic enterprise (for example a garment exporter with a head office, two factories and a cloud presence) and produce a modernisation pack: campus redesign, dual-ISP BGP edge, SD-WAN overlay, secure wireless and an automation pipeline, validated in a lab.

Deliverables

☐ Current-state assessment and requirements

☐ Campus architecture with HA design

☐ Advanced STP/MST design with protection features

☐ Multi-area OSPF design with redistribution and filtering

☐ Dual-homed BGP edge with inbound/outbound policy

☐ VRF and SD-WAN overlay design

☐ Enterprise WLAN design with roaming and 802.1X

☐ Telemetry and assurance plan with KPIs

☐ Automation pipeline (YANG/NETCONF or Ansible) with pre-checks

☐ Migration plan with rollback

Self-assessment rubric

CRITERION	DEVELOPING	PROFICIENT	EXCELLENT
Architecture	○ Ad hoc	○ Hierarchical and redundant	○ Justified trade-offs and failure domains
Layer 2	○ Defaults	○ MST/RSTP designed	○ Protected and verified
IGP	○ Single area	○ Multi-area with summaries	○ Policy-controlled redistribution, loop-safe
BGP	○ Basic peering	○ Policies applied	○ Deterministic traffic engineering
Overlays	○ None	○ VRF/VPN working	○ SD-WAN policy aligned to apps
Wireless	○ Coverage only	○ Secure design	○ Roaming, QoS and analytics planned
Assurance	○ No visibility	○ Monitoring in place	○ Telemetry-driven KPIs and alerting
Automation	○ Manual	○ Scripts	○ Tested pipeline with rollback

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Planning pages

Sketch your design, timeline or architecture here.

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EXAM PREPARATION

Practice exam · 60 questions

The official online exam draws 40 questions at random from a separate, secret bank that tests the same skills, with a 75-minute limit and a 80% pass mark. Practise with these questions first: time yourself, mark your answers, then check the key and read every explanation, including for the questions you got right.

Architect — Enterprise Network Architecture

1. A university campus has grown from two to seven buildings, each with its own distribution pair. The buildings are currently connected to each other in a partial mesh, and every new building requires new fibre runs to several others. What should the architect recommend?

   (A) Complete the full mesh so that every building connects to every other one

   (B) Introduce a dedicated core pair that every distribution pair uplinks to

   (C) Collapse all buildings into one distribution pair in the data centre

   (D) Extend the access VLANs between buildings to simplify routing

2. A leaf switch has 48 × 25G server ports and 4 × 100G uplinks to four spines. Storage traffic is congesting the uplinks. The spines have free ports and the leaves have two unused 100G uplink ports. What is the best way to lower the oversubscription ratio?

   (A) Add two spines and connect each leaf's spare uplinks to them

   (B) Add more leaves so storage servers spread across more racks

   (C) Connect pairs of leaves directly using the spare 100G uplink ports

   (D) Upgrade the server ports from 25G to 50G on each leaf

3. Users on a routed-access campus complain that calls drop for about 40 seconds when a fibre between an access switch and its distribution switch fails through a media converter that keeps the port up. What change addresses this most directly?

   (A) Increase the OSPF dead interval on both ends to avoid flapping

   (B) Change the access switch to Layer 2 and run RSTP on the uplinks

   (C) Add a floating static default route on the access switch through each uplink

   (D) Enable BFD on the routed uplinks so the IGP learns of the failure quickly

4. A manufacturer needs both uplinks from each access switch to forward traffic, and several production VLANs must remain available on access switches in different halls. Which design meets both requirements?

   (A) Routed access with ECMP uplinks and a separate gateway on every access switch

   (B) Classic Layer 2 access with RSTP and VRRP on the distribution pair

   (C) Layer 2 access with multi-chassis link aggregation to the distribution pair

   (D) A single distribution switch with one uplink per access switch

5. During a design review, an engineer proposes connecting the new internet firewalls directly to two floor access switches because they have free 10G ports. What is the strongest objection?

   (A) Access switches cannot reliably forward firewall traffic at 10 Gbit/s rates

   (B) Firewalls must always be connected to the spine layer

   (C) Internet traffic must not use any IPv4 addresses inside

   (D) Edge functions belong in their own module off the core or an edge block

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6. A bank wants rules such as 'contractors may reach only the ticketing system' to remain valid as users roam between floors and receive different IP addresses. Which approach fits best?

   (A) Per-floor VLAN access lists written around contractor IP addresses
   (B) Group-based policy assigned at authentication in a fabric campus
   (C) Static IP addresses for every contractor laptop and phone
   (D) A separate physical network built only for contractors

7. A distribution pair runs VRRP for VLAN 30. DS1 is the VRRP master, but the spanning-tree root for VLAN 30 is DS2. What is the likely effect?

   (A) Traffic crosses the inter-switch link before reaching the gateway
   (B) VRRP fails, because the master must also be the spanning-tree root
   (C) Hosts in VLAN 30 receive two different default gateways
   (D) Spanning tree blocks the VRRP hello packets between them

Switch — Advanced Layer 2

8. After a floor switch was replaced, users report that their PCs take about 30 seconds to get network access after boot, but uplinks converge instantly. `show spanning-tree` shows the host ports as `Desg LRN` for several seconds after link up, with type `P2p`. What is missing?

   (A) Loop guard configuration on the host ports
   (B) A lower bridge priority on the floor switch
   (C) Edge (PortFast) configuration on the host ports
   (D) Root guard configuration on the uplink ports

9. Two distribution switches run MST. After a change, `show spanning-tree mst configuration` shows identical names and mappings, but DS1 shows revision 4 and DS2 revision 3. What is the result?

   (A) They remain in one region because the names and mappings both match
   (B) They are in different regions and their link becomes a boundary
   (C) Only instance 0 stops working until the switches reload
   (D) DS2 automatically adopts revision 4 from its neighbour

10. A contractor's switch, configured with priority 0, was connected to a distribution switch's downlink port and became root. The port is a trunk to an access switch. Which feature on that port would have kept the original root?

    (A) BPDU guard
    (B) Loop guard
    (C) Root guard
    (D) BPDU filter

11. `show port-channel summary` on DS1 shows `Po2(SU) LACP Te1/0/1(P) Te1/0/2(P) Te1/0/3(s)`. Te1/0/3 was added yesterday. What should the engineer check first?

    (A) Whether Te1/0/3's configuration matches the bundle (VLANs, speed, mode)
    (B) Whether the LACP rate on Te1/0/3 is fast instead of slow
    (C) Whether the partner switch uses passive mode on its links
    (D) Whether Te1/0/3 has a lower LACP port priority than the two other members

12. A data-centre interconnect fibre develops a fault in one direction. Shortly afterwards a Layer 2 loop forms through the alternate port on the downstream switch. Which feature would most directly have prevented the loop?

    (A) BPDU guard on all trunk ports
    (B) Root guard on the downstream switch
    (C) Storm control on all edge and uplink ports
    (D) Loop guard on root and alternate ports

13. Users in VLAN 40 on AS5 cannot reach their gateway; VLANs 10 and 20 on the same switch work. On the distribution switch, `show interfaces trunk` lists `Vlans allowed on trunk: 10,20,30` for the link to AS5. What is the fix?

    (A) Add VLAN 40 to the allowed list on the trunk
    (B) Change the native VLAN on the trunk to 40
    (C) Create an SVI for VLAN 40 on the access switch
    (D) Enable root guard on the trunk towards AS5

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14. A hosting provider wants 40 customer servers on one /24 subnet. Customers must not reach each other at Layer 2, but all must reach the shared gateway and a shared backup server. How should the ports be configured?

    (A) Customers on community ports; gateway on an isolated port
    (B) Customers on promiscuous ports; gateway on a community port
    (C) Customers on isolated ports; gateway and backup server on promiscuous ports
    (D) All servers in one shared community, with the backup server on an isolated port

15. During an outage, logs show `Host 0050.56a1.22b3 in vlan 20 is flapping between port Te1/0/13 and port Te1/0/14` and topology changes are rising by hundreds per minute. What is the best first action?

    (A) Clear the MAC address table repeatedly until it settles
    (B) Disable spanning tree on VLAN 20 to stop the changes
    (C) Reload the root bridge to force a completely fresh election everywhere
    (D) Shut the suspected looping link to stop the loop, then investigate

IGP — OSPF and Route Control

16. R1 and R2 are directly connected in area 0. `show ip ospf neighbor` on R1 shows R2 stuck in `EXSTART/DR`. Hello and dead timers match. What is the most likely cause?

    (A) The OSPF process IDs differ between R1 and R2
    (B) The interface MTU differs between R1 and R2
    (C) R2 is configured with an OSPF priority of 255
    (D) Both routers use the same reference bandwidth

17. A store area contains small routers that must not receive the 9,000 external routes from the data centre, but each store redistributes a local static route for guest internet. Which area type should be used?

    (A) Totally NSSA
    (B) Totally stubby
    (C) Stub
    (D) Normal area

18. A router with the default OSPF reference bandwidth of 100 Mbit/s has a 1G path of three links and a 100G path of four links to the same destination. All traffic uses the 1G path. Why?

    (A) OSPF always prefers the path made of the lowest-bandwidth interfaces
    (B) The 100G links are not supported by OSPF without IS-IS
    (C) Every link of 100 Mbit/s or faster costs 1, so three hops beat four
    (D) OSPF prefers paths learned first when costs are close

19. After mutual redistribution between OSPF and IS-IS was enabled on two boundary routers, some IS-IS routes appear in IS-IS as external routes originated by the other boundary router. What is the best fix?

    (A) Remove one boundary router so that only one redistribution point remains
    (B) Increase the IS-IS seed metric for all routes redistributed from OSPF
    (C) Use policy-based routing to forward IS-IS prefixes correctly
    (D) Tag routes when redistributing and deny the opposite tag on both routers

20. An ABR summarises area 10 with `area 10 range 10.10.0.0 255.255.240.0`. Which set of subnets is fully covered by this summary?

    (A) 10.10.0.0/24 through 10.10.31.0/24
    (B) 10.10.0.0/24 through 10.10.15.0/24
    (C) 10.10.8.0/24 through 10.10.23.0/24
    (D) 10.10.0.0/24 through 10.10.7.0/24 only

21. Debug output on R1 shows `Mismatched hello parameters from 10.1.12.2` followed by `Dead R 40 C 120, Hello R 10 C 30`. What should be changed?

    (A) Make R1's hello and dead timers match its neighbour's 10 and 40 seconds
    (B) Make R1's network type broadcast so that it matches the neighbour's setting
    (C) Increase the neighbour's dead timer to 160 seconds
    (D) Change the area ID on R1 to match the neighbour

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22. A new IPv6-only branch router running OSPFv3 never forms adjacencies. `show ospfv3` reports that the process cannot start. The router has no IPv4 addresses. What is the most likely cause?

    (A) OSPFv3 needs a global unicast address on every link
    (B) No 32-bit router ID has been configured for the process
    (C) OSPFv3 requires an IPv4 address on each interface
    (D) OSPFv3 cannot run without an IS-IS process for IPv6

23. Guest traffic from 10.50.0.0/16 to TCP port 443 must leave via a cheap broadband router at 203.0.113.1, while all other traffic follows the routing table. What is the right tool?

    (A) A static default route pointing to 203.0.113.1 on the core
    (B) A route map redistributing 10.50.0.0/16 into OSPF as external type 1
    (C) A prefix list denying 10.50.0.0/16 from OSPF updates
    (D) Policy-based routing applied inbound on the guest gateway interface

BGP — BGP for the Enterprise

24. An eBGP session to ISP-B stays in Active. The ISP's router at 192.0.2.9 replies to ping from the enterprise router's outside interface. The firewall in front of the edge router was recently replaced. What should be checked first?

    (A) Whether LOCAL_PREF is set higher for the routes received from ISP-B
    (B) Whether the AS path is prepended towards ISP-B
    (C) Whether the new firewall permits TCP port 179 between the peers
    (D) Whether communities are being sent to ISP-B

25. An internal router shows the default route from E1 as `203.0.113.1 (inaccessible)` with `no best path`. E1's eBGP peer is 203.0.113.1. What is the most appropriate fix?

    (A) Set a higher LOCAL_PREF on E1 for routes from ISP-A
    (B) Configure ebgp-multihop on the internal router
    (C) Advertise the default route with a lower MED value internally
    (D) Configure next-hop-self on E1 for its iBGP neighbours

26. An enterprise multihomed to two ISPs sees its links filled with traffic between the ISPs' customers. Its outbound route maps contain only `route-map TO-ISP permit 10`. What is the correct fix?

    (A) Apply a maximum-prefix limit on each eBGP session
    (B) Match only its own prefixes and an empty AS path in outbound policy
    (C) Raise LOCAL_PREF on all routes learned from the busier of the two providers
    (D) Stop advertising the default route to internal routers

27. Paths to the same prefix: path 1 has LOCAL_PREF 100 and AS_PATH length 2; path 2 has LOCAL_PREF 150 and AS_PATH length 5. Both next hops are reachable. Which is selected and why?

    (A) Path 2, because LOCAL_PREF is compared before AS_PATH length
    (B) Path 1, because a shorter AS_PATH is always preferred before anything
    (C) Path 1, because it has the lower LOCAL_PREF value
    (D) Both, because BGP load-shares across different paths

28. A company has 30 iBGP routers in a full mesh and plans to reach 60. Operations wants to reduce configuration effort without changing AS numbers. What should be proposed?

    (A) A confederation splitting the AS into six sub-ASes
    (B) Redistributing BGP into the IGP to remove iBGP sessions
    (C) A redundant pair of route reflectors with all other routers as clients
    (D) Replacing iBGP with eBGP sessions between every pair of routers in the AS

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29. An enterprise wants inbound traffic to arrive mainly over ISP-A, using ISP-B only as backup. It already prepends three times towards ISP-B, but some large networks still send traffic through ISP-B. What is the most effective additional step?

(A) Prepend ten more times towards ISP-B to strengthen the effect of the AS path

(B) Use ISP-B's documented community to lower local preference inside ISP-B

(C) Send a lower MED to ISP-A than to ISP-B on every route

(D) Raise LOCAL_PREF on routes received from ISP-A internally

30. RR1 peers with edge routers E1 and E2 over iBGP, but the configuration has no route-reflector-client statements. E2 never learns the prefixes E1 receives from its ISP, although RR1 has them. Why?

(A) E2 needs ebgp-multihop configured to receive routes from any reflector

(B) RR1 must change the next hop before advertising to E2

(C) Without client configuration RR1 obeys the iBGP split-horizon rule

(D) E1's routes carry a CLUSTER_LIST that RR1 rejects

Overlay — Overlay — Virtualisation & Overlays

31. A company adds an IOT VRF to its campus. Transit links between distribution and core use VRF-lite, and the IoT management platform sits in a shared-services VRF. IoT devices reach each other but not the platform. What should the engineer do FIRST?

(A) Give the IOT VRF the same route distinguisher as the shared-services VRF on every hop

(B) Move the IoT management platform into the global routing table so all VRFs can see it

(C) Leak the platform prefix into IOT and the IoT subnets back, via the policy point

(D) Replace VRF-lite with a single flat VLAN for IoT devices and the management platform

32. A GRE tunnel between HQ and a branch flaps roughly every minute. The logs show the tunnel being disabled because of recursive routing. The branch advertises its WAN subnet into OSPF over the tunnel. What is the most effective fix?

(A) Increase the OSPF cost on the tunnel so that the physical path is preferred

(B) Shorten the GRE keepalive timers so the tunnel recovers more quickly each time

(C) Change the tunnel mode to IP-in-IP

(D) Keep the WAN subnet out of the overlay IGP, or use a front-door VRF

33. An IKEv2 tunnel to a partner fails during IKE_SA_INIT with a 'no proposal chosen' notification. Authentication has not yet been attempted. What is the most likely cause?

(A) The pre-shared key configured on the partner's router does not match HQ's key

(B) The partner's routing table has no return route to the tunnel subnet

(C) The peers do not share a common encryption, PRF or Diffie-Hellman group

(D) The tunnel IP MTU is larger than the path MTU between the two peers

34. On a route-based IPsec tunnel, `#pkts encaps` increases on HQ but `#pkts decaps` stays at zero, while the IKEv2 SA is established. What should be investigated next?

(A) The IKEv2 proposal and transform set configured on the HQ router

(B) The branch's return routing, NAT or filtering into its tunnel

(C) The certificate chain used by HQ to authenticate itself to the branch

(D) The GRE keepalive configuration on both of the tunnel interfaces

35. In a hub-and-spoke dynamic multipoint VPN, spoke-to-spoke traffic always passes through the hub even though direct tunnels are expected. Spokes are registered with the hub. Which configuration is most likely missing?

(A) NHRP redirect on the hub and NHRP shortcut on the spokes

(B) A route distinguisher on the hub's multipoint tunnel interface

(C) A separate point-to-point GRE tunnel between every pair of spokes

(D) IPsec NAT traversal on the hub's internet-facing interface

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36. A new VXLAN EVPN fabric passes all ping tests between servers, but database replication and backups between racks stall. Which check should come FIRST?

(A) The BGP EVPN route-target import configuration on every leaf switch

(B) A full-size do-not-fragment ping between VTEP loopbacks

(C) Spanning-tree root placement on the spines

(D) The anycast gateway MAC address configured on each leaf switch

37. A leaf receives a Type 2 EVPN route for a remote server in VNI 20020 but does not install it, and the servers cannot communicate. What is the most likely cause?

(A) Its route targets do not match the local import policy for VNI 20020

(B) The underlay MTU on the spine links is set to 1,500 bytes

(C) The local leaf has no Type 5 route for the tenant VRF's subnet prefix in BGP

(D) The remote leaf is using a different UDP source port for VXLAN

38. An SD-WAN branch has MPLS and broadband. Video calls to a cloud collaboration service hairpin through the data centre and perform poorly. Security policy requires all internet traffic to be inspected. What design best improves the experience?

(A) Send all internet traffic directly from the branch and remove inspection for speed

(B) Increase the MPLS bandwidth so the hairpinned video has more capacity available

(C) Local breakout for the approved service, with cloud or edge security inspection

(D) Move video traffic into the bulk SLA class so it uses the cheapest available path

Wireless — Wireless — Enterprise Wireless

39. A voice WLAN shows strong signal everywhere in a survey, but handset users report one-way audio near the edges of cells. AP transmit power is set to maximum. What is the most likely cause?

(A) DFS radar events moving the APs to new channels during every call

(B) 802.11k neighbour reports listing too many candidate APs for the handsets

(C) WMM mapping voice into the video access category instead of voice

(D) Link asymmetry: handsets hear the APs, but the APs miss the handsets

40. A university lecture hall needs Wi-Fi for 300 laptops and phones. The design team proposes 160 MHz channels on 5 GHz for maximum speed. What is the best recommendation?

(A) Use narrower channels for reuse, and wider ones only on 6 GHz

(B) Approve 160 MHz on 5 GHz because wider channels always increase total capacity

(C) Disable 5 GHz and serve all students on 2.4 GHz

(D) Use one high-power AP at the front of the hall to avoid co-channel interference

41. Clinicians' voice badges pause for about half a second when roaming on an 802.1X SSID; the RADIUS server is in a remote data centre. Which change most directly reduces roam time?

(A) Increase the AP transmit power on every floor so that the badges roam less often

(B) Enable 802.11r Fast BSS Transition after testing the badges for compatibility

(C) Move the voice SSID to 2.4 GHz so the badges stay connected longer

(D) Disable protected management frames on the clinical voice SSID to speed up roams

42. A security audit finds that staff laptops connect to a look-alike SSID in the car park and submit credentials. The corporate SSID uses PEAP-MSCHAPv2. What is the most effective remediation?

(A) Hide the corporate SSID so that it is no longer broadcast in beacons

(B) Increase the minimum password length for every user account

(C) Reduce AP transmit power so the SSID does not reach the car park

(D) EAP-TLS with device certificates and enforced server validation

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43. A new building will use 6 GHz radios. The team plans to reuse the existing WPA2-Enterprise SSID on all bands. What is the issue?

(A) 6 GHz radios only support personal security with a shared passphrase

(B) WPA2-Enterprise requires 320 MHz channels when it is used on 6 GHz

(C) 6 GHz requires WPA3 or OWE, so the SSID cannot use 6 GHz

(D) 6 GHz radios cannot use RADIUS

44. A voice SSID carries traffic marked DSCP EF from managed handsets, but calls degrade when the cell is busy with downloads. The AP maps all traffic to best effort. What should be configured?

(A) Map DSCP EF to user priority 1 so voice uses the background category

(B) Disable WMM so that all traffic shares a single fair queue on the AP

(C) Raise the AP transmit power

(D) Map DSCP EF to user priority 6, the AC_VO access category

45. A retailer wants to count visitors in each store using Wi-Fi analytics and compare footfall week by week. What limitation should be explained first?

(A) Randomised MACs make counts estimates; one phone may appear as several

(B) Wi-Fi analytics can only see devices that are associated to a guest SSID

(C) Location engines only work when the APs operate in the 2.4 GHz band

(D) Footfall cannot be measured with Wi-Fi without UWB tags on every visitor

Assure — Assure — Assurance, Security & Troubleshooting

46. A NOC uses SNMPv2c with the community string 'public' across 800 devices. An audit requires authenticated and encrypted management traffic. What is the best approach?

(A) Change the community string to a long random value on every device

(B) Migrate to SNMPv3 authPriv users with read-only views and source ACLs

(C) Move SNMP polling to a dedicated VLAN and keep SNMPv2c unchanged

(D) Replace SNMP polling with debug-level syslog sent to the collector

47. A WAN link is congested every morning. Interface graphs confirm high utilisation but not its cause. Which data source should the engineer use to identify the top conversations?

(A) SNMP interface counters polled at a much shorter interval during the busy period

(B) Syslog messages from the router during the busy period

(C) IPFIX flow records exported from the router for the busy period

(D) Control-plane policing counters on the WAN router

48. Customers report one-minute outages on a ring several times a week. Five-minute SNMP polling shows nothing. What change gives the NOC visibility of these events?

(A) 30-second SNMP polling everywhere

(B) A daily configuration backup compared with the previous day's backup

(C) On-change state telemetry plus 10–30 s counter sampling

(D) Debug logging enabled on all ring routers and sent to the collector

49. A branch uses a tracked default route with an ICMP probe to the primary ISP and a floating static route to the backup ISP. During a primary ISP outage, the branch keeps flapping between the two ISPs. What is the most likely cause?

(A) The floating static route has a lower administrative distance than the tracked route

(B) The probe frequency is too low to detect the outage within a few seconds

(C) The probe uses ICMP instead of a UDP jitter operation with a responder

(D) The probe target is reachable via the backup path, so the probe recovers

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50. After a routing-policy change, users at one branch cannot reach the data centre. A VRF-aware traceroute from the branch reaches the DC core and then times out. What is the next best step?

(A) Replace the branch router because the traceroute does not complete
(B) Check that the DC has a route back to the branch in that VRF
(C) Check the branch's Wi-Fi channel plan for co-channel interference
(D) Reboot the DC core to clear any stale routing state from memory

51. An organisation wants to enforce 802.1X on 4,000 access ports without disrupting users or printers. What is the best rollout approach?

(A) Monitor mode first, fix supplicant and MAB gaps, then low-impact and closed
(B) Enable closed mode on all ports in one weekend and fix failures as they are reported
(C) Use multi-host mode everywhere so one authenticated device opens each port
(D) Rely on MAB for all devices because it needs no supplicant configuration

52. Security asks how a camera authenticated by MAB should be treated compared with an EAP-TLS laptop. What is the best answer?

(A) As equal trust, since both methods are accepted by the same RADIUS server
(B) As higher trust, because cameras rarely move between switch ports
(C) As untrusted, so it must be placed on the guest SSID and VRF
(D) Lower trust: narrow policy plus profiling, since MACs can be spoofed

53. During a flood towards a core router's own addresses, OSPF and BGP sessions drop although links are not saturated. Which combination gives the best protection?

(A) An edge infrastructure ACL plus CoPP protecting routing protocols
(B) Longer OSPF and BGP hold timers to keep sessions up during floods
(C) A router with a faster CPU so it can process the flood traffic
(D) Blocking all ICMP on every interface of the core routers

Program — Program — Network Automation & NetDevOps

54. A team must apply a multi-part BGP change to 60 routers so that each router gets all of it or none of it, with automatic rollback if access is lost. All routers support NETCONF with a candidate datastore. Which method fits best?

(A) A series of RESTCONF PATCH requests, one per configuration section
(B) NETCONF edit-config to candidate, validate, then a confirmed commit
(C) A gNMI Subscribe request in on-change mode for each BGP neighbour
(D) CLI commands sent over SSH and checked with regular expressions

55. A RESTCONF GET for interface GigabitEthernet0/0/1 returns 404 although the interface exists. The URL ends in `interface=GigabitEthernet0/0/1`. What is the most likely issue?

(A) RESTCONF cannot read interfaces whose names contain digits
(B) The request used JSON instead of XML for the Accept header
(C) The slashes in the key value were not percent-encoded as %2F
(D) The interface must first be locked with a NETCONF lock operation

56. A Python script adds an ACL entry to edge routers. Run twice, it created duplicates and broke the ACL. What is the best fix?

(A) Add a warning comment at the top of the script about running it twice
(B) Diff intended against current state and apply only the difference
(C) Schedule the script to run only during weekend maintenance windows
(D) Rewrite the script in another language with better error handling

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57. A pull request changes the GUEST VRF import list. Lint passes but an intent test that forbids GUEST importing CORP routes fails. The change is urgent. What should happen?

(A) Merge the change now and disable the failing test until next week's release

(B) Apply the change manually on the devices and skip the pipeline

(C) Delete the intent test because it is blocking an urgent change

(D) Keep the gate: investigate, then fix the data or formally change the rule

58. A NetDevOps team wants to limit the impact of a bad change that passed all lab tests. Which pipeline practice helps most?

(A) Canary waves with automated post-checks and rollback

(B) A larger number of linting rules applied to the YAML files

(C) A faster template rendering stage in the build environment

(D) Two approvers on every merge request

59. An operations team proposes that event-driven automation should immediately shut any access uplink that flaps more than five times. What is the best first implementation?

(A) Shut the uplink automatically from day one because uplinks are redundant

(B) Let an AI assistant decide whether to shut each uplink and apply it directly

(C) Collect diagnostics and open a ticket first; add shutdown later with safeguards

(D) Handle all flaps manually and avoid automation for interface events

60. An engineer wants to paste full router configurations into an external AI tool to help troubleshoot. What guardrail applies first?

(A) Redact secrets and customer data, and use only approved tools

(B) Paste the configurations in smaller pieces to fit the input size

(C) Convert the configurations to JSON before pasting them into the tool

(D) Ask the tool to delete the configurations after it has answered

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EXAM PREPARATION

Practice exam answers & explanations

1. B With seven blocks a full mesh needs 21 links and keeps growing quadratically; a core makes growth linear and isolates each block. Collapsing everything creates one huge failure domain, and stretching VLANs spreads Layer 2 failures across buildings.

2. A Today the ratio is 1,200G ÷ 400G = 3:1; six uplinks give 1,200G ÷ 600G = 2:1. Leaf-to-leaf links break the fabric model, more leaves do not add uplink capacity, and faster server ports make oversubscription worse.

3. D The port stays up, so OSPF waits for its 40-second dead interval. BFD detects path loss in under a second. Longer timers make it worse; returning to Layer 2 adds spanning tree without solving detection; static routes would not detect the failure either.

4. C Multi-chassis aggregation keeps VLANs able to span switches while both uplinks forward as one bundle. Routed access confines VLANs to one switch, classic RSTP blocks an uplink, and a single switch has no redundancy.

5. D Edges such as internet, WAN and DMZ are separate modules so they can be secured, scaled and changed independently of user access. Port speed and spine placement are not the issue.

6. B Group-based policy follows identity rather than address, so rules stay valid as users move. Address-based ACLs break when addresses change, static addressing does not scale, and a separate physical network is costly and rigid.

7. A Frames follow the spanning tree towards DS2 and then cross to DS1 to be routed. It still works, but it is inefficient; aligning root and master fixes it.

8. C Non-edge ports with no RSTP peer cannot complete proposal/agreement and wait through timers. Edge ports forward immediately. Loop guard and root guard do not speed convergence, and priority affects root placement only.

9. B The region identifier is name, revision and mapping digest; any difference creates separate regions. Revision numbers are not synchronised automatically.

10. C Root guard on designated downlinks blocks superior BPDUs (root-inconsistent) while allowing normal BPDUs. BPDU guard would shut the trunk to a legitimate access switch, loop guard addresses missing BPDUs, and filter removes protection.

11. A Suspended members are incompatible with the bundle, usually through mismatched configuration. Rate and port priority do not cause suspension, and a passive partner still bundles with an active side.

12. D When BPDUs stop on a unidirectional link, the alternate port ages them out and moves to forwarding. Loop guard holds it in loop-inconsistent state. Storm control limits damage but does not prevent the loop.

13. A VLAN 40 is not carried by the trunk. Making it native would carry it untagged but breaks consistency; an SVI on the access switch would not bring the VLAN to the gateway.

14. C Isolated ports reach only promiscuous ports, which suits customer separation. Promiscuous ports for shared services let every customer reach them.

15. D MAC flapping with constant topology changes indicates a loop; isolating the link restores service while evidence is collected. Clearing tables does not stop the loop, disabling spanning tree can worsen it, and reloading the root causes a wider outage.

16. B ExStart/Exchange problems are classically caused by MTU mismatch (or duplicate router IDs). Process IDs are locally significant, priority affects DR election, and matching reference bandwidth is desirable.

17. A Totally NSSA blocks type 3, 4 and 5 LSAs (replacing them with a default) but allows type 7 externals inside the area. Stub types forbid redistribution; a normal area keeps all externals.

18. C With reference 100 Mbit/s, both 1G and 100G cost 1, so the path cost is 3 versus 4. Raising the reference bandwidth consistently fixes this.

19. D Tags identify a route's origin and prevent it from re-entering its home domain. Removing a boundary router loses redundancy; metrics alone do not prevent feedback; PBR acts on packets, not routes.

20. B 255.255.240.0 is /20, which covers 16 /24s from 10.10.0.0 to 10.10.15.0. The /24s up to .31 would need a /19; a range starting at .8 is not aligned. The 0–7 range is covered but is not the full set.

21. A R is received and C is configured: the neighbour uses 10/40, R1 uses 30/120. The timers must match on both sides; the area ID and network type are not indicated by this message.

22. B OSPFv3 still uses a 32-bit router ID; with no IPv4 address to derive one from, it must be configured. OSPFv3 runs over link-local addresses and does not need IPv4 interfaces.

23. D PBR matches packets by source and port and overrides the routing table only for them. A static default would move all traffic; redistribution and prefix lists change route advertisements, not per-packet forwarding.

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24. C Active means TCP is not completing. Ping proves basic reachability, so a filter on TCP 179 is the likely culprit. The other options are policy settings that only matter once the session is up.

25. D The internal router cannot reach the external next hop. Next-hop-self replaces it with E1's loopback, which the IGP carries. Attribute changes cannot make an unreachable next hop valid.

26. B A permit-all outbound policy re-advertises each provider's routes to the other, making the enterprise a transit AS. Max-prefix, local preference and internal defaults do not stop the leak.

27. A After next-hop reachability (and any implementation-specific weight), LOCAL_PREF is the first attribute compared; higher wins, so AS_PATH length is never reached.

28. C A full mesh of 60 would need 1,770 sessions; two reflectors need about 117. Confederations require reconfiguring every router into sub-ASes; redistributing a full BGP table into the IGP is unsafe; eBGP everywhere changes the design entirely.

29. B Networks that prefer ISP-B by local preference never compare AS-path length; a provider community changes the decision inside ISP-B itself. Extra prepends have diminishing returns, MED is not compared across providers, and internal LOCAL_PREF controls outbound traffic only.

30. C A router only reflects iBGP-learned routes to peers marked as clients (or from clients to non-clients). Without it, normal iBGP rules stop RR1 passing E1's routes to E2. Multihop is for eBGP, next hop changes are not required, and no cluster list exists before reflection.

31. C Inter-VRF reachability needs routes leaked in both directions at a controlled point. RDs never control import, the global table is not visible from a VRF, and flattening the network removes segmentation.

32. D The tunnel destination is being learned through the tunnel. Separating transport and overlay routing removes the loop; cost changes lose to longer prefixes, and timers or encapsulation changes do not address the cause.

33. C IKE_SA_INIT negotiates algorithms and DH. A PSK mismatch appears later in IKE_AUTH; routing and MTU problems appear after the SA is established.

34. B A working SA with traffic leaving but none returning points to the far side not sending protected traffic back. Crypto parameters are already agreed.

35. A Redirect tells the spoke a better path exists; shortcut lets it resolve the peer and install the direct route. Static spoke-to-spoke tunnels defeat the purpose of the design.

36. B Size-dependent failures point to MTU: VXLAN adds 50 bytes, so the underlay must carry more than the host MTU. Route-target problems break all traffic, not only large packets.

37. A Received but not installed means import policy rejected it, and RTs are that policy. MTU affects data, Type 5 carries prefixes, and source ports vary by design.

38. C Local breakout for trusted SaaS removes the hairpin, while consistent inspection meets the security requirement. More MPLS bandwidth does not remove the added latency.

39. D High AP power with low client power creates links that work in one direction only. Matching power and fixing placement resolves it.

40. A High density is limited by contention; more, narrower channels allow more APs to work in parallel. 6 GHz has room for wider channels.

41. B FT removes the full EAP exchange with the remote RADIUS server from each roam. Power and band changes do not shorten the key exchange; disabling PMF weakens security.

42. D Certificate-based authentication with enforced server validation removes the credential-harvesting attack. Hidden SSIDs, password policy and power changes do not stop impersonation.

43. C 6 GHz operation requires WPA3 or OWE with PMF; legacy WPA2 SSIDs cannot be broadcast there.

44. D WMM prioritisation requires voice to land in AC_VO; EF maps to UP 6 per RFC 8325 guidance.

45. A Randomised MACs affect counting accuracy. Probe-based analytics can see unassociated devices, but identities are unstable, so treat counts as trends.

46. B Only SNMPv3 authPriv provides authentication and encryption. A new community string is still sent in clear text.

47. C Flow records show who talked to whom and how much. Counters, syslog and CoPP statistics cannot attribute traffic to conversations.

48. C On-change subscriptions report state changes immediately and fine-grained sampling catches short events, without the load of aggressive polling.

49. D Once the primary route is withdrawn, probes follow the backup default and succeed. Pin the probe target to the primary path with a host route.

50. B Forward traffic reaches the core, so the return path is the prime suspect. Following the path in both directions is faster and safer than reboots or replacements.

51. A A phased rollout reveals failures before enforcement. Multi-host and MAB-only designs give weak assurance, and a big-bang closed mode risks widespread outages.

52. D MAB proves only a MAC address. Profiling and least-privilege policy limit what a spoofed MAC could reach.

53. A The iACL stops most traffic to device addresses at the border and CoPP protects what reaches the CPU. Timers, hardware and blanket ICMP blocking do not address the cause.

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54. B Candidate plus confirmed commit gives an atomic change with automatic rollback. Separate PATCH calls commit individually; Subscribe is telemetry.

55. C Reserved characters in list keys must be percent-encoded; otherwise the server interprets them as path separators.

56. B Idempotency comes from comparing desired and actual state. Comments, timing and language choice do not fix the logic.

57. D The test protects a security boundary. Bypassing or deleting it, or changing devices out of band, creates risk and drift.

58. A Canary waves with post-checks catch production-only issues on a few devices and reverse them. More linting or approvers cannot detect runtime behaviour.

59. C Read-only automation adds value safely; any automatic change needs conditions such as a healthy redundant path and rate limits. Unreviewed AI actions bypass guardrails.

60. A Configurations contain secrets and sensitive data. Redaction and approved, controlled tools come before any analysis.

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NOTES

My notes

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